

电子电路与系统基础(A1)---电阻电路

第13讲：三种组态

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清华大学电子工程系

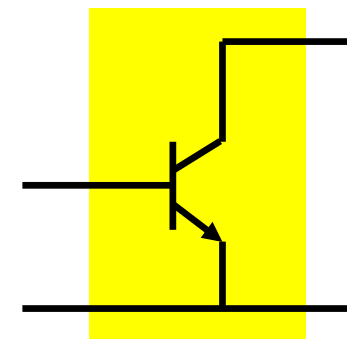
A班课程内容安排

第一学期：电阻	序号	第二学期：动态
电路定律	1	电容电感
电阻电源	2	一阶时域
分析方法	3	正弦稳态
网络参量	4	一阶时频
典型网络	5	二阶时域
电子基础	6	二阶时频
二极管	7	二阶应用
MOSFET	8	寄生效应
BJT	9	频率特性
反相电路	10	作业选讲
数字门	11	负反馈
放大器	12	正反馈
三种组态	13	振荡器
差分放大	14	期末复习
期末复习	15	总复习

三种组态晶体管的恒流区模型 内容

■ 三种组态

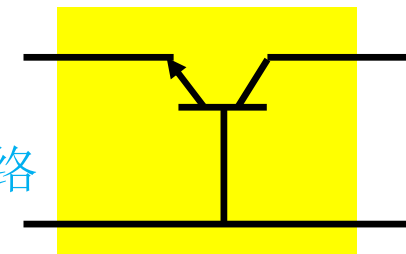
- 晶体管的三个端点，取其中之一作为公共端点形成二端口网络
 - 发射极为公共端点：共射组态：Common Emitter
 - 基极为公共端点：共基组态：Common Base
 - 集电极为公共端点：共集组态：Common Collector



■ 三组组态可以共用一个通用跨导器模型

- 考虑 r_{be} 、 r_{ce} 影响的跨导器模型
- 输入电阻、输出电阻、增益----作业选讲

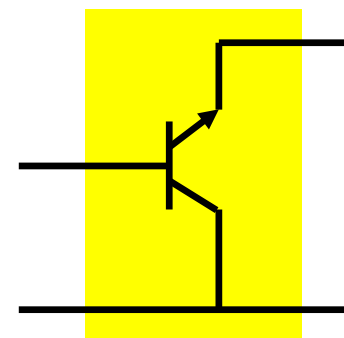
单向网络/双向网络



■ 理想晶体管 ($r_{be} \rightarrow \infty, r_{ce} \rightarrow \infty$) 三种组态的理想模型

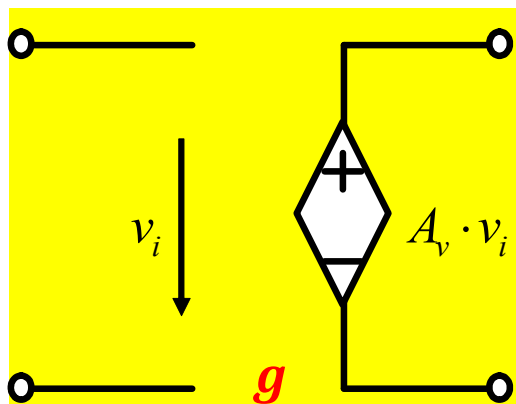
- CE: 理想跨导模型：理想压控流源模型
- CB: 电流缓冲模型：流控流源模型
- CC: 电压缓冲模型：压控压源模型

单向网络
单向网络
单向网络



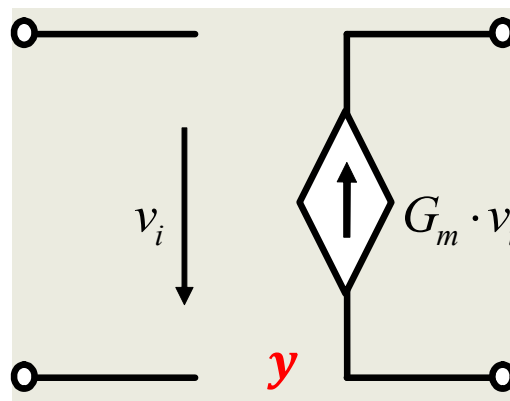
■ 一个应用举例：什么时候采用理想晶体管模型？

四种理想受控源



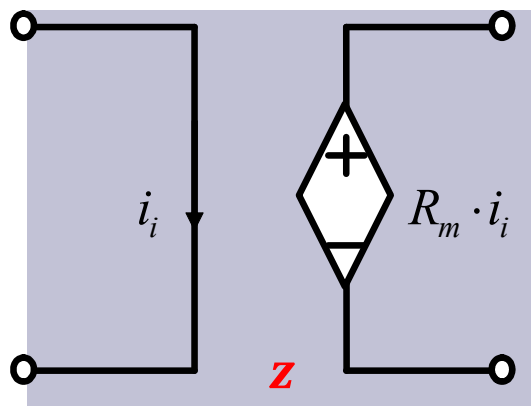
VCVS

$$\begin{bmatrix} i_1 \\ v_2 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ A_v & 0 \end{bmatrix} \begin{bmatrix} v_1 \\ i_2 \end{bmatrix}$$



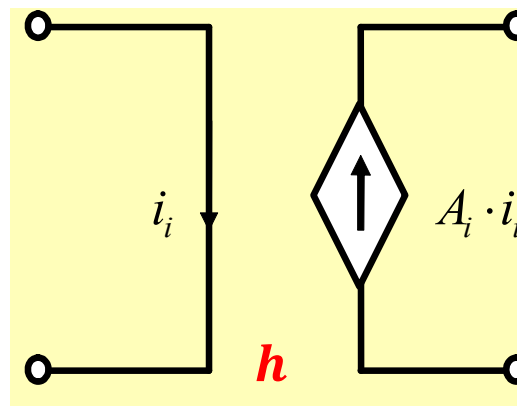
VCCS

$$\begin{bmatrix} i_1 \\ i_2 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ -G_m & 0 \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \end{bmatrix}$$



CCVS

$$\begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ R_m & 0 \end{bmatrix} \begin{bmatrix} i_1 \\ i_2 \end{bmatrix}$$



CCCS

$$\begin{bmatrix} v_1 \\ i_2 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ -A_i & 0 \end{bmatrix} \begin{bmatrix} i_1 \\ v_2 \end{bmatrix}$$

理想压控源
输入阻抗无穷大（开路）

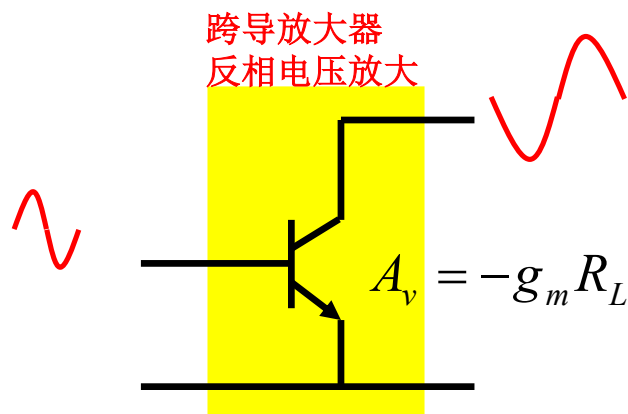
理想流控源
输入阻抗为零（短路）

理想受控电压源
输出阻抗为零

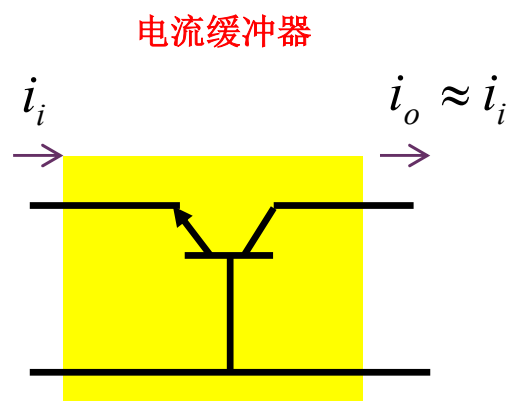
理想受控电流源
输出阻抗无穷大

晶体管的三种组态的直观理解

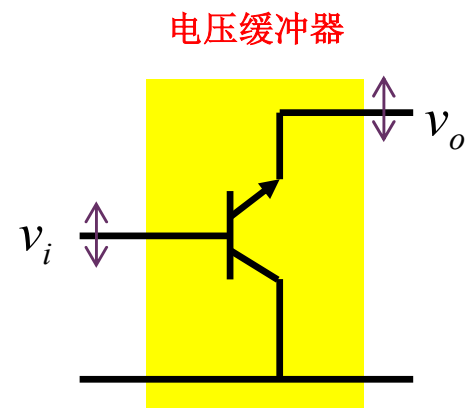
同相放大



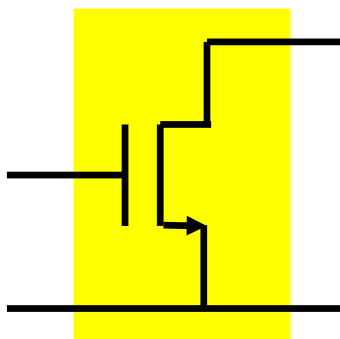
Common Emitter
CE: 共射组态



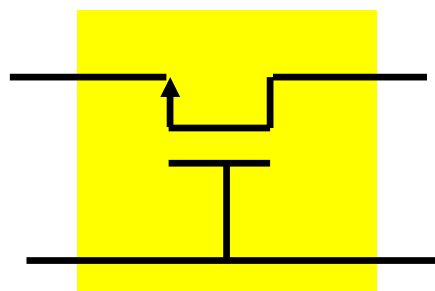
Common Base
CB: 共基组态



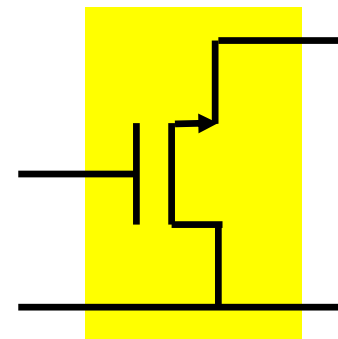
Common Collector
CC: 共集组态



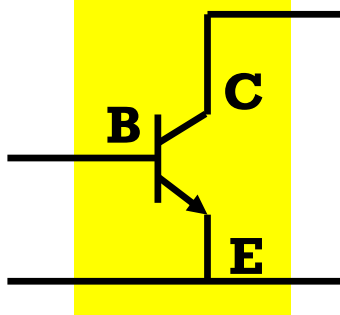
Common Source
CS: 共源组态



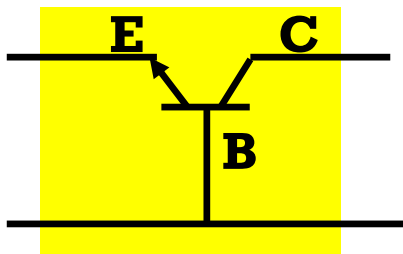
Common Gate
CG: 共栅组态



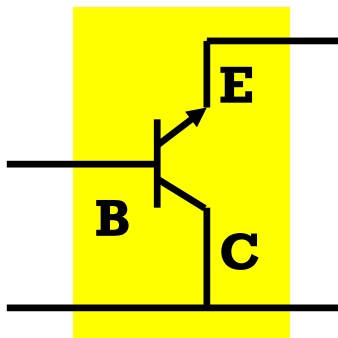
Common Drain
CD: 共漏组态



Common Emitter

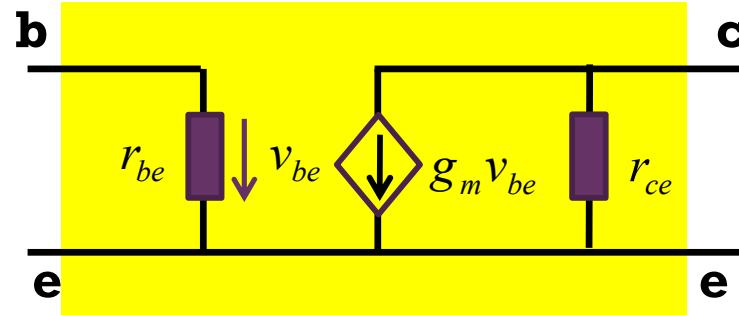


Common Base

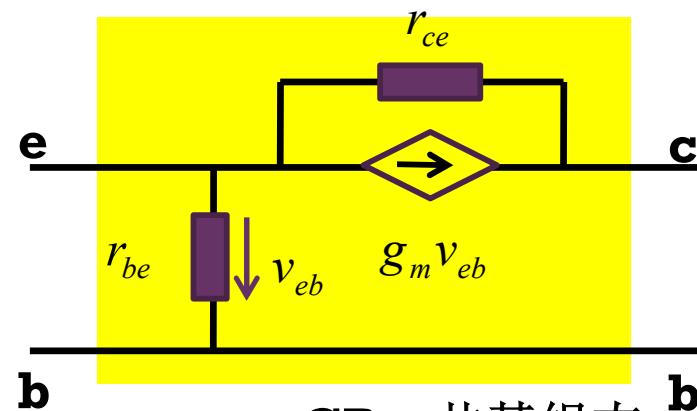


Common Collector

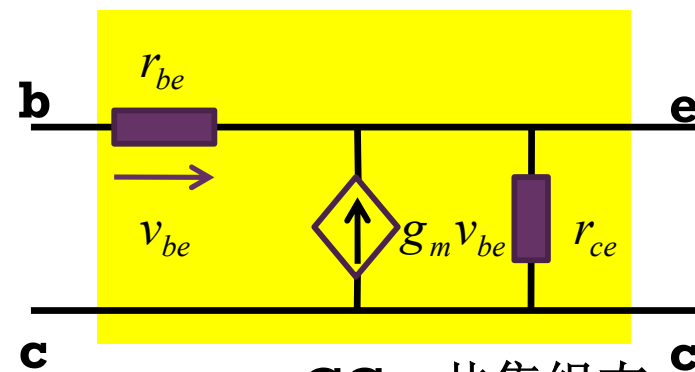
三种组态的通用跨导器模型



CE: 共射组态



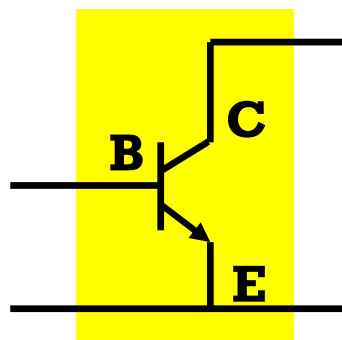
CB: 共基组态



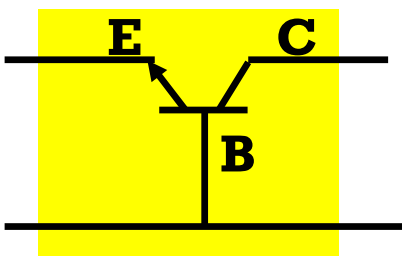
CC: 共集组态

作业选讲

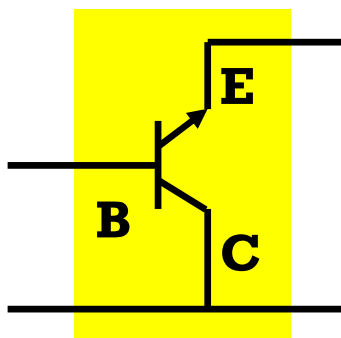
作业5.8 三种组态小信号放大器



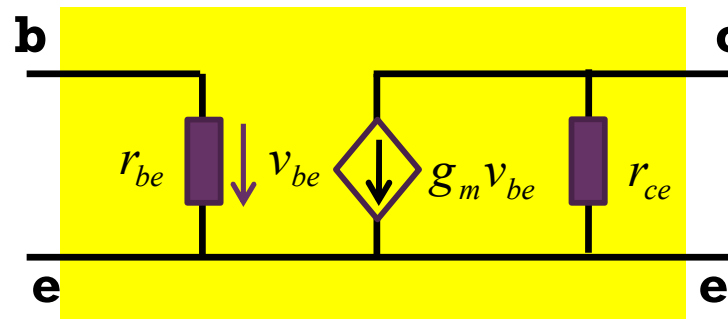
Common Emitter



Common Base



Common Collector

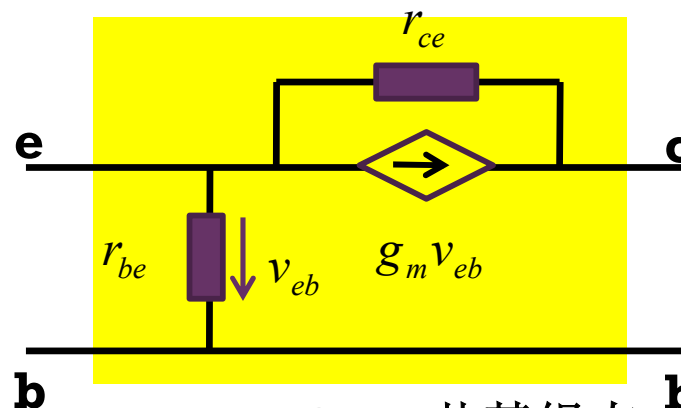


CE: 共射组态

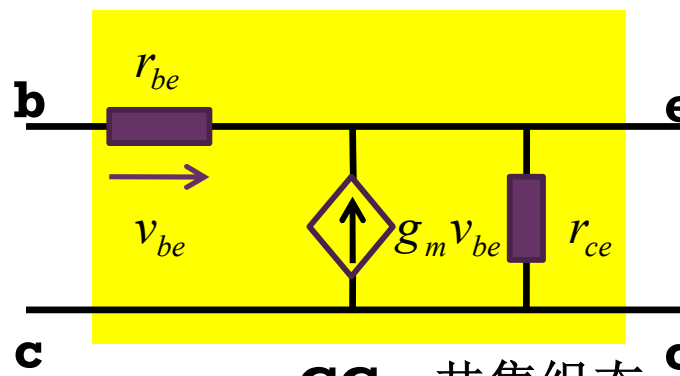
$$g_m = 40\text{mS}$$

$$r_{be} = 10\text{k}\Omega$$

$$r_{ce} = 100\text{k}\Omega$$



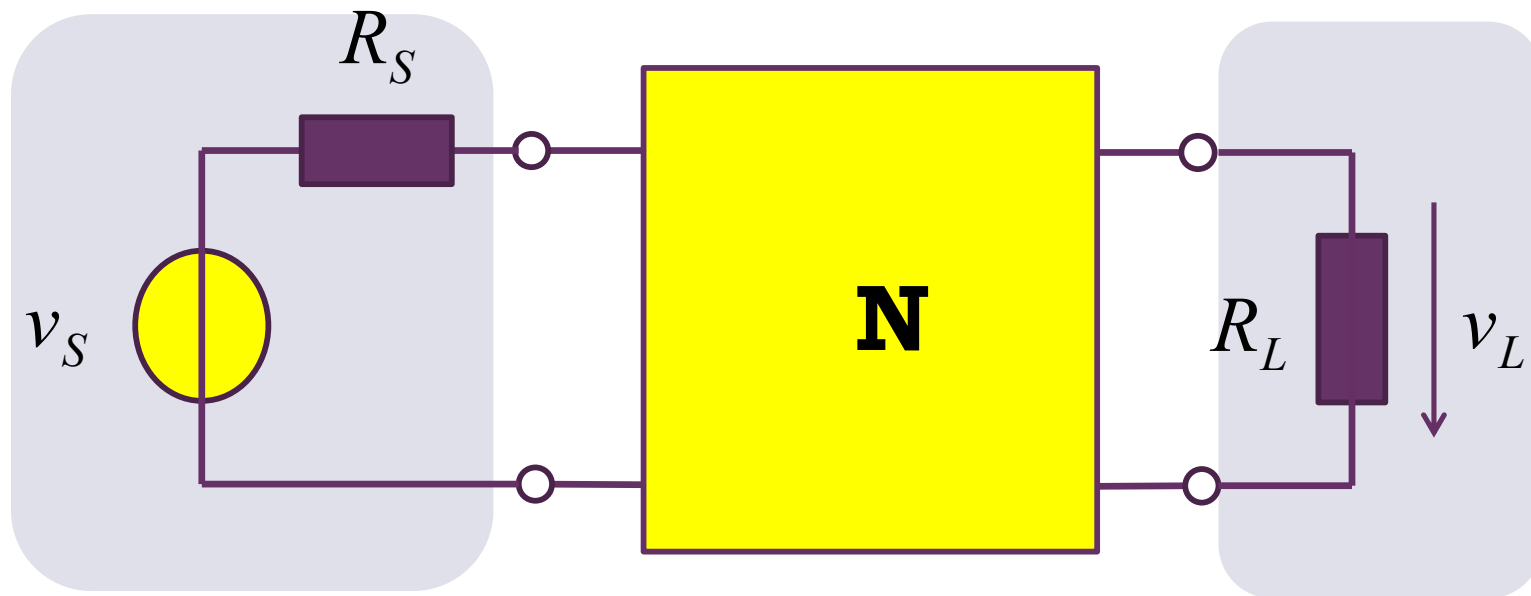
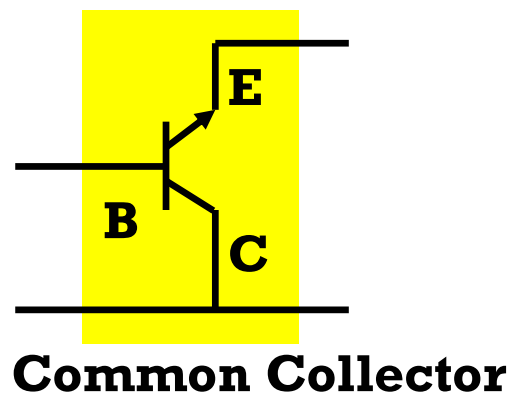
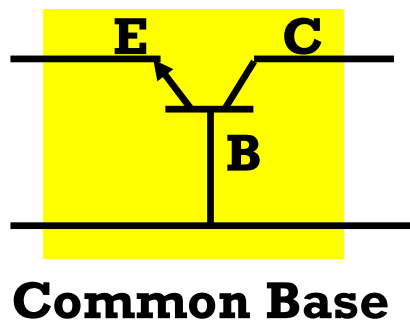
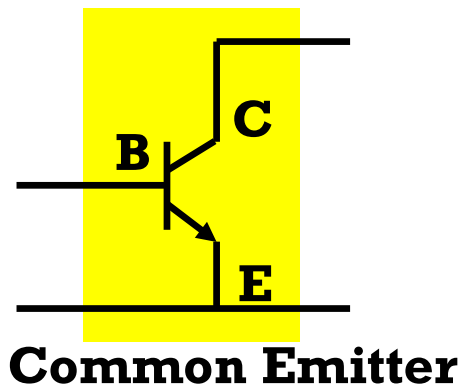
CB: 共基组态



CC: 共集组态

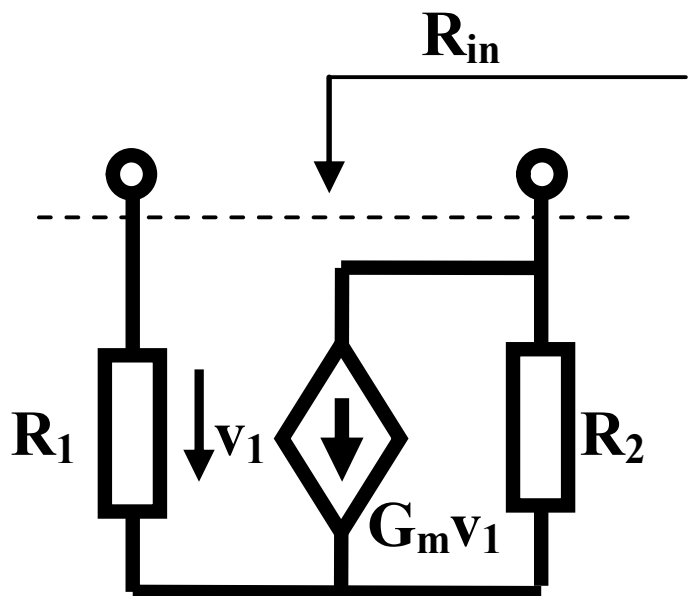
晶体管放大器小信号分析

假设不存在外加反馈回路的单晶体管放大电路



- 求三种组态晶体管放大器的输入电阻，输出电阻，电压传递函数表达式，代入具体数值求其输入电阻、输出电阻和电压放大倍数
($R_S=50\Omega, R_L=1\text{k}\Omega$)

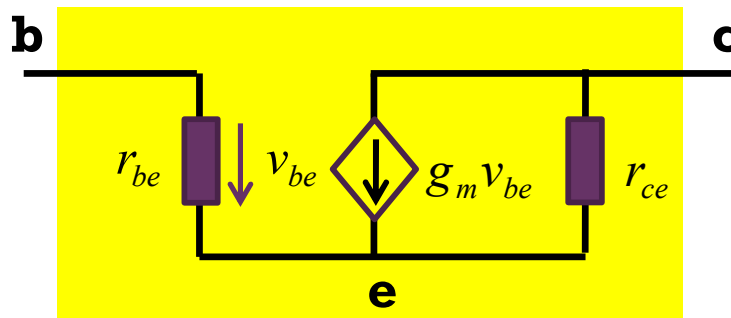
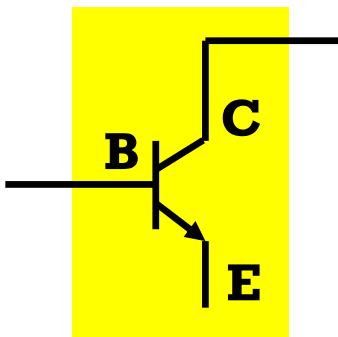
bc端口阻抗 (作业5.7)



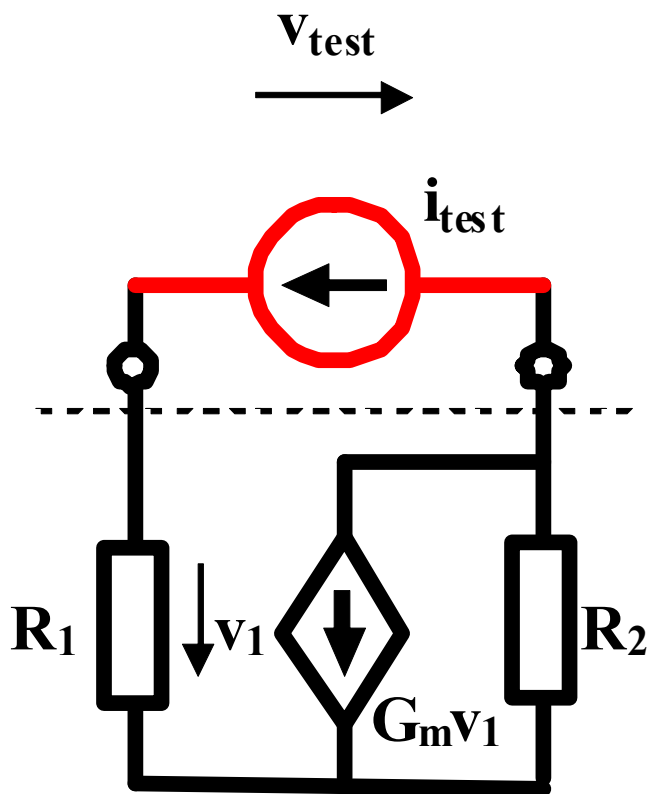
用加流求压法证明:

$$R_{in} = R_1 \langle G_m \rangle R_2 = R_1 + R_2 + G_m R_1 R_2$$

牢记这个结论：经常会用

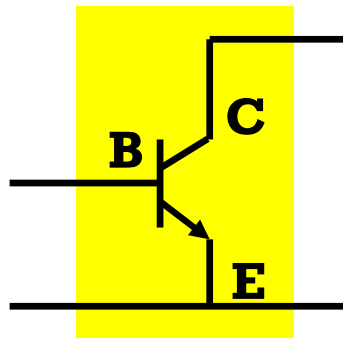


加流求压



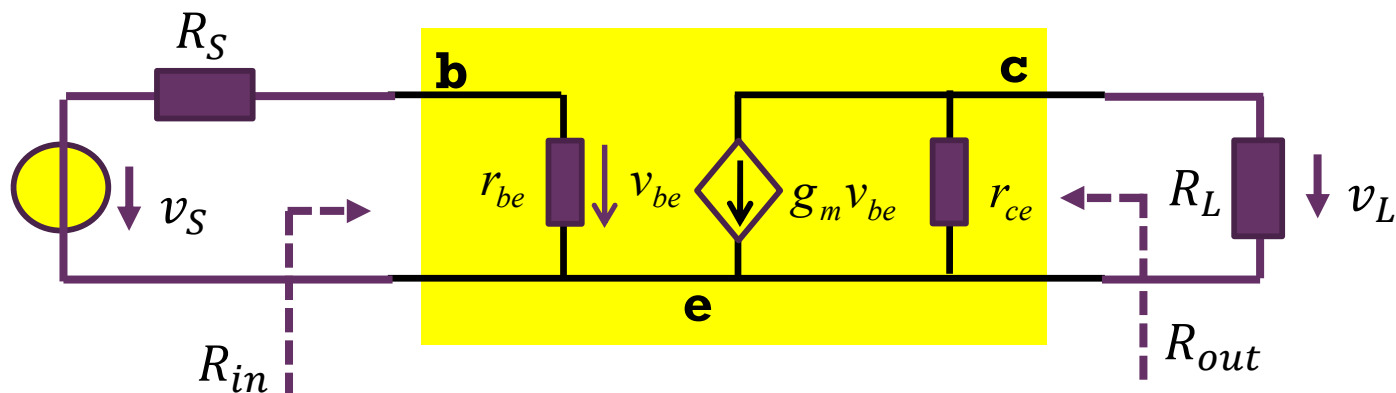
$$\begin{aligned} v_{test} &= i_{test} R_1 + (i_{test} + G_m v_1) R_2 \\ &= i_{test} R_1 + (i_{test} + G_m i_{test} R_1) R_2 \end{aligned}$$

$$\begin{aligned} R_{in} &= \frac{v_{test}}{i_{test}} = R_1 + (1 + G_m R_1) R_2 \\ &= R_1 + R_2 + G_m R_1 R_2 \end{aligned}$$



Common Emitter

CE组态晶体管放大器分析



$$R_{in} = r_{be} = 10k\Omega$$

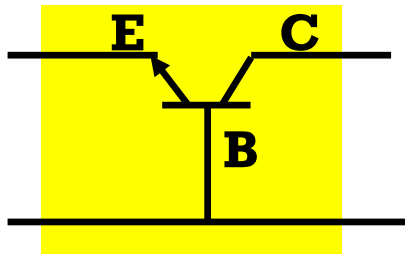
$$R_{out} = r_{ce} = 100k\Omega$$

单向网络

$$H = A_v = \frac{r_{ce} R_L}{r_{ce} + R_L} (-g_m) \frac{r_{be}}{r_{be} + R_S}$$

输出回路 本征跨 输入回路
总电阻 导增益 分压系数

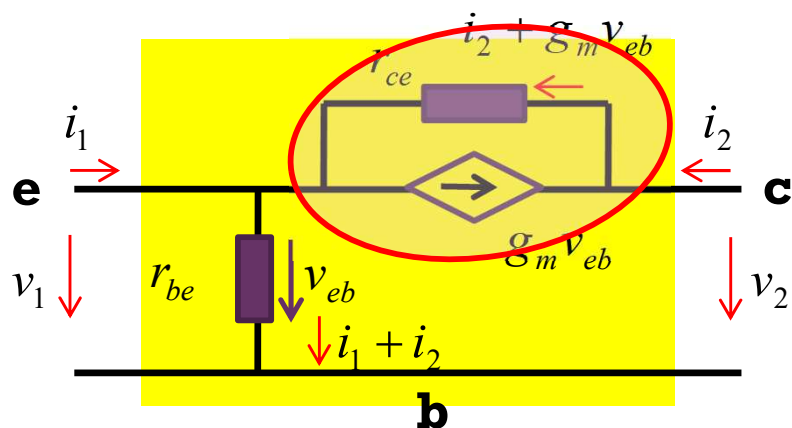
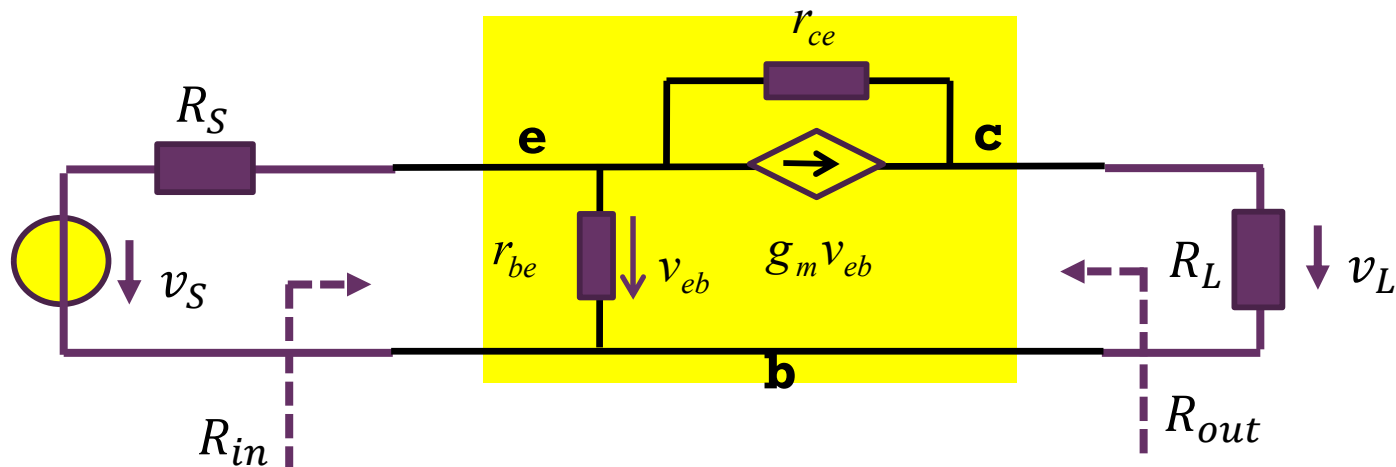
$$\begin{aligned} &= (100k\Omega || 1k\Omega) \times (-40mS) \times \frac{10k\Omega}{10k\Omega + 50\Omega} \\ &= 990\Omega \times (-40mS) \times 0.995 \\ &= -39.4 = 31.9dB \text{ 反相放大电压增益} \end{aligned}$$



CB组态晶体管放大器分析

Common Base

回路电流法，
结点电压法，
等效电路法，
第3讲电路定理一节中都有，
这里只考察网络参量法



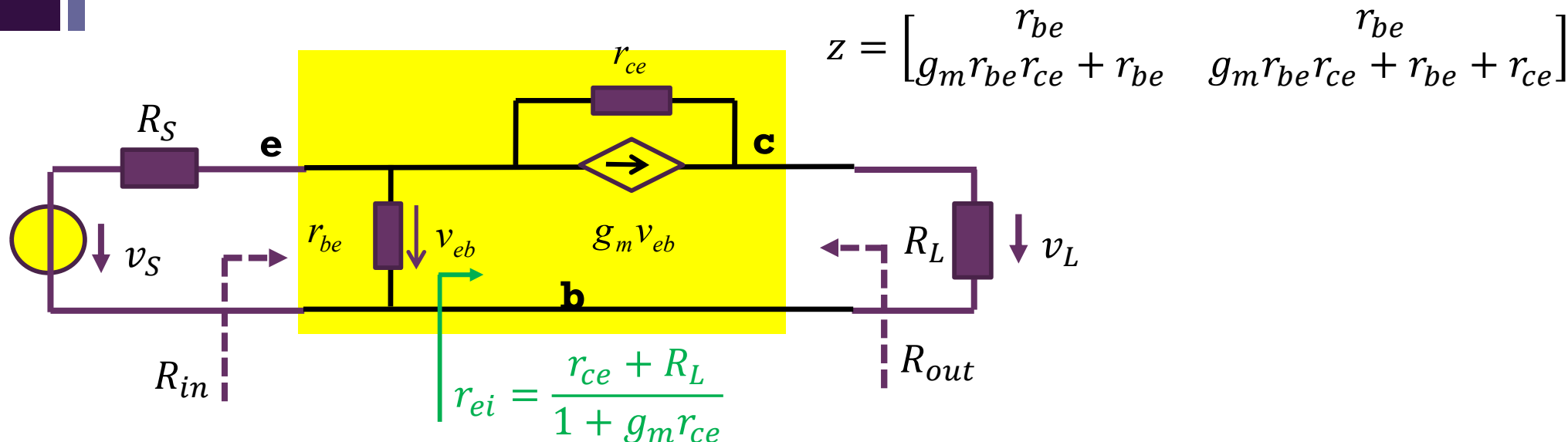
$$v_1 = (i_1 + i_2)r_{be} = r_{be}i_1 + r_{be}i_2$$

$$v_2 = (i_2 + g_m v_{eb})r_{ce} + v_{eb} = i_2 r_{ce} + (g_m r_{ce} + 1)v_1$$

$$= (g_m r_{ce} + 1)r_{be}i_1 + (g_m r_{be}r_{ce} + r_{be} + r_{ce})i_2$$

$$z = \begin{bmatrix} r_{be} & r_{be} \\ g_m r_{be} r_{ce} + r_{be} & g_m r_{be} r_{ce} + r_{be} + r_{ce} \end{bmatrix}$$

输入电阻和输出电阻



$$R_{in} = z_{in} = z_{11} - \frac{z_{12}z_{21}}{z_{22} + R_L} = r_{be} - \frac{r_{be}(g_m r_{ce} + 1)r_{be}}{g_m r_{be} r_{ce} + r_{be} + r_{ce} + R_L} = \frac{(r_{ce} + R_L)r_{be}}{g_m r_{be} r_{ce} + r_{be} + r_{ce} + R_L}$$

$$= \frac{(r_{ce} + R_L)r_{be}}{r_{be}(g_m r_{ce} + 1) + r_{ce} + R_L} = \frac{\frac{r_{ce} + R_L}{1 + g_m r_{ce}} r_{be}}{r_{be} + \frac{r_{ce} + R_L}{1 + g_m r_{ce}}} = r_{be} \parallel \frac{r_{ce} + R_L}{1 + g_m r_{ce}} = 10k \parallel 25.24 = 25.18\Omega$$

发射极对地阻抗 $1/g_m = 25\Omega$

$$R_{out} = z_{out} = z_{22} - \frac{z_{21}z_{12}}{z_{11} + R_S} = g_m r_{be} r_{ce} + r_{be} + r_{ce} - \frac{r_{be}(g_m r_{ce} + 1)r_{be}}{r_{be} + R_S}$$

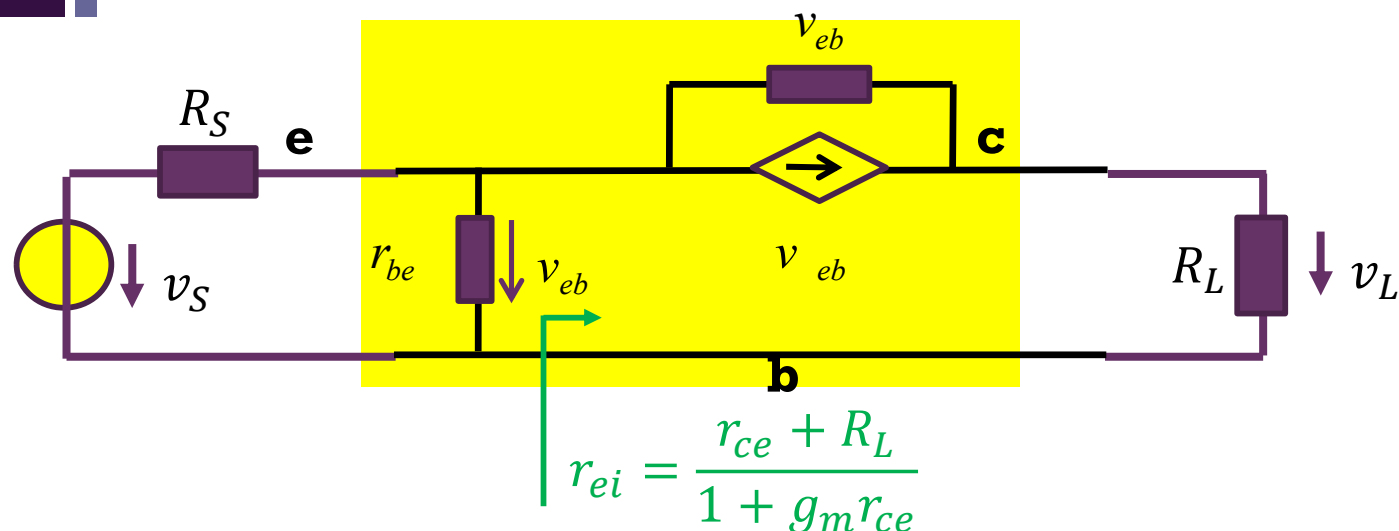
$$= g_m r_{be} r_{ce} \left(1 - \frac{r_{be}}{r_{be} + R_S}\right) + r_{be} \left(1 - \frac{r_{be}}{r_{be} + R_S}\right) + r_{ce} = g_m (r_{be} \parallel R_S) r_{ce} + r_{be} \parallel R_S + r_{ce}$$

$$= 199k + 49.75 + 100k = 299.05k\Omega$$

bc端口阻抗

电压放大倍数

$$z = \begin{bmatrix} r_{be} & r_{be} \\ g_m r_{be} r_{ce} + r_{be} & g_m r_{be} r_{ce} + r_{be} + r_{ce} \end{bmatrix}$$



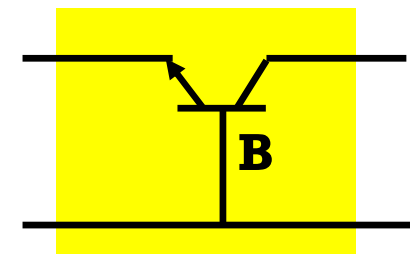
$$H = A_v = \frac{z_{21} R_L}{(z_{22} + R_L)(z_{11} + R_S) - z_{21} z_{12}}$$

$$= \frac{(g_m r_{ce} + 1) r_{be} R_L}{(g_m r_{be} r_{ce} + r_{be} + r_{ce} + R_L)(r_{be} + R_S) - r_{be} (g_m r_{ce} + 1) r_{be}}$$

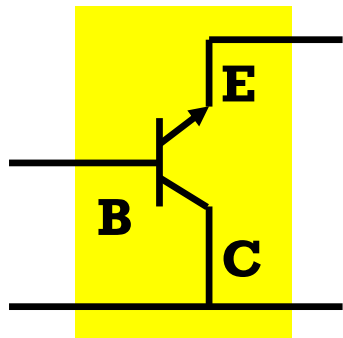
$$= \frac{(g_m r_{ce} + 1) r_{be} R_L}{(g_m r_{ce} + 1) r_{be} R_S + (r_{be} + R_S)(r_{ce} + R_L)}$$

$$= R_L \cdot \frac{r_{be} \parallel R_S}{r_{be} \parallel R_S + r_{ei}} \cdot \frac{1}{R_S}$$

$$= 13.27 = 22.46dB \text{ 同相放大电压增益}$$

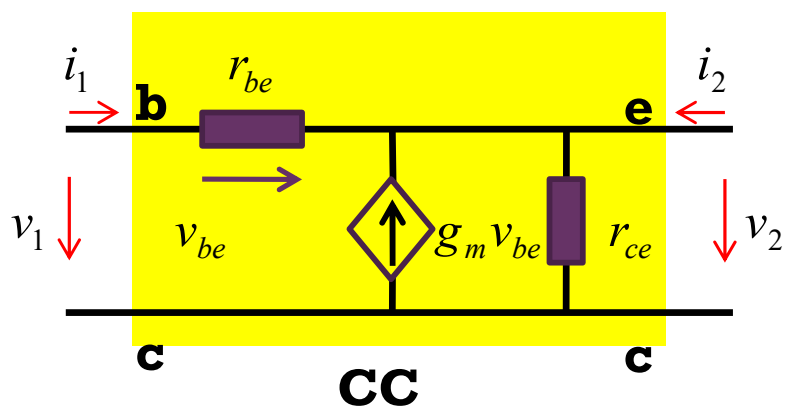
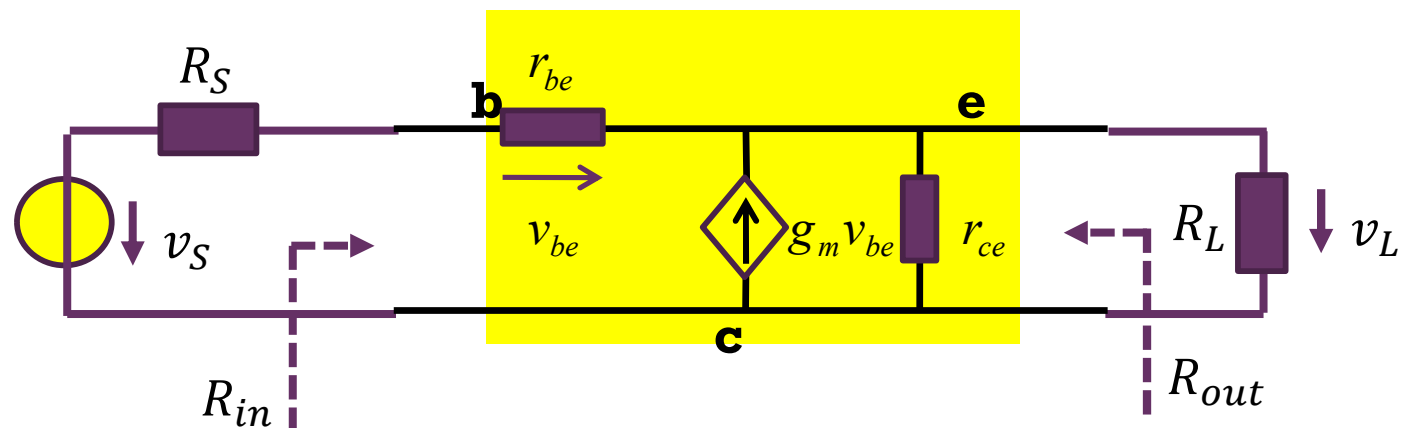


Common Base



Common Collector

CC组态晶体管放大器分析



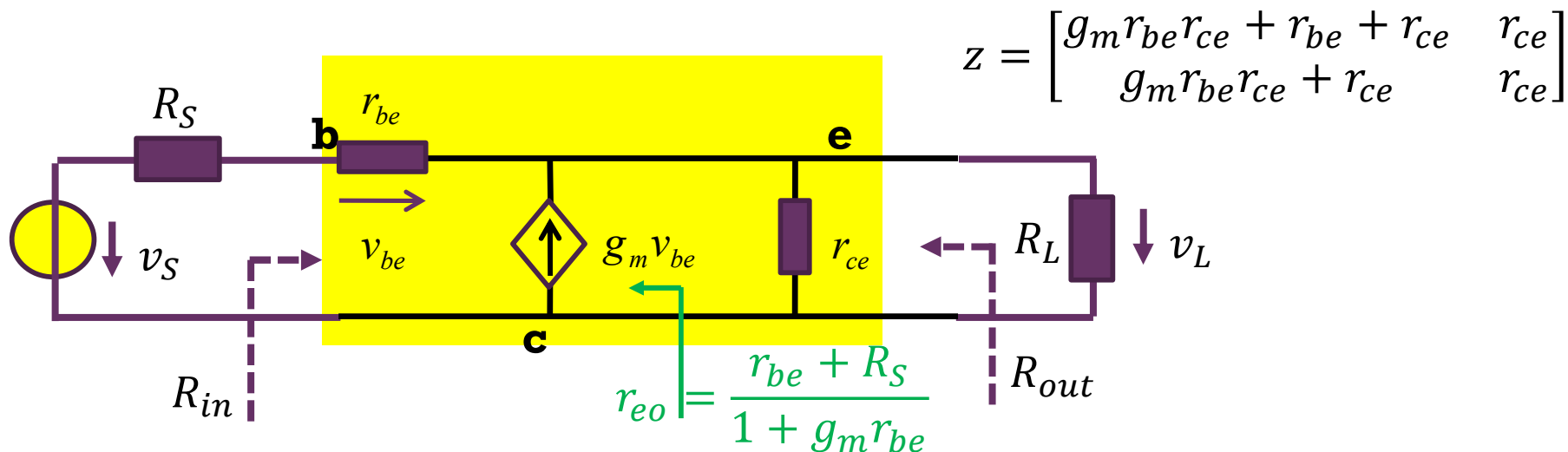
$$v_2 = r_{ce} (i_2 + i_1 + g_m v_{be}) = r_{ce} (i_2 + i_1 + g_m r_{be} i_1)$$

$$= (1 + g_m r_{be}) r_{ce} i_1 + r_{ce} i_2$$

$$v_1 = i_1 r_{be} + v_2 = (r_{be} + r_{ce} + g_m r_{be} r_{ce}) i_1 + r_{ce} i_2$$

$$z = \begin{bmatrix} g_m r_{be} r_{ce} + r_{be} + r_{ce} & r_{ce} \\ g_m r_{be} r_{ce} + r_{ce} & r_{ce} \end{bmatrix}$$

输入电阻和输出电阻



$$\begin{aligned}
 R_{in} = z_{in} &= z_{11} - \frac{z_{12}z_{21}}{z_{22} + R_L} = g_m r_{be} r_{ce} + r_{be} + r_{ce} - \frac{r_{ce}(g_m r_{be} + 1)r_{ce}}{r_{ce} + R_L} \\
 &= g_m r_{be} r_{ce} \left(1 - \frac{r_{ce}}{r_{ce} + R_L}\right) + r_{be} + r_{ce} \left(1 - \frac{r_{ce}}{r_{ce} + R_L}\right) = g_m r_{be} (r_{ce} \parallel R_L) + r_{be} + r_{ce} \parallel R_L \\
 &= 396k + 10k + 990 = 407k\Omega
 \end{aligned}$$

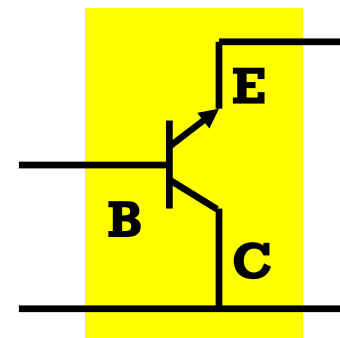
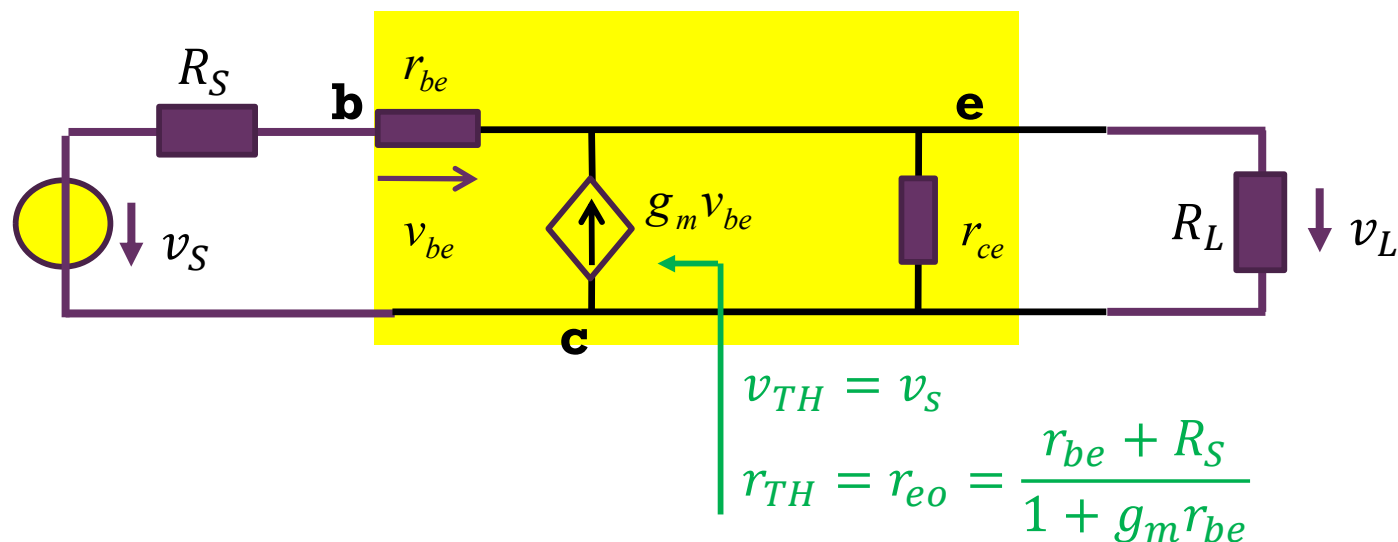
bc端口阻抗

$$\begin{aligned}
 R_{out} = z_{out} &= z_{22} - \frac{z_{21}z_{12}}{z_{11} + R_S} = r_{ce} - \frac{r_{ce}(g_m r_{be} + 1)r_{ce}}{g_m r_{be} r_{ce} + r_{be} + r_{ce} + R_S} = \frac{(r_{be} + R_S)r_{ce}}{g_m r_{be} r_{ce} + r_{be} + r_{ce} + R_S} \\
 &= \frac{(r_{be} + R_S)r_{ce}}{r_{ce}(g_m r_{be} + 1) + r_{be} + R_S} = \frac{\frac{r_{be} + R_S}{1 + g_m r_{be}} r_{ce}}{r_{ce} + \frac{r_{be} + R_S}{1 + g_m r_{be}}} = r_{ce} \parallel \frac{r_{be} + R_S}{1 + g_m r_{be}} = 100k \parallel 25.06 = 25.06\Omega
 \end{aligned}$$

发射极对地阻抗 $1/g_m = 25\Omega$

电压放大倍数

$$z = \begin{bmatrix} g_m r_{be} r_{ce} + r_{be} + r_{ce} & r_{ce} \\ g_m r_{be} r_{ce} + r_{ce} & r_{ce} \end{bmatrix}$$

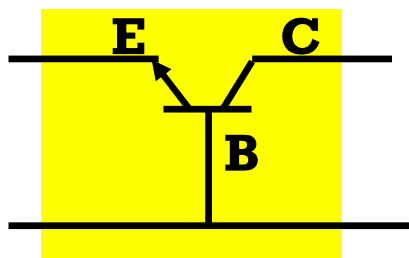


Common Collector

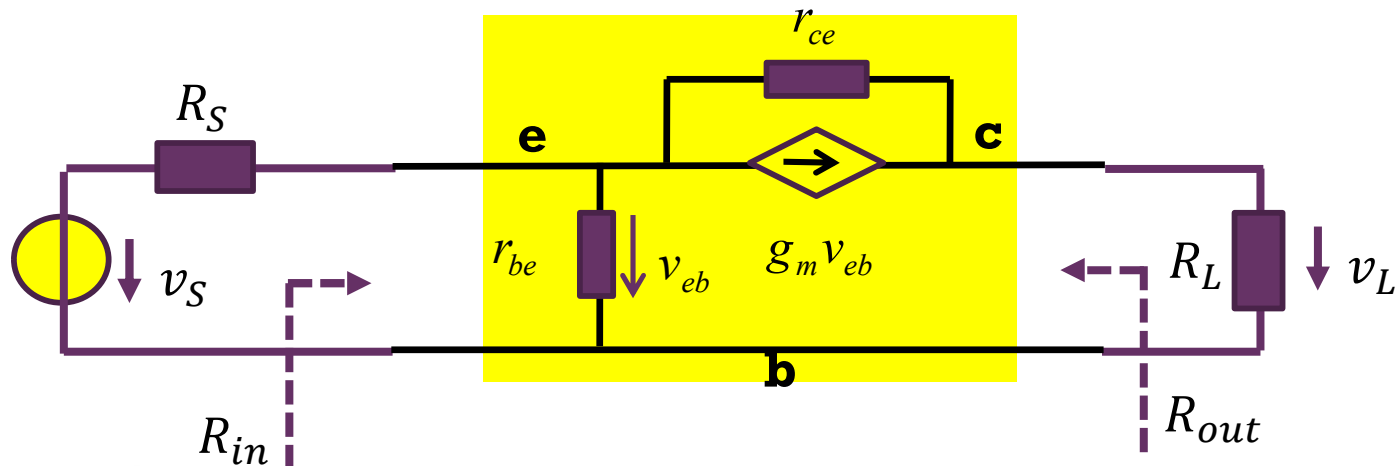
$$\begin{aligned}
 H = A_v &= \frac{z_{21} R_L}{(z_{22} + R_L)(z_{11} + R_S) - z_{21} z_{12}} \\
 &= \frac{(g_m r_{be} + 1) r_{ce} R_L}{(r_{ce} + R_L)(g_m r_{be} r_{ce} + r_{be} + r_{ce} + R_S) - r_{ce} (g_m r_{be} + 1) r_{ce}} \\
 &= \frac{(g_m r_{be} + 1) r_{ce} R_L}{(g_m r_{be} + 1) r_{ce} R_L + (r_{be} + R_S)(r_{ce} + R_L)} = \frac{r_{ce} \parallel R_L}{r_{ce} \parallel R_L + r_{eo}}
 \end{aligned}$$

$= 0.9753 = -0.22\text{dB}$ 电压缓冲？（同相放大电压增益近似为 $1=0\text{dB}$ ）

输入阻抗和输出阻抗总结



Common Base



$$R_{in} = r_{be} \parallel \frac{r_{ce} + R_L}{1 + g_m r_{ce}}$$

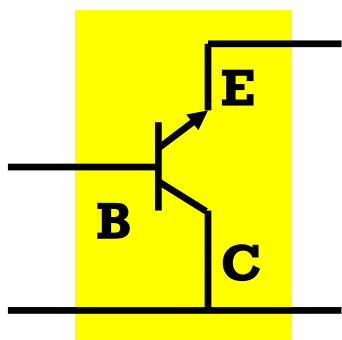
$$= 25.18\Omega$$

发射极对地阻抗 $1/g_m = 25\Omega$

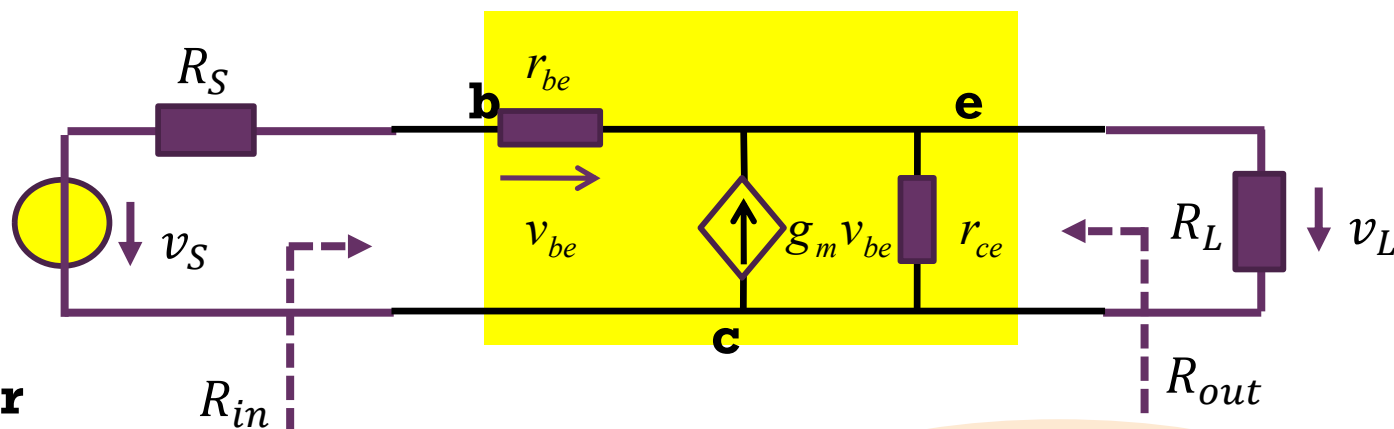
$$R_{out} = g_m (r_{be} \parallel R_S) r_{ce} + r_{be} \parallel R_S + r_{ce}$$

$$= 299k\Omega$$

bc端口阻抗



Common Collector



$$R_{in} = g_m r_{be} (r_{ce} \parallel R_L) + r_{be} + r_{ce} \parallel R_L$$

$$= 407k\Omega$$

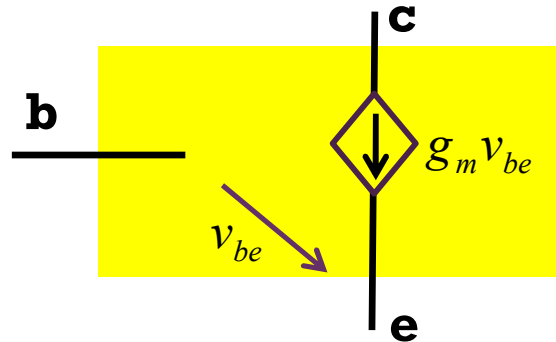
bc端口阻抗

$$R_{out} = r_{ce} \parallel \frac{r_{be} + R_S}{1 + g_m r_{be}}$$

$$= 25.06\Omega$$

发射极对地阻抗 $1/g_m = 25\Omega$

电压增益总结



理想晶体管结论

$$r_{be} \rightarrow \infty$$

$$r_{ce} \rightarrow \infty$$

$$A_{v,CE} = \frac{r_{ce}R_L}{r_{ce} + R_L} (-g_m) \frac{r_{be}}{r_{be} + R_S}$$

$$= -39.4$$

$$\approx -g_m R_L = -40$$

$R_S \ll r_{be}$ $R_L \ll r_{ce}$
反相放大电压增益

$$A_{v,CB} = \frac{(g_m r_{ce} + 1) r_{be} R_L}{(g_m r_{ce} + 1) r_{be} R_S + (r_{be} + R_S)(r_{ce} + R_L)} = R_L \cdot \frac{r_{be} \parallel R_S}{r_{be} \parallel R_S + r_{ei}} \cdot \frac{1}{R_S}$$

$$= 13.27$$

$$\approx \frac{g_m r_{ce} r_{be} R_L}{g_m r_{ce} r_{be} R_S + r_{be} r_{ce}} = \frac{g_m}{1 + g_m R_S} R_L = 13.33$$

$$R_L \ll r_{ce}$$

同相放大电压增益

$$A_{v,CC} = \frac{(g_m r_{be} + 1) r_{ce} R_L}{(g_m r_{be} + 1) r_{ce} R_L + (r_{be} + R_S)(r_{ce} + R_L)} = \frac{r_{ce} \parallel R_L}{r_{ce} \parallel R_L + r_{eo}}$$

$$= 0.9753$$

$$\approx \frac{g_m r_{be} r_{ce} R_L}{g_m r_{be} r_{ce} R_L + r_{be} r_{ce}} = \frac{g_m}{1 + g_m R_L} R_L = 0.9756$$

$$R_S \ll r_{be}$$

同相放大电压增益

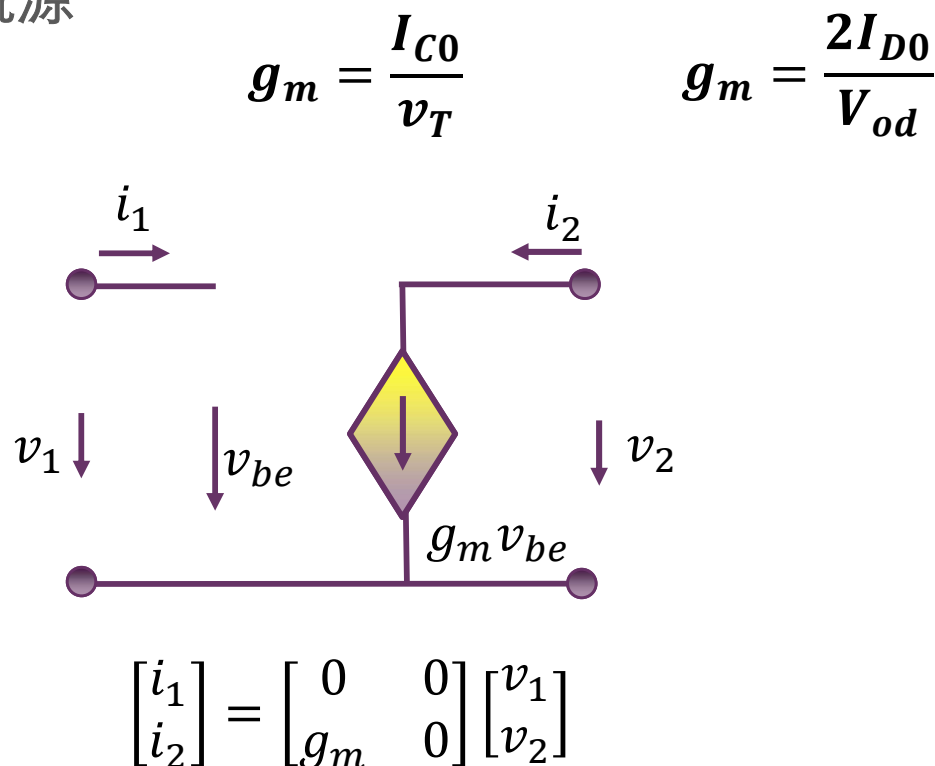
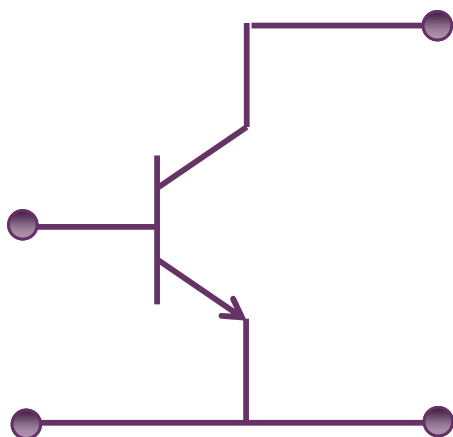
理想晶体管

- 电流增益和厄利电压被抽象为无穷大的晶体管为理想晶体管

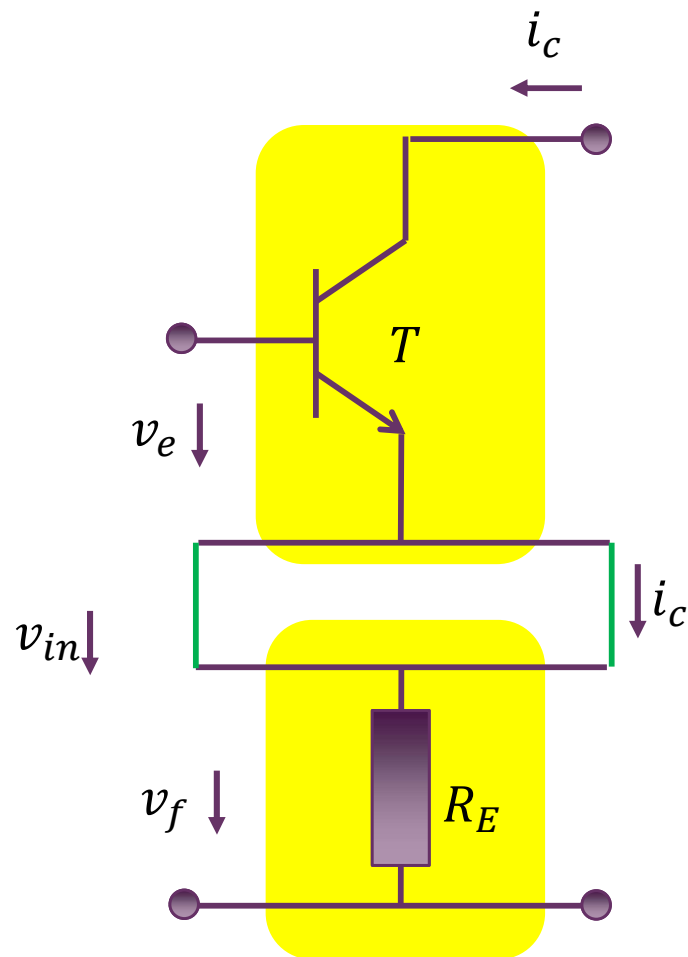
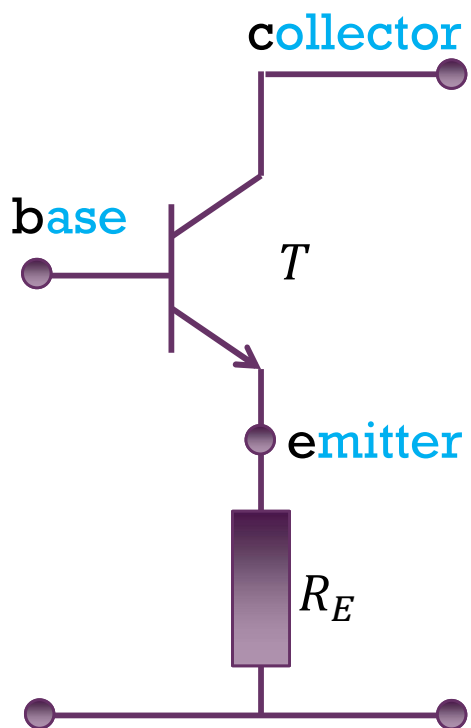
- $\beta \rightarrow \infty, V_A \rightarrow \infty$

- $r_{be} = \beta \frac{1}{g_m} = \beta \frac{v_T}{I_{C0}} \rightarrow \infty, r_{ce} = \frac{V_A}{I_{C0}} \rightarrow \infty$

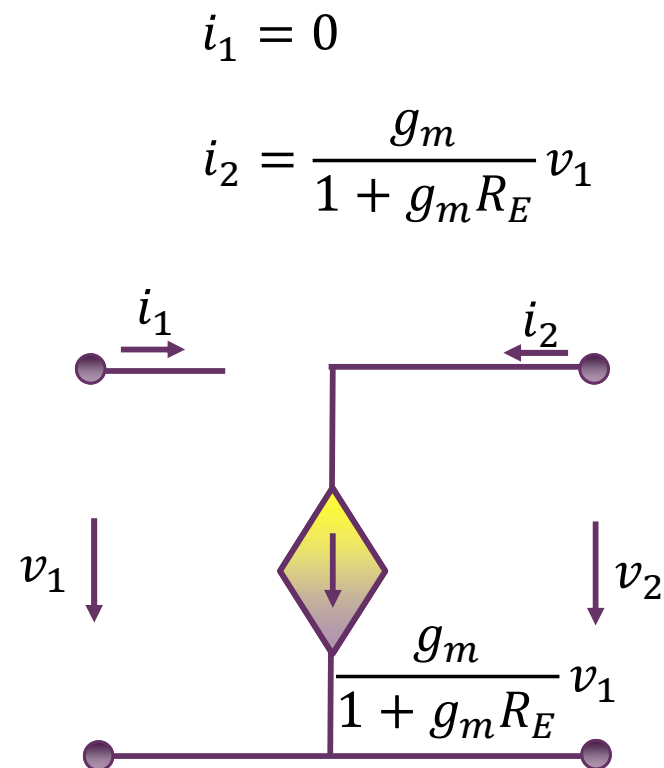
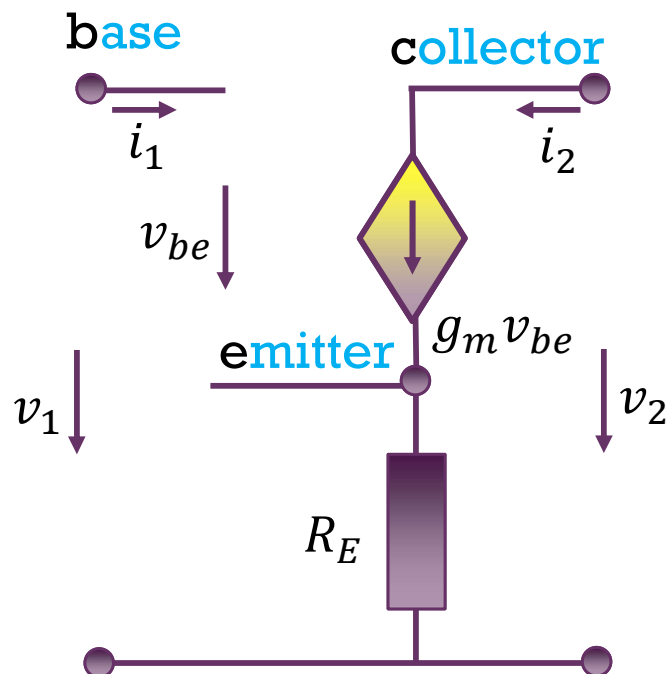
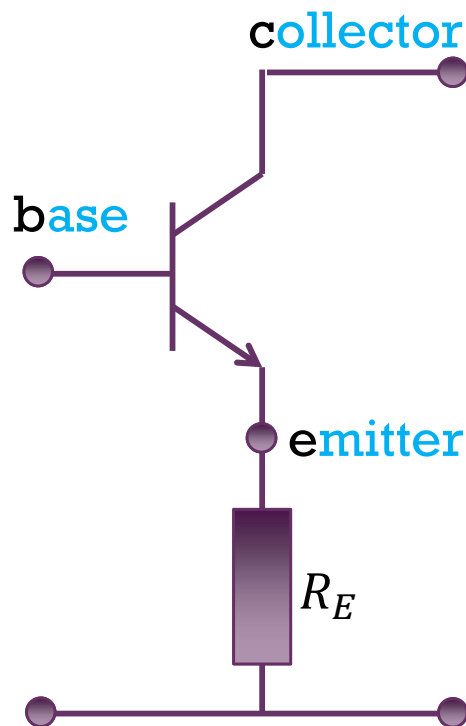
- 理想晶体管CE组态为理想压控流源



串串负反馈



复合串串负反馈电阻后理想晶体管还是理想晶体管



$$i_2 = g_m v_{be} = g_m (v_b - v_e)$$

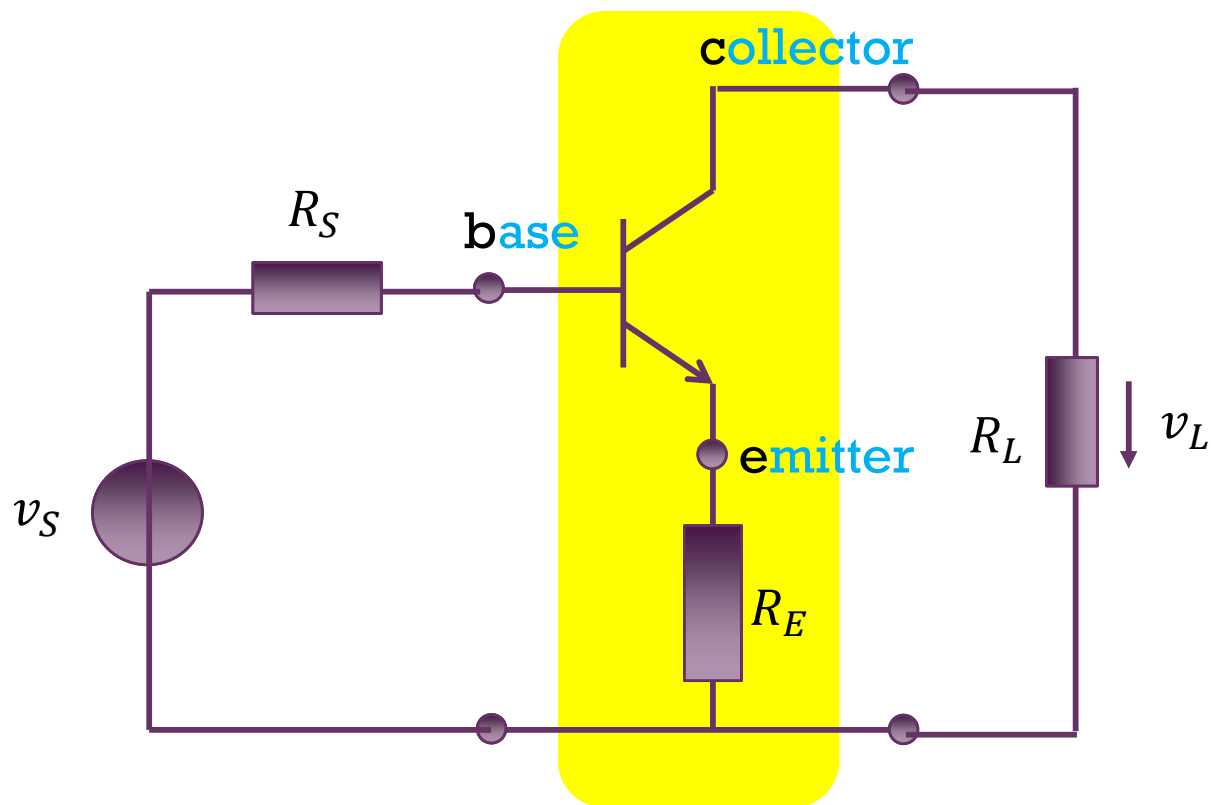
$$= g_m (v_1 - i_2 R_E) = g_m v_1 - i_2 g_m R_E$$

$$\begin{bmatrix} i_1 \\ i_2 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ \frac{g_m}{1 + g_m R_E} & 0 \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \end{bmatrix}$$

$$i_2 = \frac{g_m}{1 + g_m R_E} v_1 = g_{mf} v_1$$

结论：理想晶体管加串联负反馈电阻 $\mathbf{R_E}$ 后仍然是理想晶体管，只不过跨导增益发生变小而已，同时变得稳定了

CE组态负反馈放大器



$$A_v = -g_{mf} R_L = -\frac{g_m}{1 + g_m R_E} R_L \stackrel{g_m R_E \gg 1}{\approx} -\frac{R_L}{R_E}$$

实验电路：折中方案

旁路电容：消除交流负反馈，增益很高，但不稳定、线性度差、带宽窄

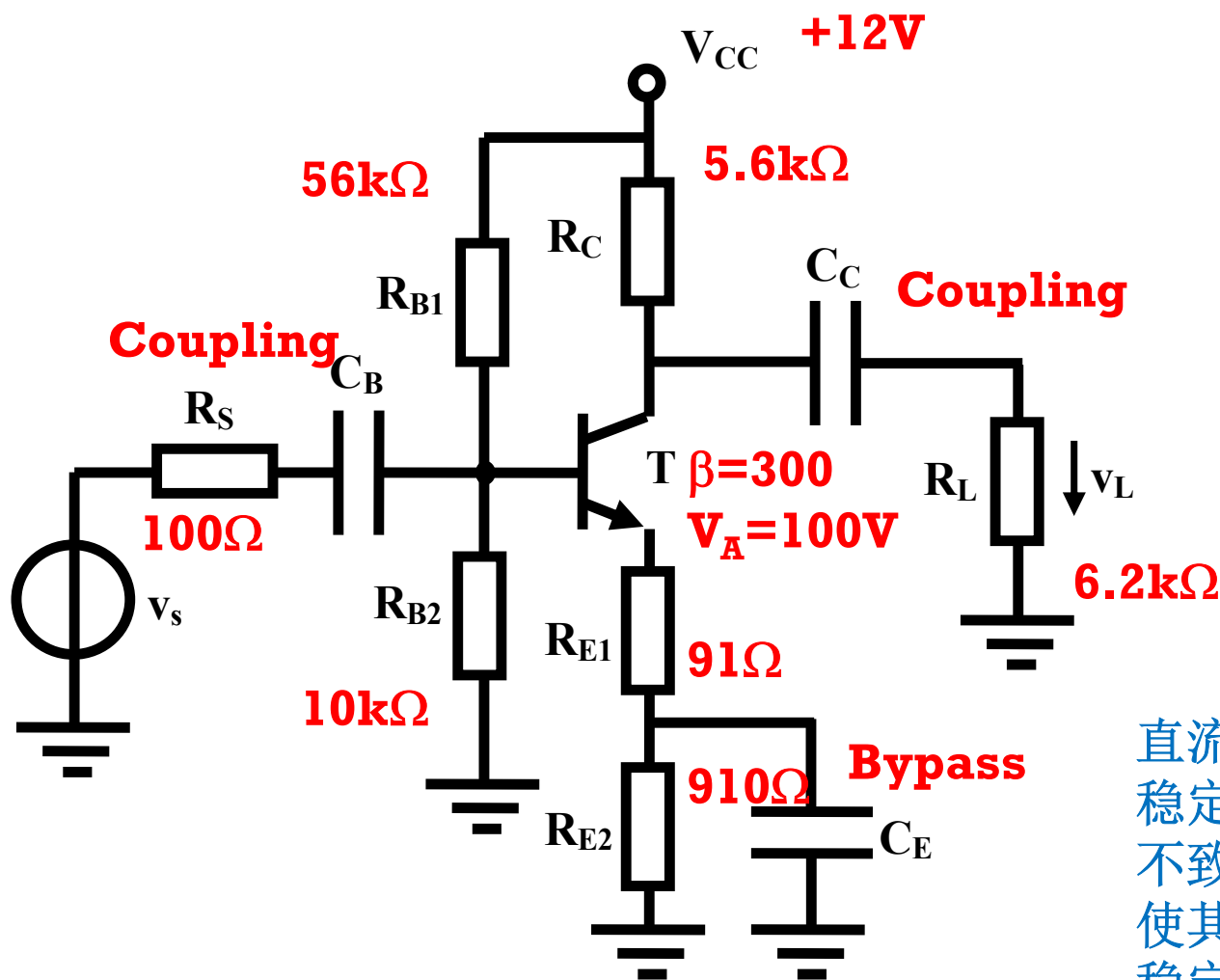
$$G_{m0} = -g_m$$

无旁路电容：交流负反馈导致增益下降，但十分稳定、线性度高、带宽宽

$$G_{m0} = -\frac{g_m}{1 + g_m R_E}$$

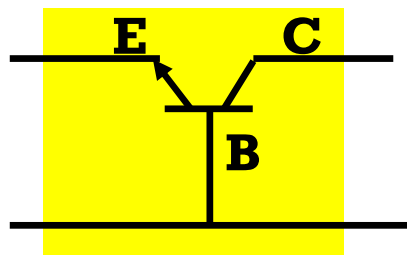
折中方案：增益有所下降，稳定性有所增加

$$G_{m0} = -\frac{g_m}{1 + g_m R_{E1}}$$

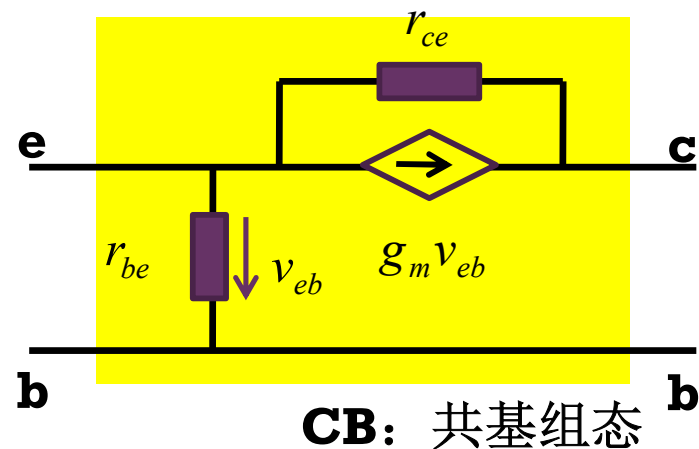


直流负反馈很深，确保直流工作点的高稳定性；交流负反馈较浅，使电压增益不致太低，同时提供串串交流负反馈：使其更接近于理想压控流源，提高增益稳定性，提高放大器带宽，提高线性度

CB组态的理想晶体管



Common Base

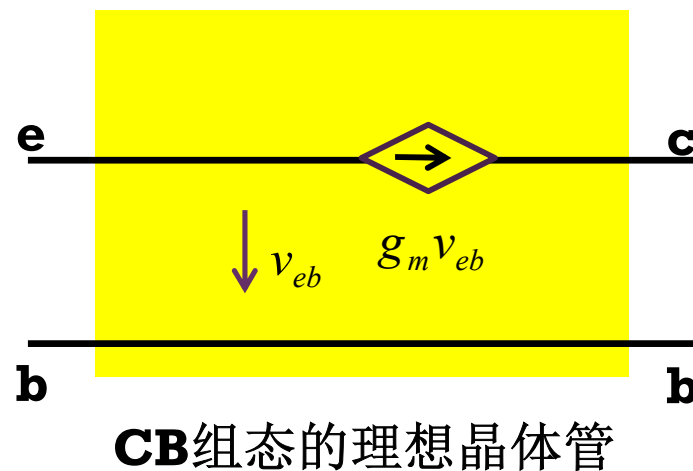


端口伏安特性方程

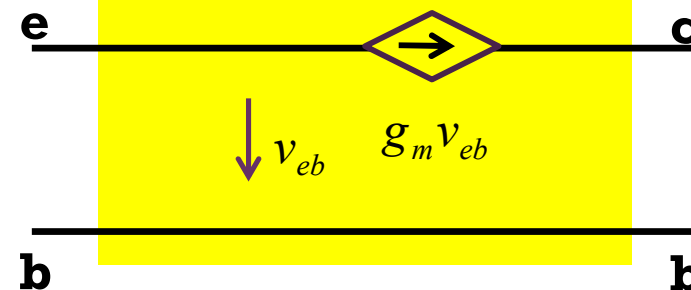
$$i_1 = g_m v_1$$

$$i_2 = -i_1 = -g_m v_1$$

y参量表述



CB组态理想晶体管 电路模型

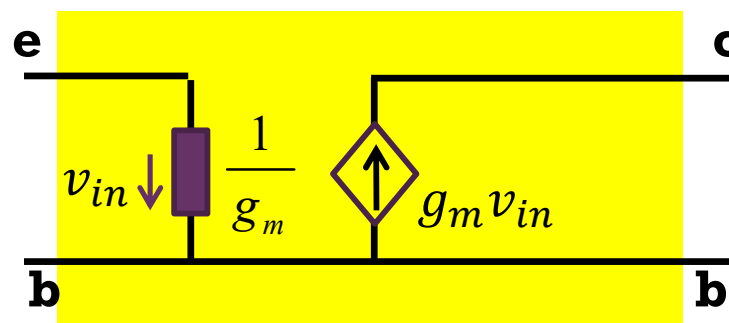


端口伏安特性方程

$$i_1 = g_m v_1$$

$$i_2 = -g_m v_1$$

y参量表述



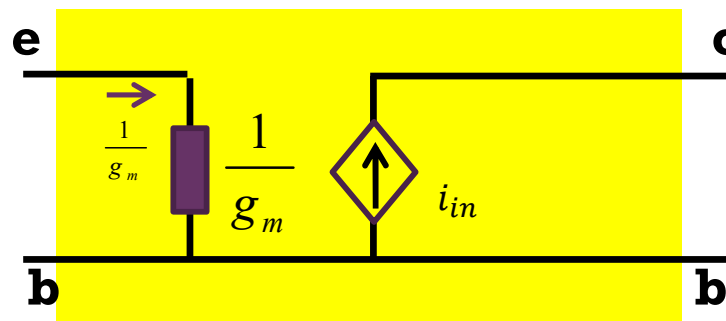
理想晶体管**CB**组态**y**参量电路模型

偏离理想压控流源较远

$$v_1 = \frac{1}{g_m} i_1$$

$$i_2 = -i_1$$

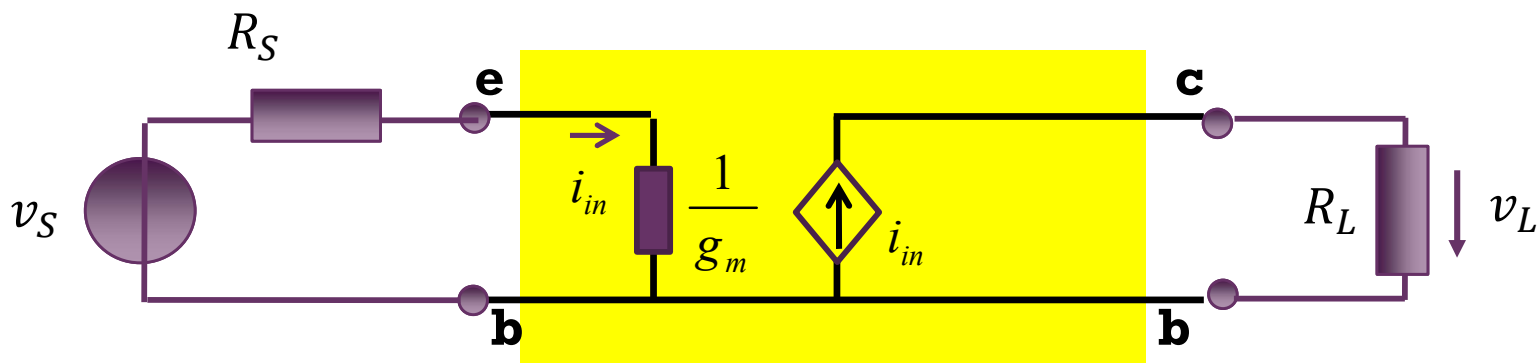
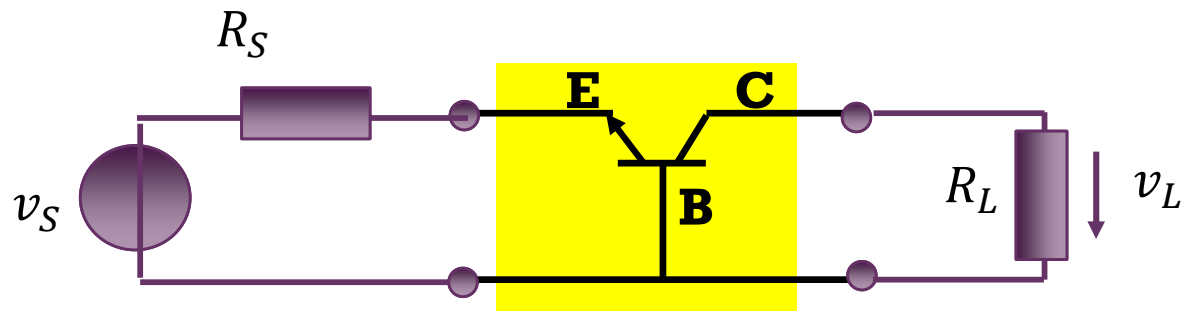
h参量表述



理想晶体管**CB**组态**h**参量电路模型

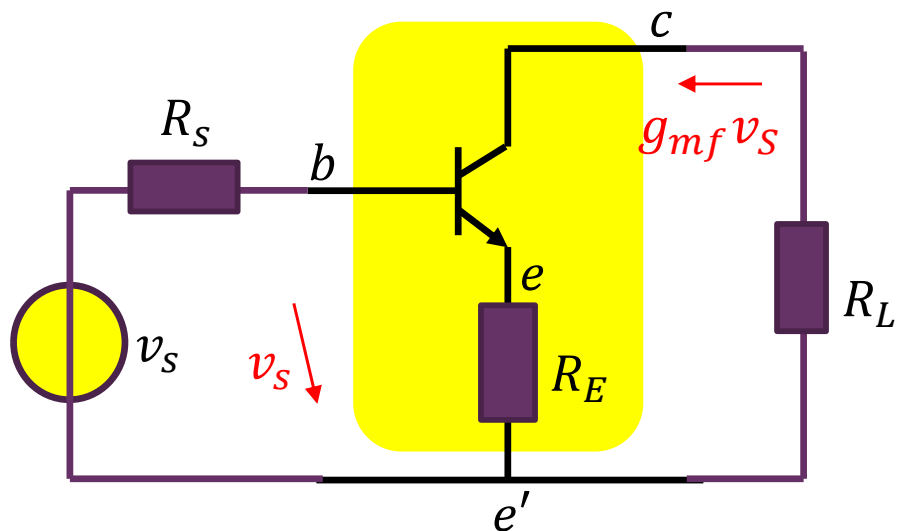
电流缓冲器模型：接近于理想流控流源

CB组态放大器理想模型结论

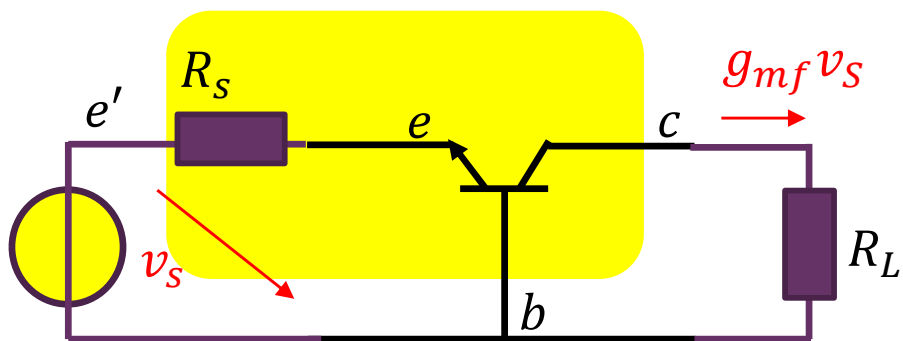


$$v_L = R_L i_{in} = R_L \frac{v_S}{R_S + \frac{1}{g_m}} = \frac{g_m}{1 + g_m R_S} R_L v_S = g_{mf} R_L v_S$$

简单理解：把信源内阻视为复合理想晶体管内部的负反馈电阻

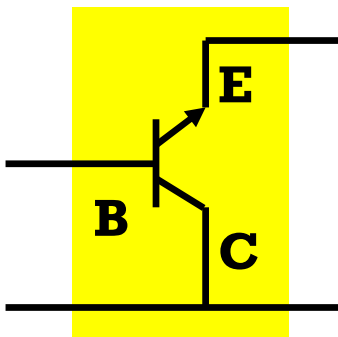


$$A_{v,CE} = -g_{mf}R_L = -\frac{g_m}{1 + g_m R_E} R_L$$

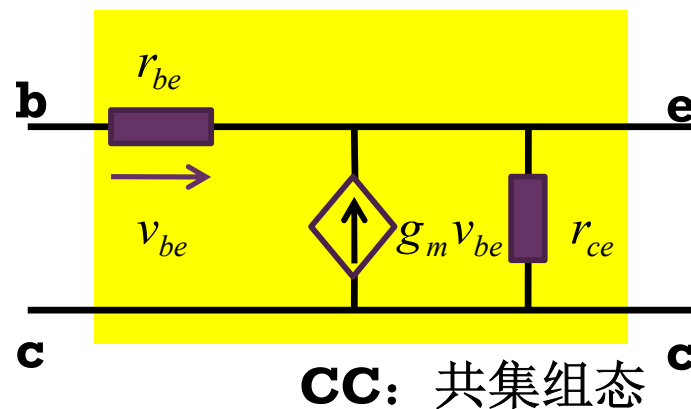


$$A_{v,CB} = +g_{mf}R_L = \frac{g_m}{1 + g_m R_S} R_L$$

CC组态的理想晶体管



Common Collector

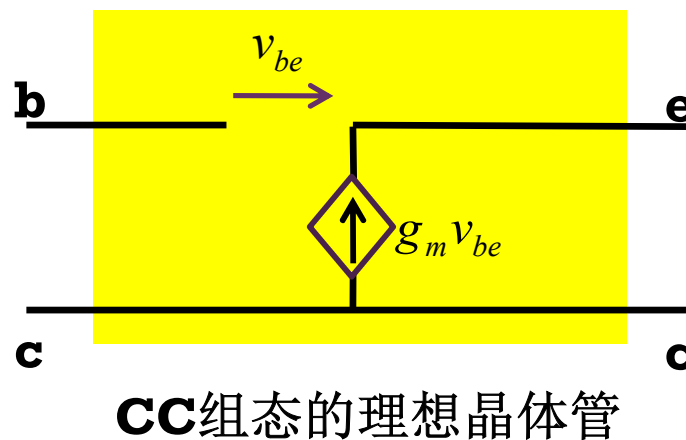


端口伏安特性方程

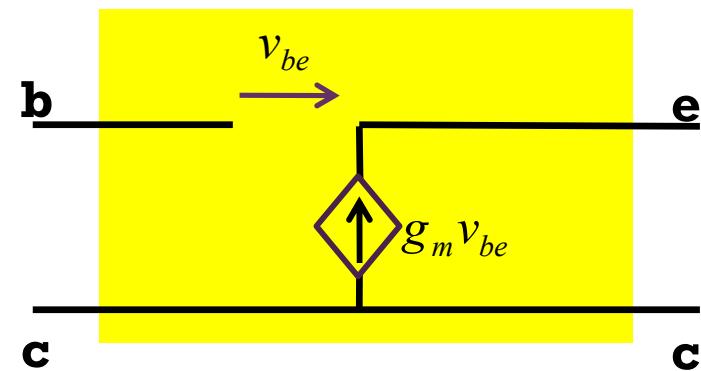
$$i_1 = 0$$

$$i_2 = -g_m v_{be} = -g_m v_1 + g_m v_2$$

y参量表述



CC组态理想晶体管 电路模型



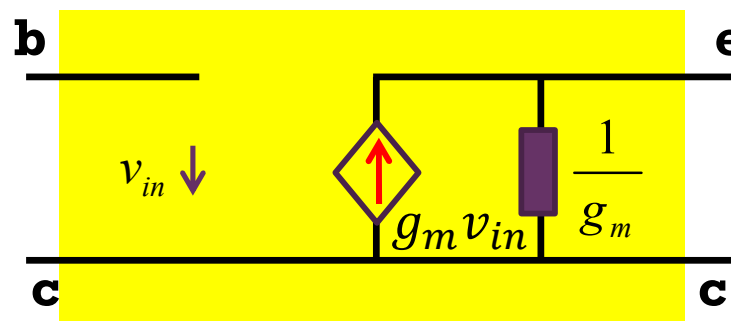
CC组态的理想晶体管

端口伏安特性方程

$$i_1 = 0$$

$$i_2 = -g_m v_{be} = -g_m v_1 + g_m v_2$$

y参量表述



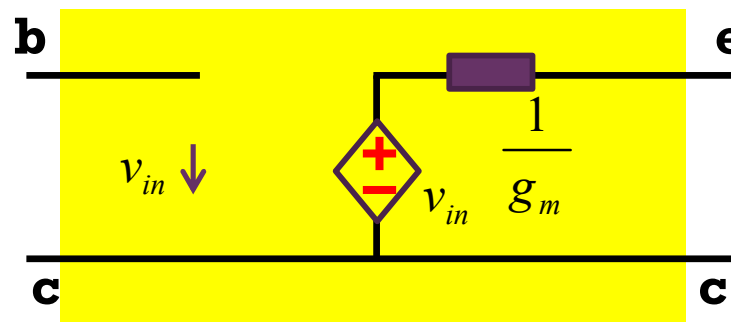
理想晶体管CC组态y参量电路模型

偏离理想压控流源较远

$$i_1 = 0$$

$$v_2 = v_1 + \frac{1}{g_m} i_2$$

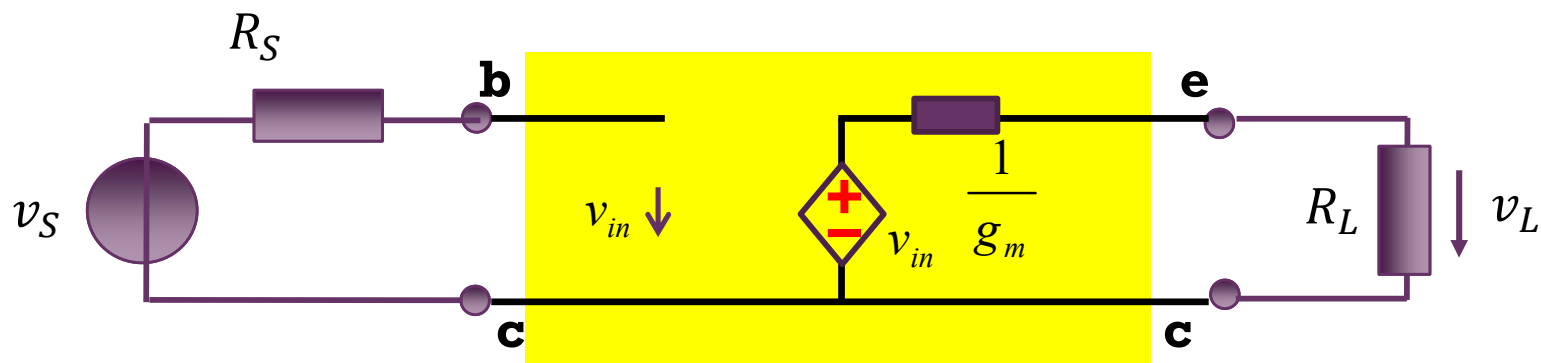
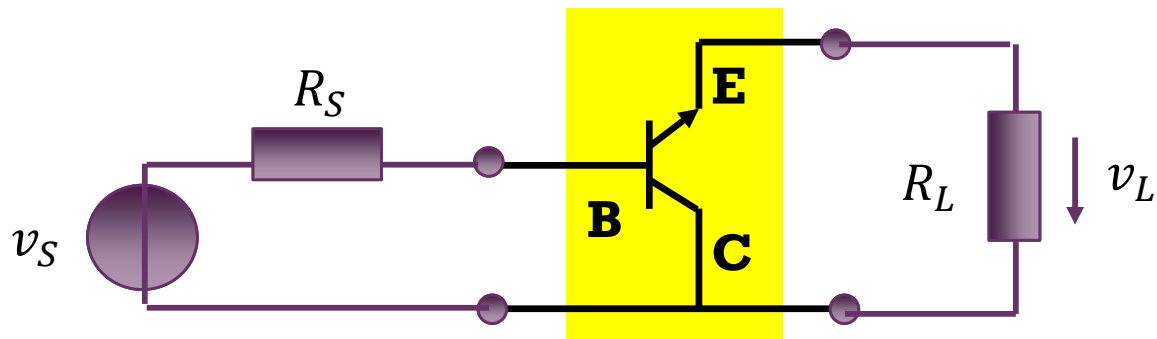
g参量表述



理想晶体管CC组态g参量电路模型

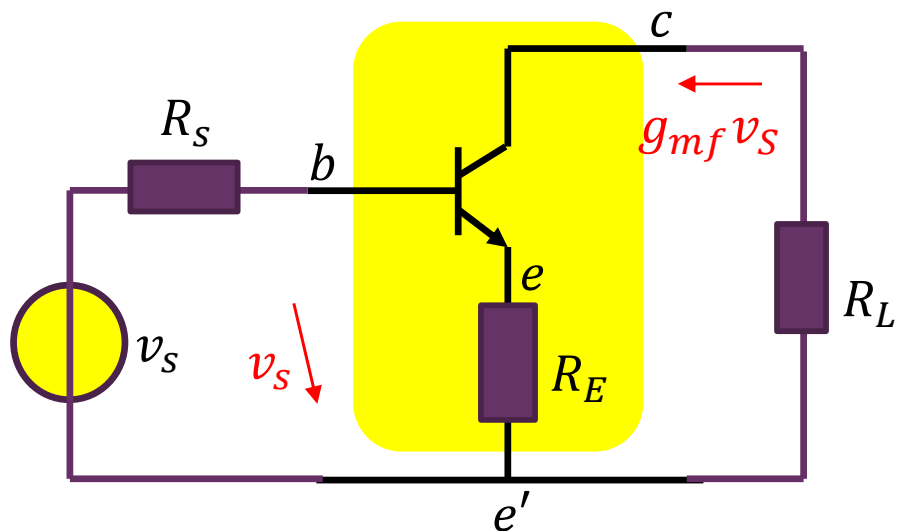
电压缓冲器模型：接近理想压控压源

CC组态放大器

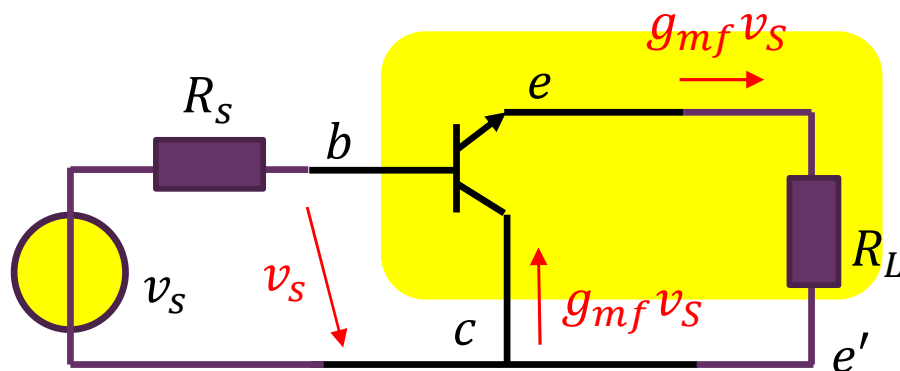


$$v_L = \frac{R_L}{R_L + \frac{1}{g_m}} i_{in} = \frac{g_m}{1 + g_m R_L} R_L v_S = g_{mf} R_L v_S$$

简单理解：把负载电阻视为复合理想晶体管内部的负反馈电阻



$$A_{v,CE} = -g_{mf} R_L = -\frac{g_m}{1 + g_m R_E} R_L$$

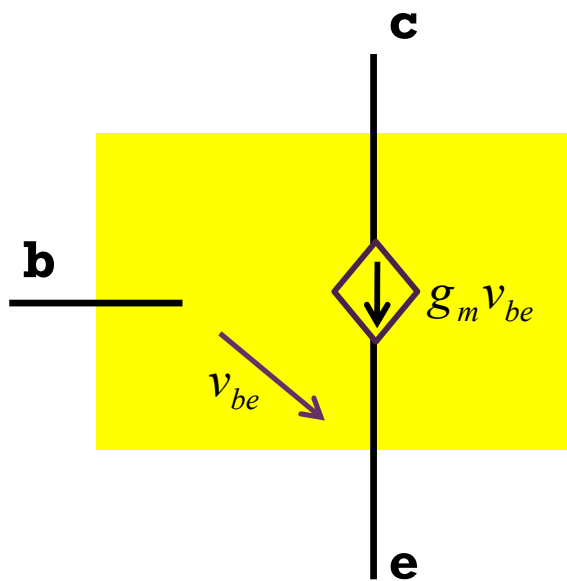


$$A_{v,CC} = +g_{mf} R_L = \frac{g_m}{1 + g_m R_L} R_L$$

理想晶体管小结

- 晶体管三种组态放大器小信号分析时，只要 $R_S \ll r_{be}$ (CE、CC)， $R_L \ll r_{ce}$ (CE、CB)，就可以采用理想晶体管模型，下述三种模型完全等价，三种组态均可通用

$$R_S \ll r_{be}, R_L \ll r_{ce}$$



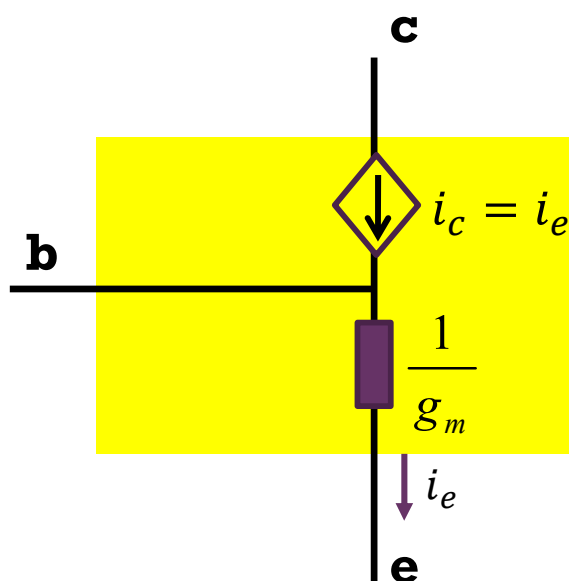
π 模型

三种组态可通用

CE组态最适

单向网络模型：理想压控流源

$$R_L \ll r_{ce}$$



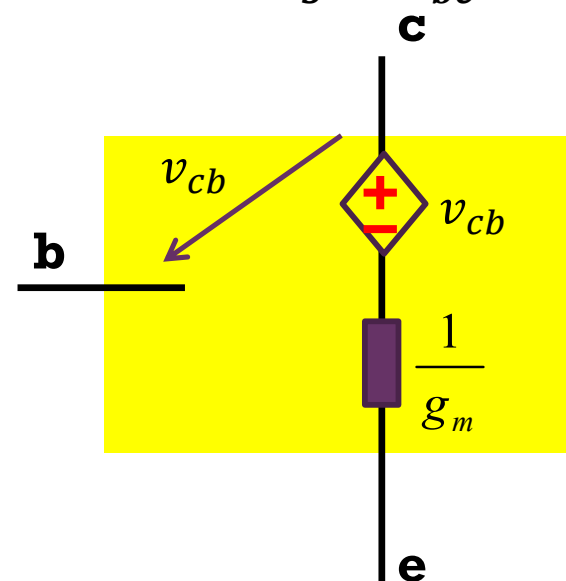
T 模型

三种组态可通用

CB组态最适

单向网络模型：电流缓冲器

$$R_S \ll r_{be}$$

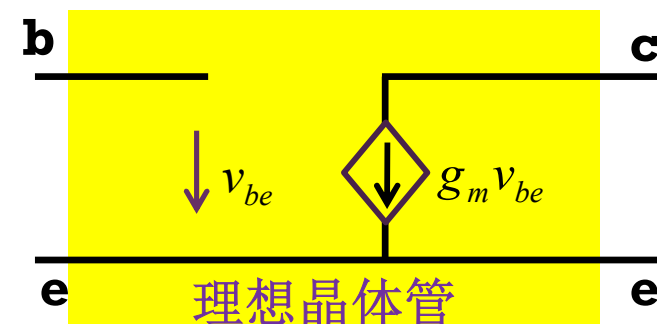
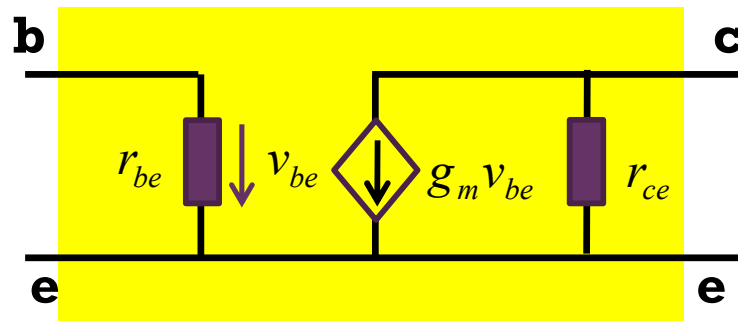
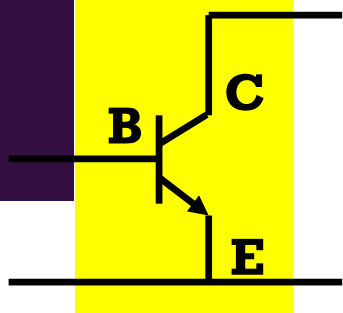


?模型

三种组态可通用

CC组态最适

单向网络模型：电压缓冲器



CE组态小结

- CE组态是单向网络，分析十分简单
 - 输入电阻和负载电阻无关，输出电阻和信源内阻无关
 - 总传递函数等于分传递函数之积，电压增益=(输出回路电阻)*(本征跨导增益)*(输入回路分压系数)

$$r_{in} = r_{be}$$

$$r_{out} = r_{ce}$$

$$R_S \ll r_{be}, R_L \ll r_{ce}$$

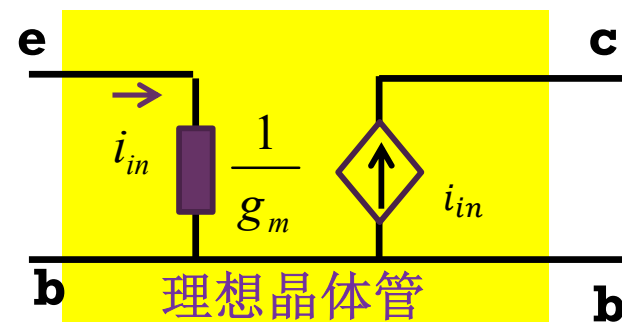
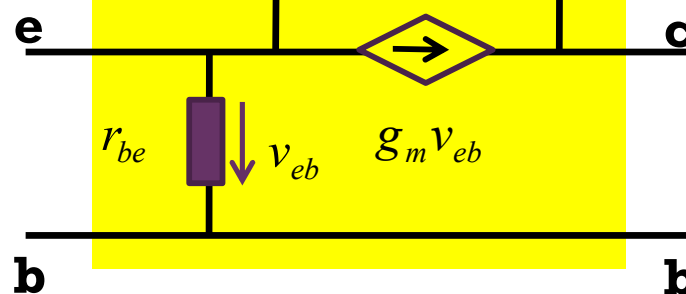
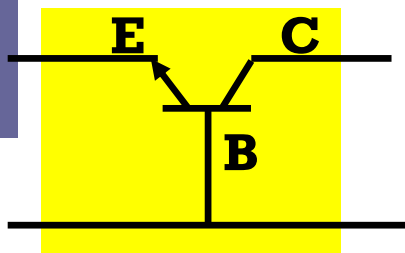
r_{in} 很大

r_{out} 很大

$$A_{v,CE} = \frac{r_{ce} R_L}{r_{ce} + R_L} (-g_m) \frac{r_{be}}{r_{be} + R_S}$$

$$A_{v,CE} \sim -g_m R_L$$

$$= \frac{r_{ce}}{r_{ce} + R_L} (-g_m R_L) \frac{r_{be}}{r_{be} + R_S}$$



CB组态小结

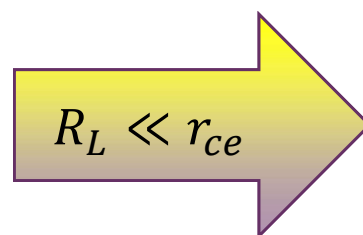
■ CB组态是双向网络

■ 输入电阻和负载电阻有关，输出电阻和信源内阻有关

■ 输出对输入的反向作用来自 r_{ce}

■ 当 $R_L \ll r_{ce}$ ，则可单向化处理，用理想晶体管CB组态电流缓冲器单向网络模型抽象

$$r_{in} = r_{be} \parallel \frac{r_{ce} + R_L}{1 + g_m r_{ce}} = r_{be} \parallel r_{ei}$$



$$r_{in} \sim \frac{1}{g_m} \quad \text{发射极对地阻抗}$$

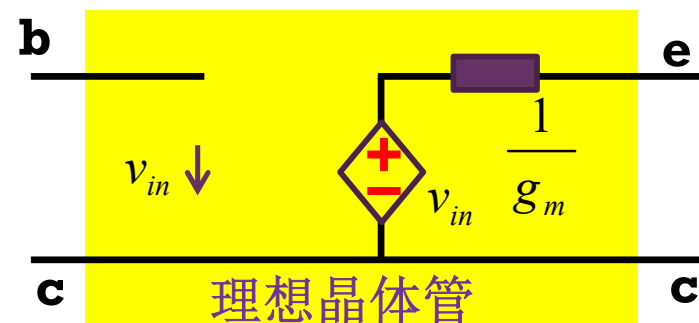
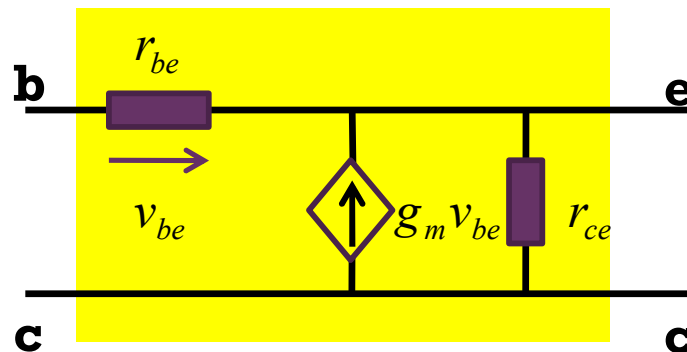
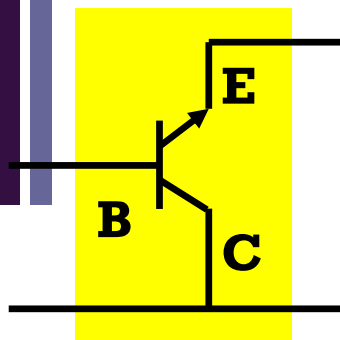
$$r_{out} = r_{be} \parallel R_S + r_{ce} + g_m (r_{be} \parallel R_S) r_{ce}$$

r_{out} 为bc阻抗，极大

$$A_{v,CB} = \frac{(g_m r_{ce} + 1) r_{be} R_L}{(g_m r_{ce} + 1) r_{be} R_S + (r_{be} + R_S)(r_{ce} + R_L)}$$

$$A_{v,CB} \sim g_{mf} R_L = \frac{g_m}{1 + g_m R_S} R_L$$

$$= R_L \cdot \frac{r_{be} \parallel R_S}{r_{be} \parallel R_S + r_{ei}} \cdot \frac{1}{R_S}$$



CC组态小结

- CC组态是双向网络，
 - 输入电阻和负载电阻有关，输出电阻和信源内阻有关
 - 输出对输入的反向作用来自 r_{be}
 - 当 $R_S \ll r_{be}$ ，则可单向化处理，用理想晶体管CC组态电压缓冲器单向网络模型抽象

$$r_{in} = r_{be} + r_{ce} || R_L + g_m r_{be} (r_{ce} || R_L)$$

$$r_{out} = r_{ce} || \frac{r_{be} + R_S}{1 + g_m r_{be}} = r_{ce} || r_{eo}$$

$$R_S \ll r_{be}$$

r_{in} 为bc阻抗，极大

$r_{out} \sim \frac{1}{g_m}$ 发射极对地阻抗

$$A_{v,CC} = \frac{(g_m r_{be} + 1) r_{ce} R_L}{(g_m r_{be} + 1) r_{ce} R_L + (r_{be} + R_S)(r_{ce} + R_L)}$$

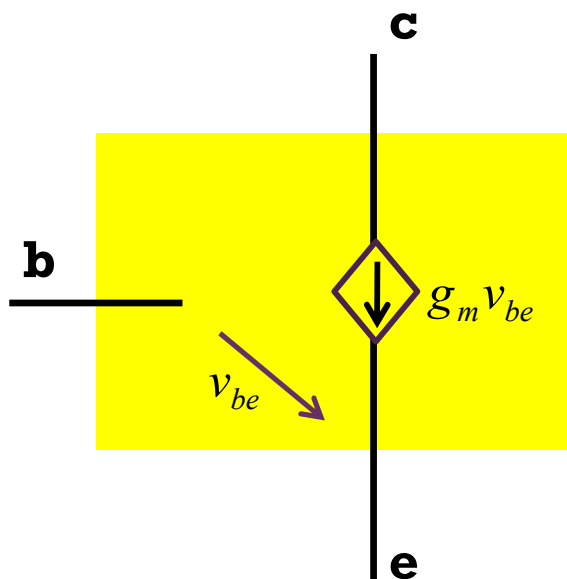
$$= \frac{r_{ce} || R_L}{r_{ce} || R_L + r_{eo}}$$

$$A_{v,CC} \sim g_{mf} R_L = \frac{g_m}{1 + g_m R_L} R_L$$

理想晶体管特性

$$r_{bc} = \infty$$

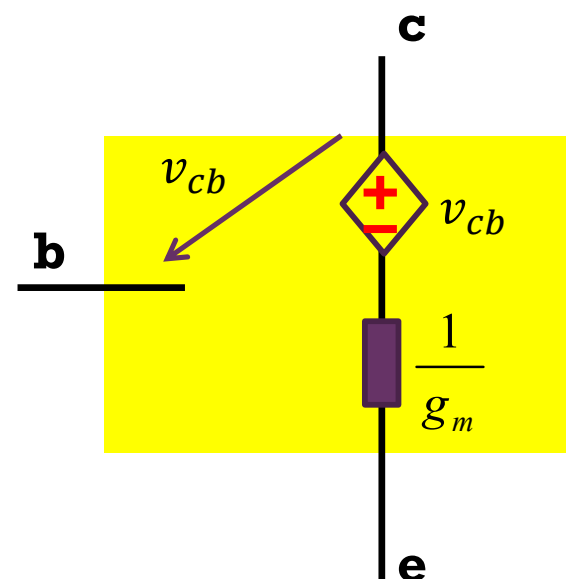
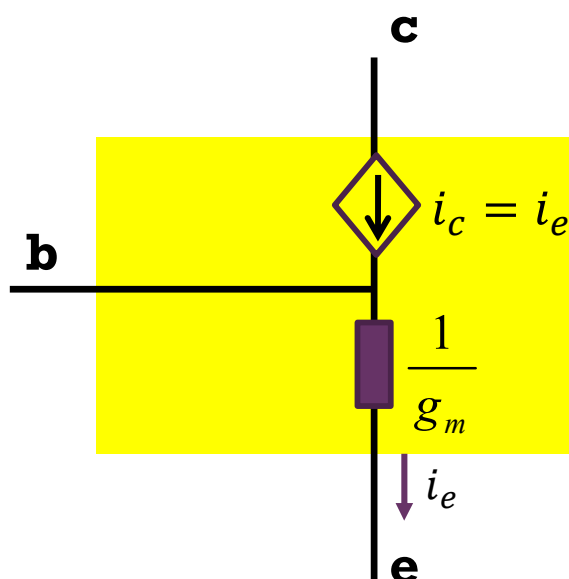
$$r_{bc} = r_{1,be} + r_{2,ce} + g_m r_{1,be} r_{2,ce}$$



$$i_b = 0$$

$$i_c = i_e = g_m v_{be}$$

$$r_e = \frac{1}{g_m}$$

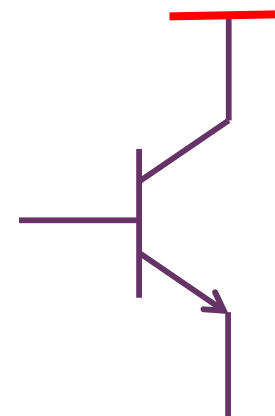
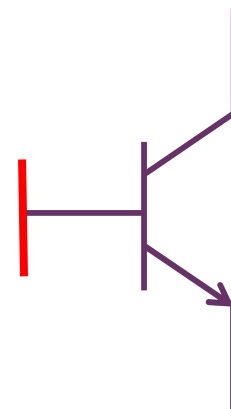
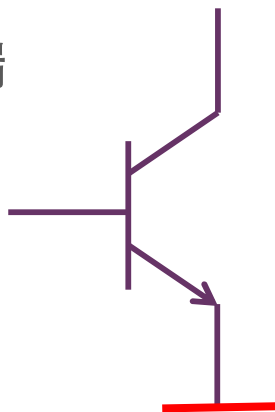


集电极电流等于发射极电流：描述如下事实
发射极发射的载流子几乎全部被集电极收集
源极提供的载流子通过沟道全部在漏极漏走

发射极看入阻抗（对地阻抗）为 $\frac{1}{g_m}$

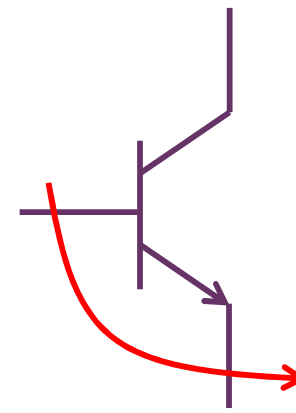
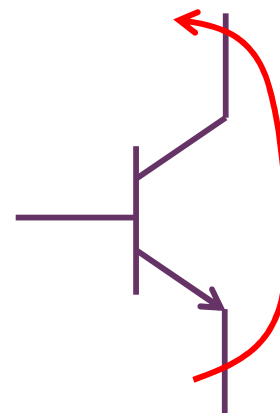
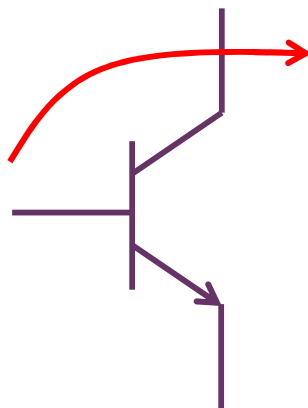
晶体管组态判定方法

- 看哪个端点交流接地
 - 谁接地，该端就是公共端



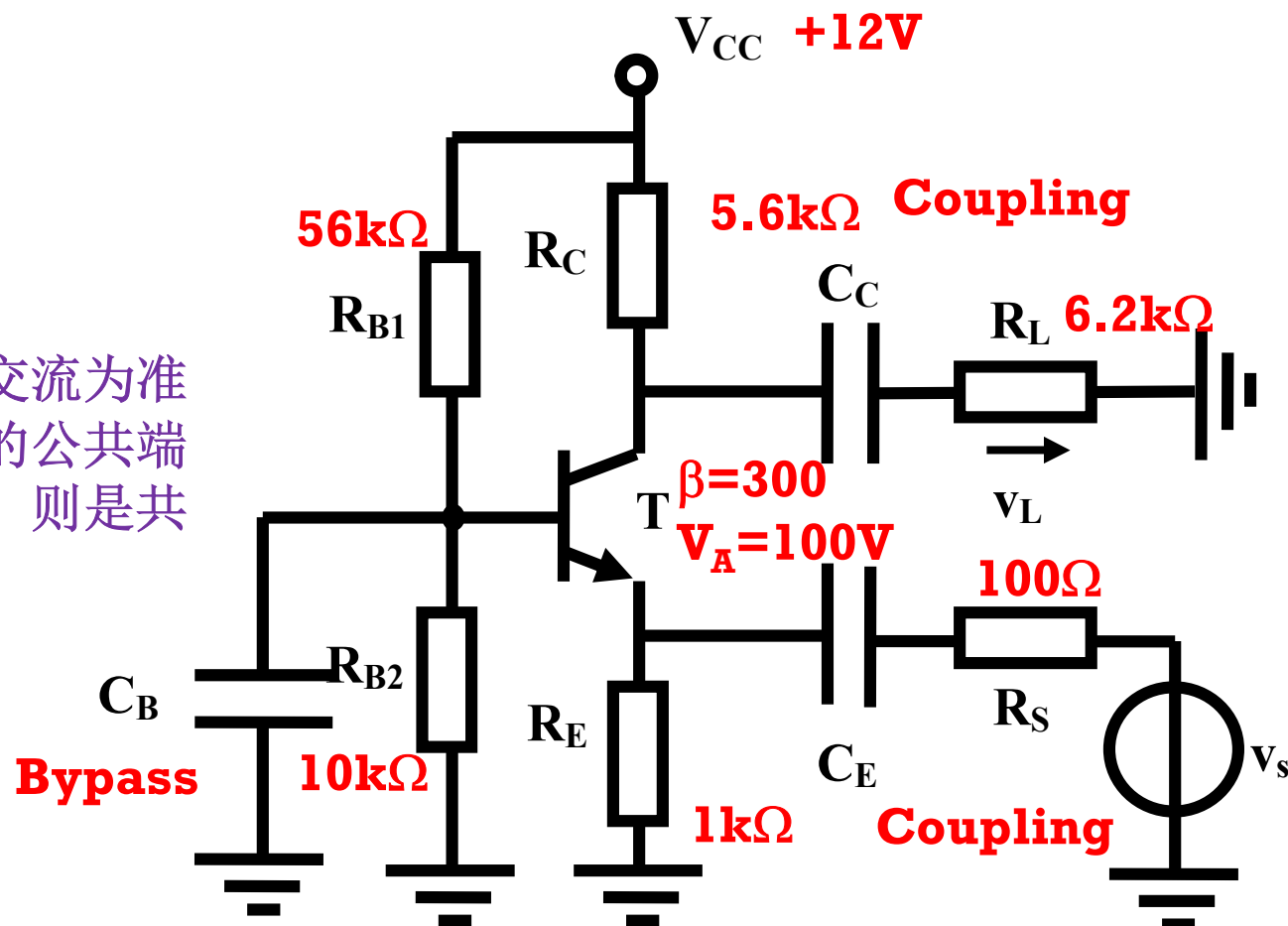
- 看信号放大路径，信号如何流动

- 信号从B到C，就是共E
- 信号从B到E，就是共C
- 信号从E到C，就是共B



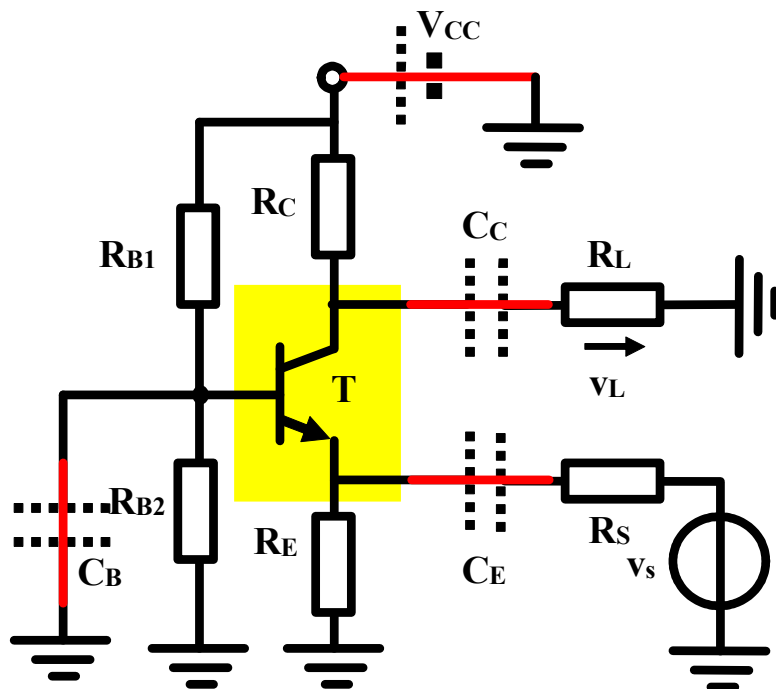
CB组态放大器分析例

组态是以交流为准
交流分析的公共端
点是基极，则是共
基组态

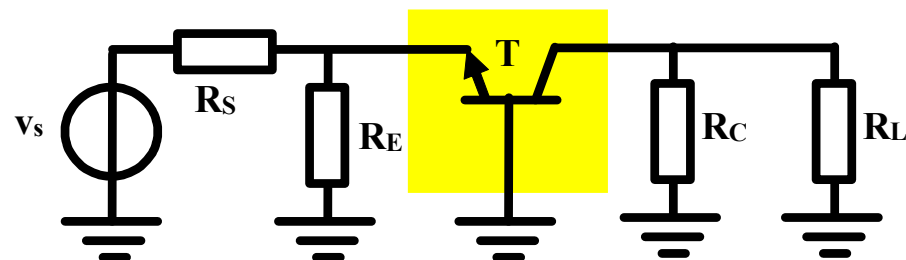


$$A_v = \frac{v_L}{v_s} = ?$$

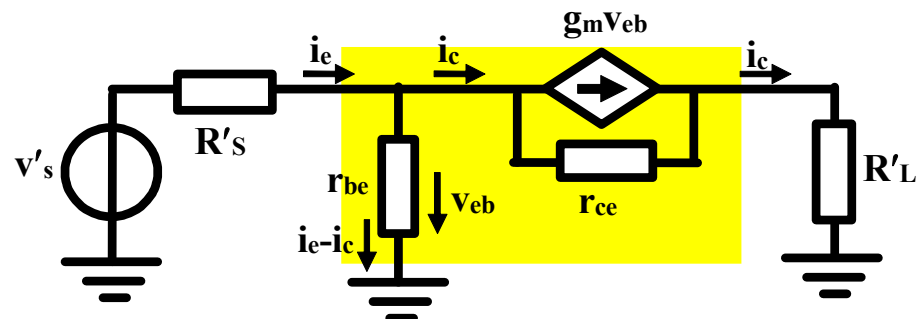
交流小信号分析



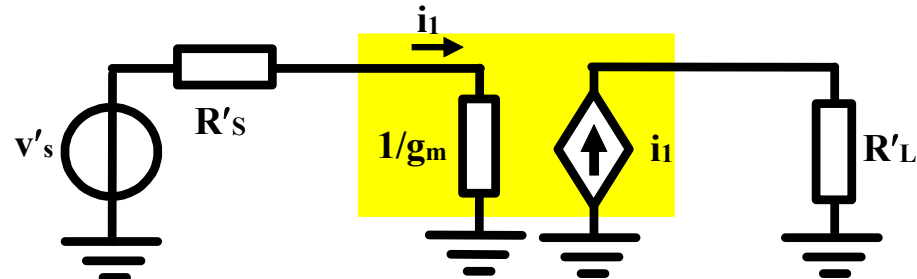
(a) 耦合电容、直流偏置电压源交流短路



(b) 交流小信号分析电路



(c) 晶体管采用通用跨导器模型



(d) 晶体管采用 CB 组态电流缓冲器模型

$$v'_s = \frac{R_E}{R_E + R_S} v_s = \frac{1k}{1k + 0.1k} v_s = 0.909 v_s$$

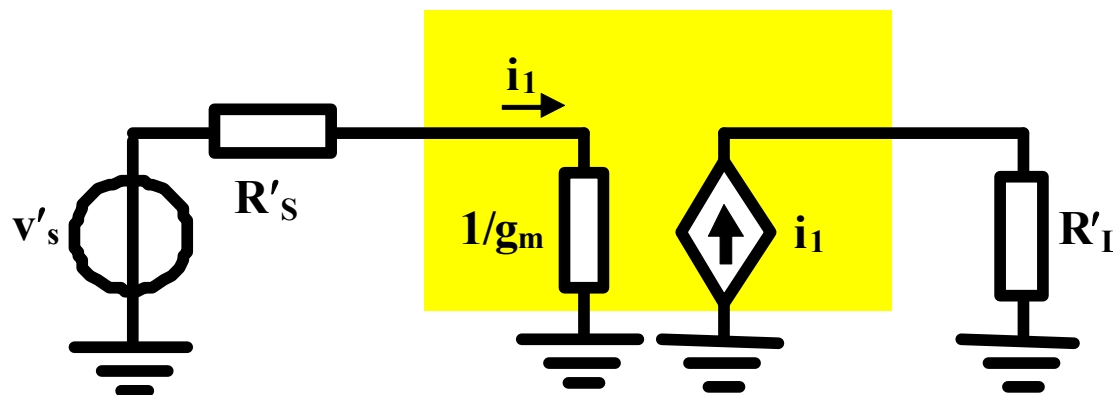
$$R'_S = R_S \parallel R_E = \frac{R_E R_S}{R_E + R_S} = \frac{1k \times 0.1k}{1k + 0.1k} = 90.9\Omega$$

$$R'_L = R_L \parallel R_C = 6.2k\Omega \parallel 5.6k\Omega = 2.94k\Omega$$

满足理想晶体管抽象条件 采用CB组态最适模型

$$R'_L = 2.94k\Omega \ll r_{ce} = 92.6k\Omega$$

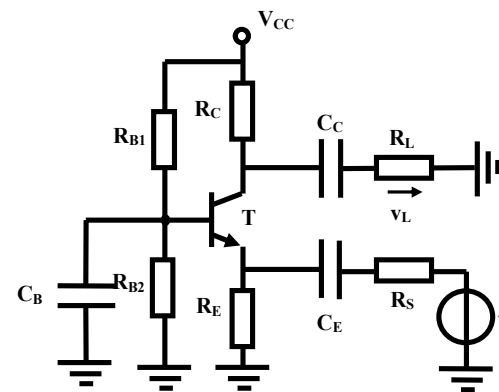
我们总是喜欢简单模型：
结论简洁，易于记忆



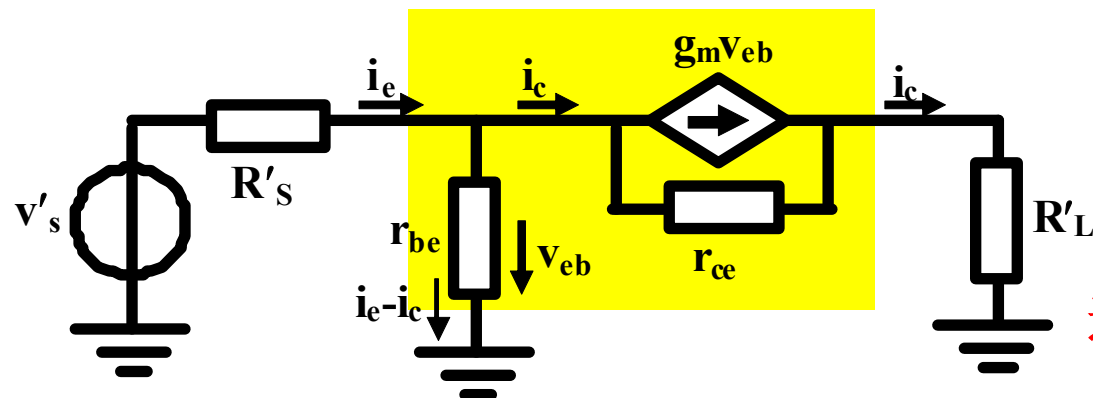
$$i_1 = \frac{v'_s}{R'_S + \frac{1}{g_m}} = \frac{g_m}{1 + g_m R'_S} v'_s = g_{mf} v'_s$$

$$v_L = i_1 R_L = \frac{g_m R'_L}{1 + g_m R'_S} v'_s = g_{mf} R'_L v'_s$$

$$= \frac{41.5m \times 2.94k}{1 + 41.5m \times 0.091k} \times 0.909 v_s = \frac{122}{1 + 3.77} \times 0.909 v_s = 23.2$$



27.3dB的同相电压放大



通用跨导器模型

$$v'_s = R'_S i_e + (i_e - i_c) r_{be}$$

$$v'_s = R'_S i_e + (i_c - g_m (i_e - i_c) r_{be}) r_{ce} + i_c R'_L$$

$$v_L = i_c R'_L = \frac{(1 + g_m r_{ce}) r_{be} R'_L}{R'_S (r_{be} + r_{ce} + g_m r_{be} r_{ce} + R'_L) + r_{be} (r_{ce} + R'_L)} v'_s = R'_L \cdot \frac{r_{be} || R'_S}{r_{be} || R'_S + r_{ei}} \cdot \frac{v'_s}{R'_S}$$

$$= \frac{(1 + 41.5m \times 92.6k) \times 7.22k \times 2.94k}{90.9 \times (7.22k + 92.6k + 41.5m \times 7.22k \times 92.6k + 2.94k) + 7.22k \times (92.6k + 2.94k)} v'_s$$

$$= 25.3 \times v'_s = 25.3 \times 0.909 v_s = 23 v_s$$

同相电压放大
27.2dB

和理想晶体管模型结论**27.3dB**比没有本质区别，但理想晶体管模型极度简单

$$\sim \frac{g_m R'_L}{1 + g_m R'_S} v'_s$$

三种组态 小结

- 理想晶体管CE组态为理想压控流源，CB组态为电流缓冲器，CC组态为电压缓冲器，上述三个理想模型对三种组态通用
- 在如下条件满足前提下，则应采用理想晶体管模型进行分析，可获得足够简单且足够精确的原理性结论
 - CE: $R_S \ll r_{be}$, $R_L \ll r_{ce}$: $A_{v0} = -g_m R_L$; $A_{v0} = -g_{mf} R_L = -\frac{g_m}{1+g_m R_E} R_L$
 - CB: $R_L \ll r_{ce}$ $A_{v0} = +g_{mf} R_L = +\frac{g_m}{1+g_m R_S} R_L$
 - CC: $R_S \ll r_{be}$ $A_{v0} = +g_{mf} R_L = +\frac{g_m}{1+g_m R_L} R_L$
- 大多数应用情况下，发射极对地阻抗用理想晶体管结论 $r_e = 1/g_m$ 足够用，bc端口阻抗 $r_{bc} = r_{1,be} + r_{2,ce} + g_m r_{1,be} r_{2,ce}$ 极大可建模为开路
 - 作业：研究输入阻抗、输出阻抗随对方端口所接负载变化而变化的规律

本讲内容在教材中的章节位置

- P342: 晶体管放大器分析
- P353: 三种组态

数字门电路作业选讲

作业11.1 加法器设计

以上为课堂授课内容
以下为录像自学内容

■ 二进制加法运算

$$\begin{array}{r} 10110100 \\ + 11101101 \\ \hline 110100001 \end{array}$$

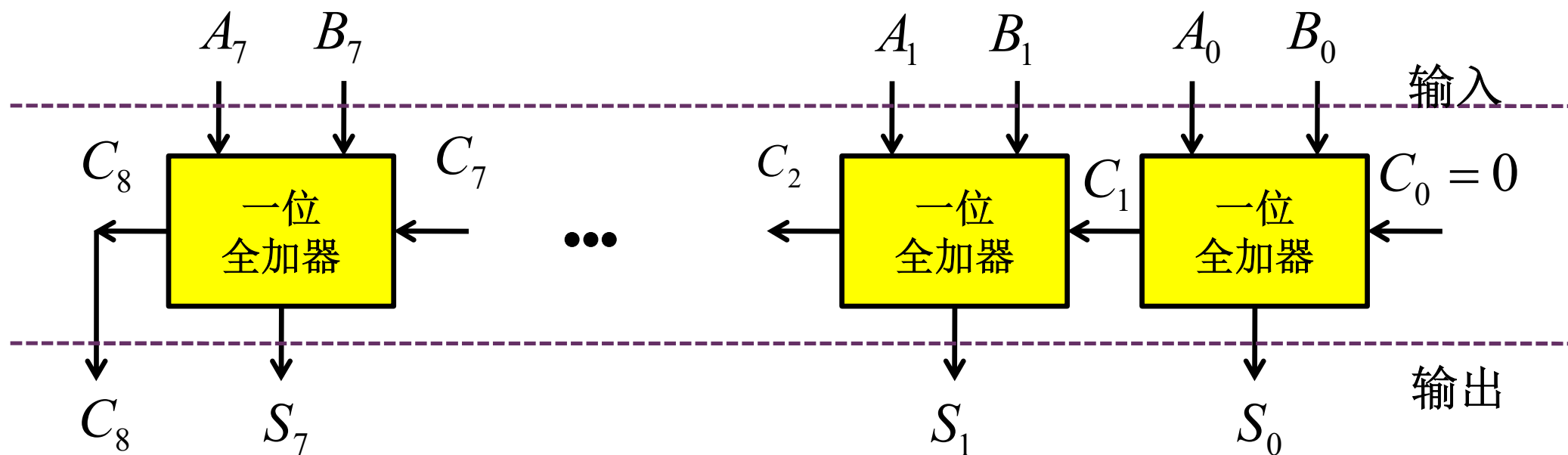
$$\begin{array}{r} 180 \\ + 237 \\ \hline 417 \end{array}$$

	C_8	C_7	C_6	C_5	C_4	C_3	C_2	C_1	0
	0	A_7	A_6	A_5	A_4	A_3	A_2	A_1	A_0
+	0	B_7	B_6	B_5	B_4	B_3	B_2	B_1	B_0
	C_8	S_7	S_6	S_5	S_4	S_3	S_2	S_1	S_0

对每一位 i ，都是 $\mathbf{A_i+B_i+C_i}$ ，产生和 $\mathbf{S_i}$ 以及进位 $\mathbf{C_{i+1}}$

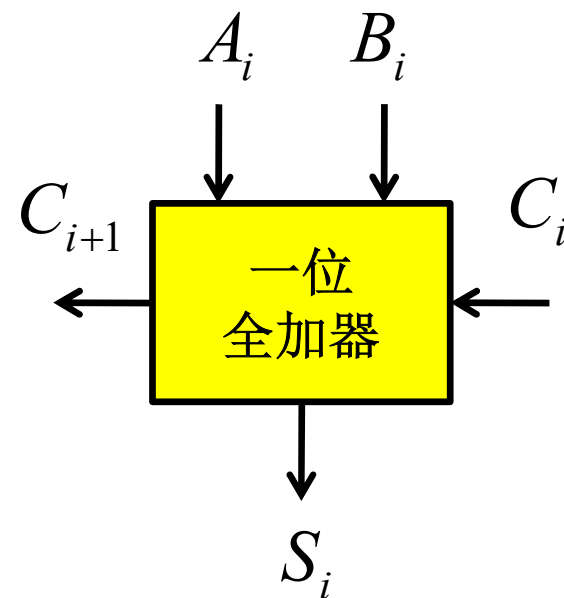
加法器系统框架

- 一位一位地加，同时考虑进位
 - 8个一位全加器级联，即可实现两个8位二进制数的相加功能



一位全加器设计

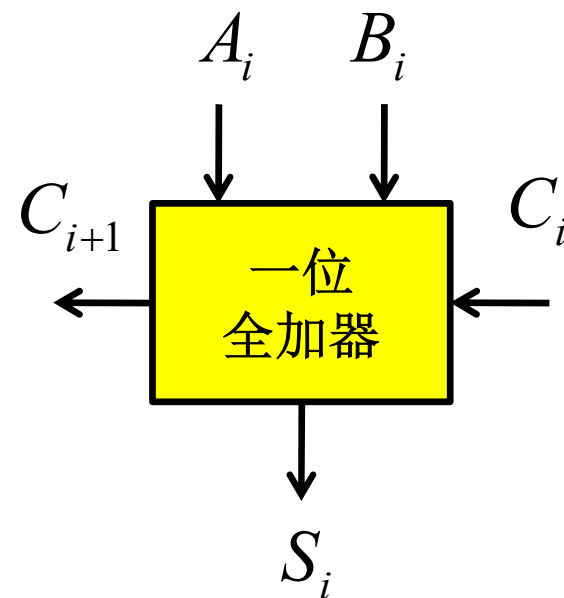
A_i	B_i	C_i	C_{i+1}	S_i
0	0	0		
0	0	1		
0	1	0		
0	1	1		
1	0	0		
1	0	1		
1	1	0		
1	1	1		



- 填写真值表
- 用卡诺图化简
- 用CMOS晶体管实现
 - 教材例题跟做一遍

真值表就是对功能的具体描述

A_i	B_i	C_i	C_{i+1}	S_i
0	0	0	0	0
0	0	1	0	1
0	1	0	0	1
0	1	1	1	0
1	0	0	0	1
1	0	1	1	0
1	1	0	1	0
1	1	1	1	1



$$S_i = \overline{A_i} \cdot \overline{B_i} \cdot C_i + \overline{A_i} \cdot B_i \cdot \overline{C_i} + A_i \cdot \overline{B_i} \cdot \overline{C_i} + A_i \cdot B_i \cdot C_i$$

$$C_{i+1} = \overline{A_i} \cdot B_i \cdot C_i + A_i \cdot \overline{B_i} \cdot C_i + A_i \cdot B_i \cdot \overline{C_i} + A_i \cdot B_i \cdot C_i$$

卡诺图化简

$$S_i = \overline{A_i} \cdot \overline{B_i} \cdot C_i + \overline{A_i} \cdot B_i \cdot \overline{C_i} + A_i \cdot \overline{B_i} \cdot \overline{C_i} + A_i \cdot B_i \cdot C_i$$

C \ AB	00	01	11	10
0	0	1	0	1
1	1	0	1	0

$$C_{i+1} = \overline{A_i} \cdot B_i \cdot C_i + A_i \cdot \overline{B_i} \cdot C_i + A_i \cdot B_i \cdot \overline{C_i} + A_i \cdot B_i \cdot C_i$$

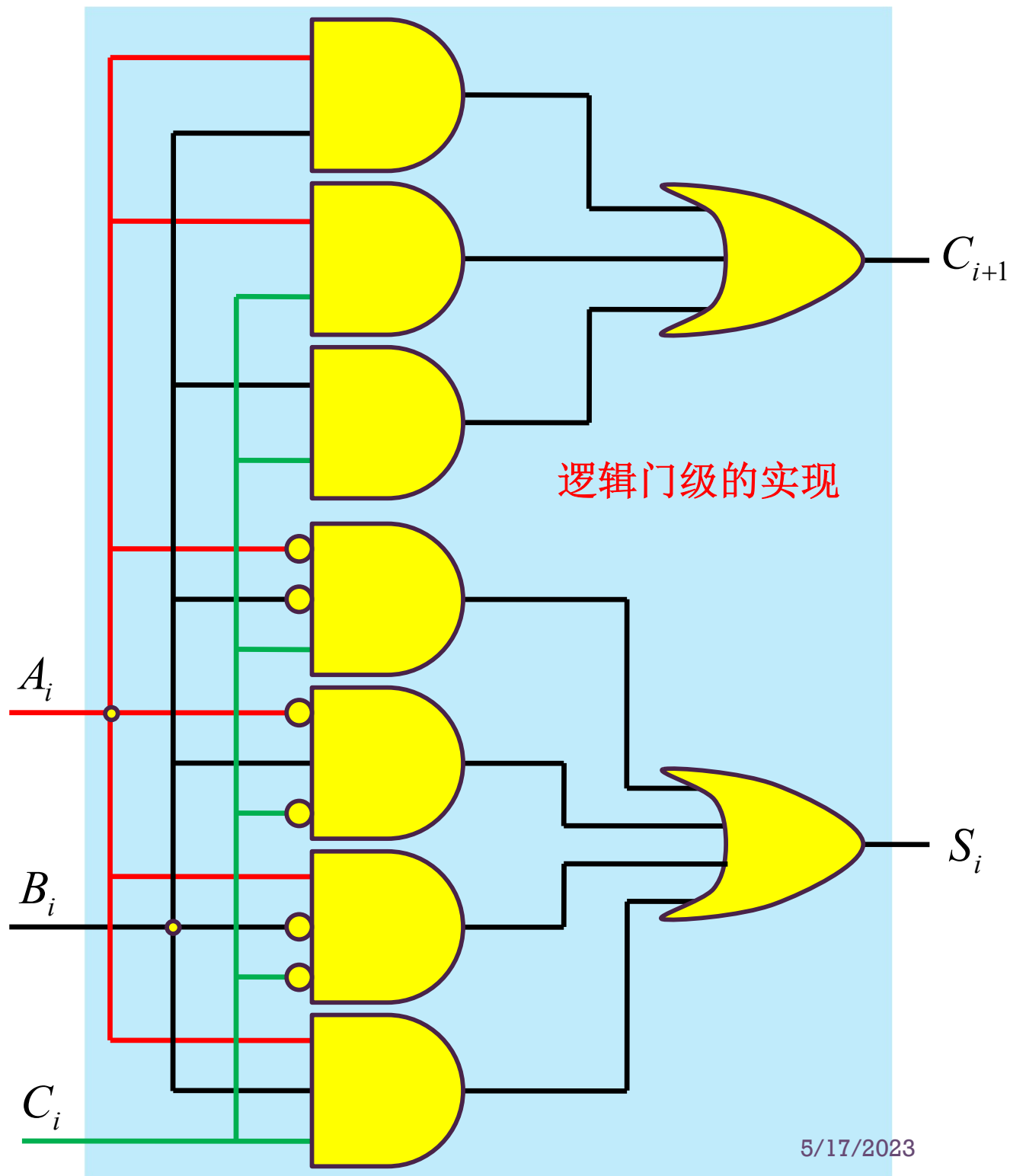
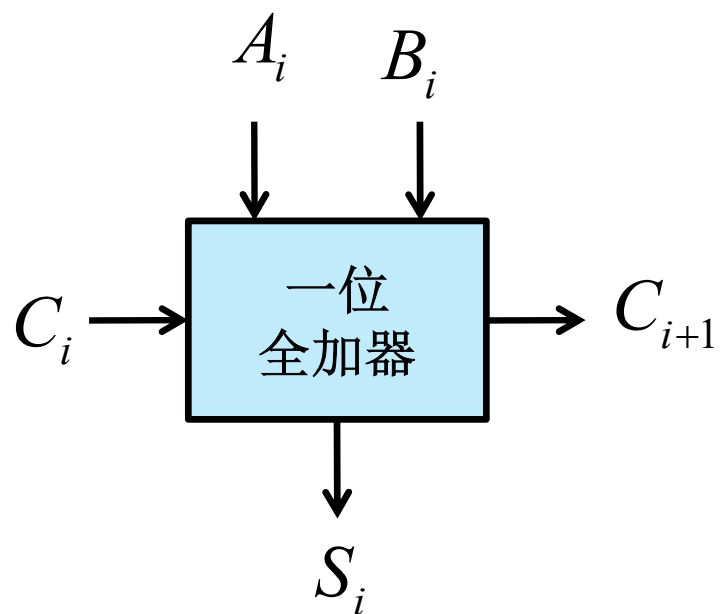
C \ AB	00	01	11	10
0	0	0	1	0
1	0	1	1	1

$$C_{i+1} = A_i B_i + B_i C_i + A_i C_i$$

逻辑级实现

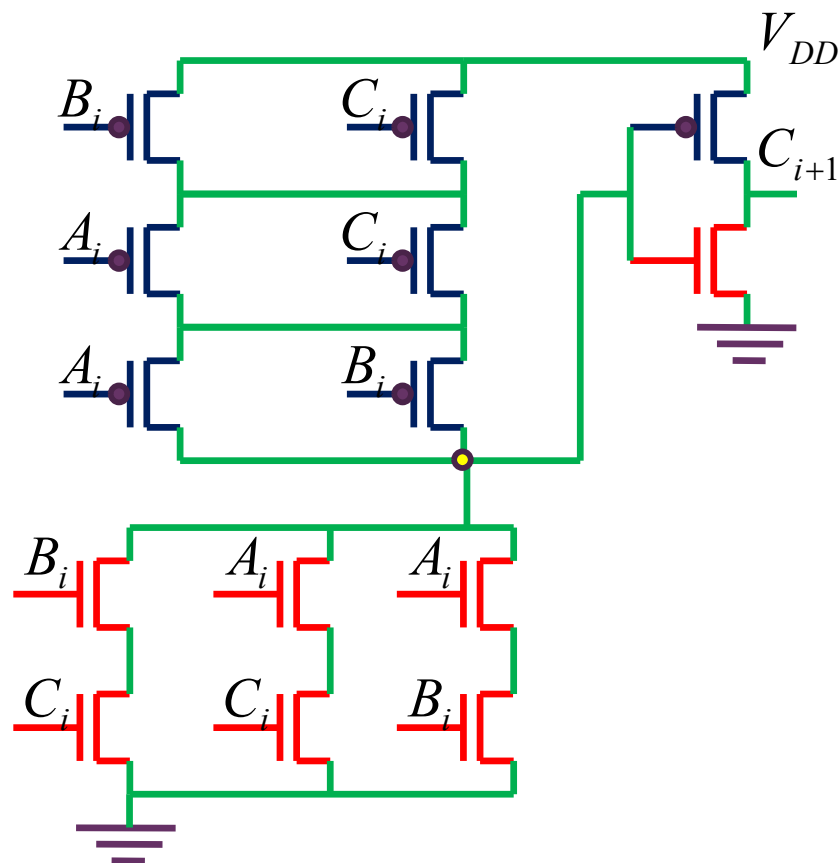
$$S_i = \overline{A_i} \cdot \overline{B_i} \cdot C_i + \overline{A_i} \cdot B_i \cdot \overline{C_i} + A_i \cdot \overline{B_i} \cdot \overline{C_i} + A_i \cdot B_i \cdot C_i$$

$$C_{i+1} = B_i \cdot C_i + A_i \cdot C_i + A_i \cdot B_i$$



进位逻辑的晶体管级实现

$$C_{i+1} = B_i \cdot C_i + A_i \cdot C_i + A_i \cdot B_i$$



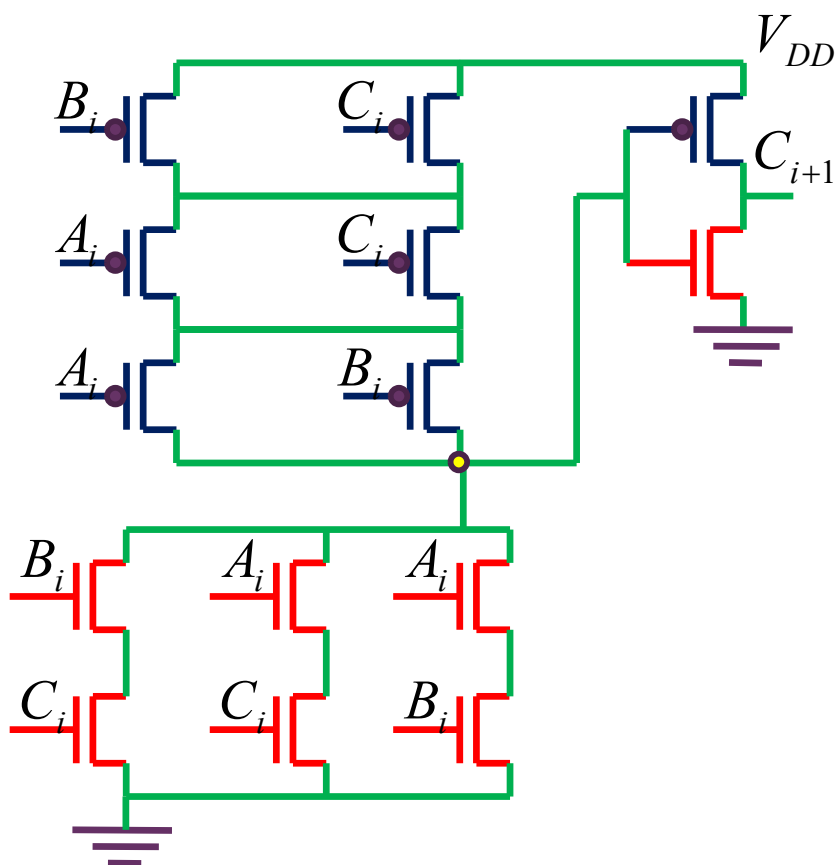
14个晶体管，5层垒叠

逻辑表达式整理后可降低晶体管个数

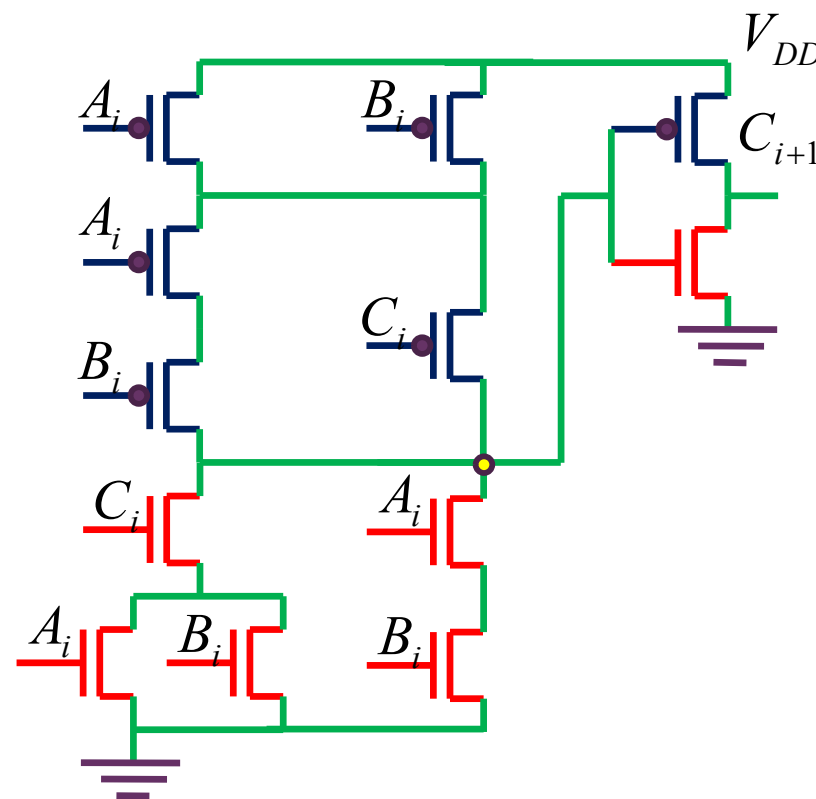
一个输入对应两个晶体管：减少输入个数

$$C_{i+1} = B_i \cdot C_i + A_i \cdot C_i + A_i \cdot B_i$$

$$C_{i+1} = (A_i + B_i)C_i + A_i B_i$$



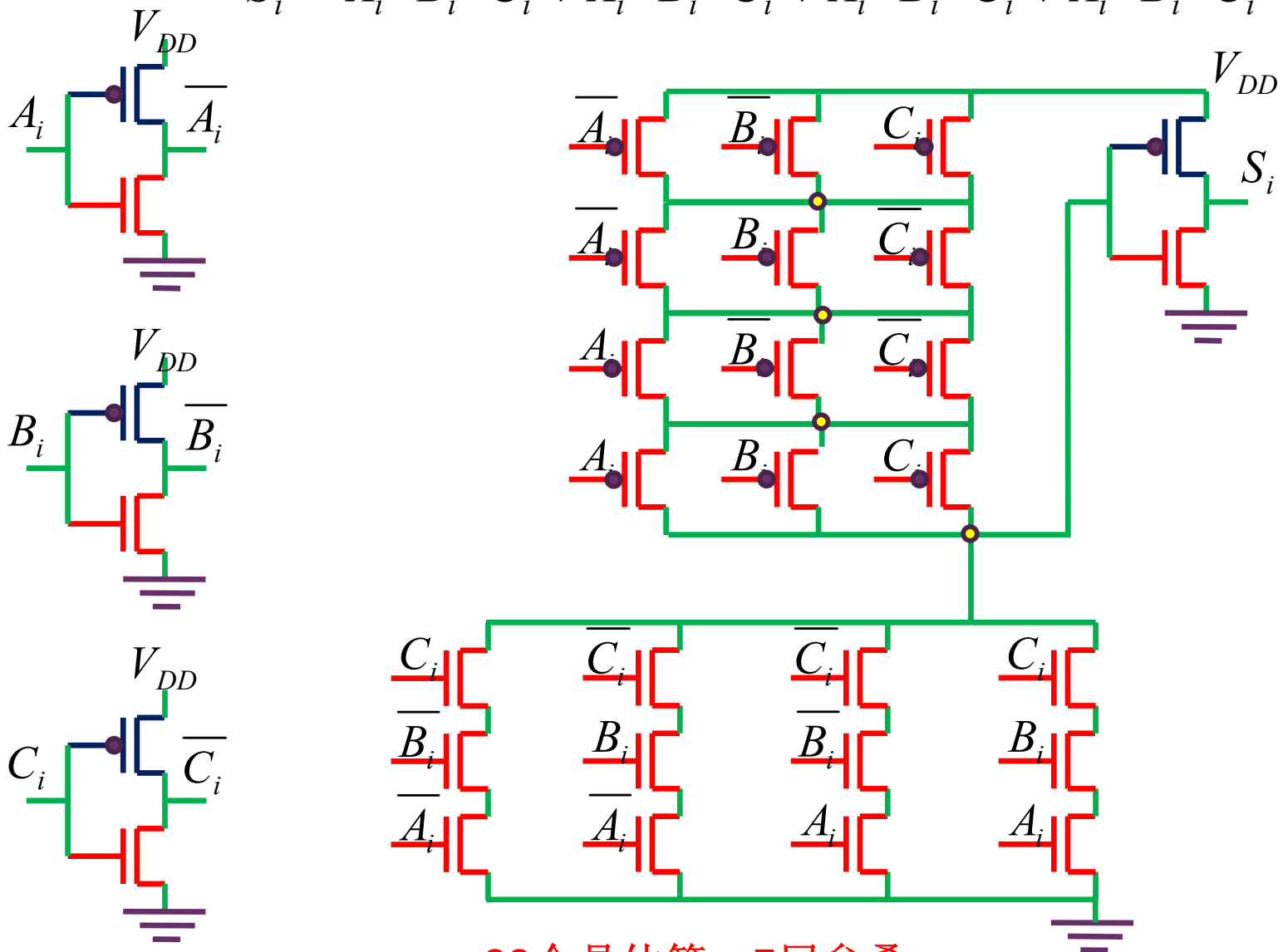
14个晶体管，5层垒叠



12个晶体管，5层垒叠

和位逻辑晶体管级实现

$$S_i = \overline{A_i} \cdot \overline{B_i} \cdot C_i + \overline{A_i} \cdot B_i \cdot \overline{C_i} + A_i \cdot \overline{B_i} \cdot \overline{C_i} + A_i \cdot B_i \cdot C_i$$



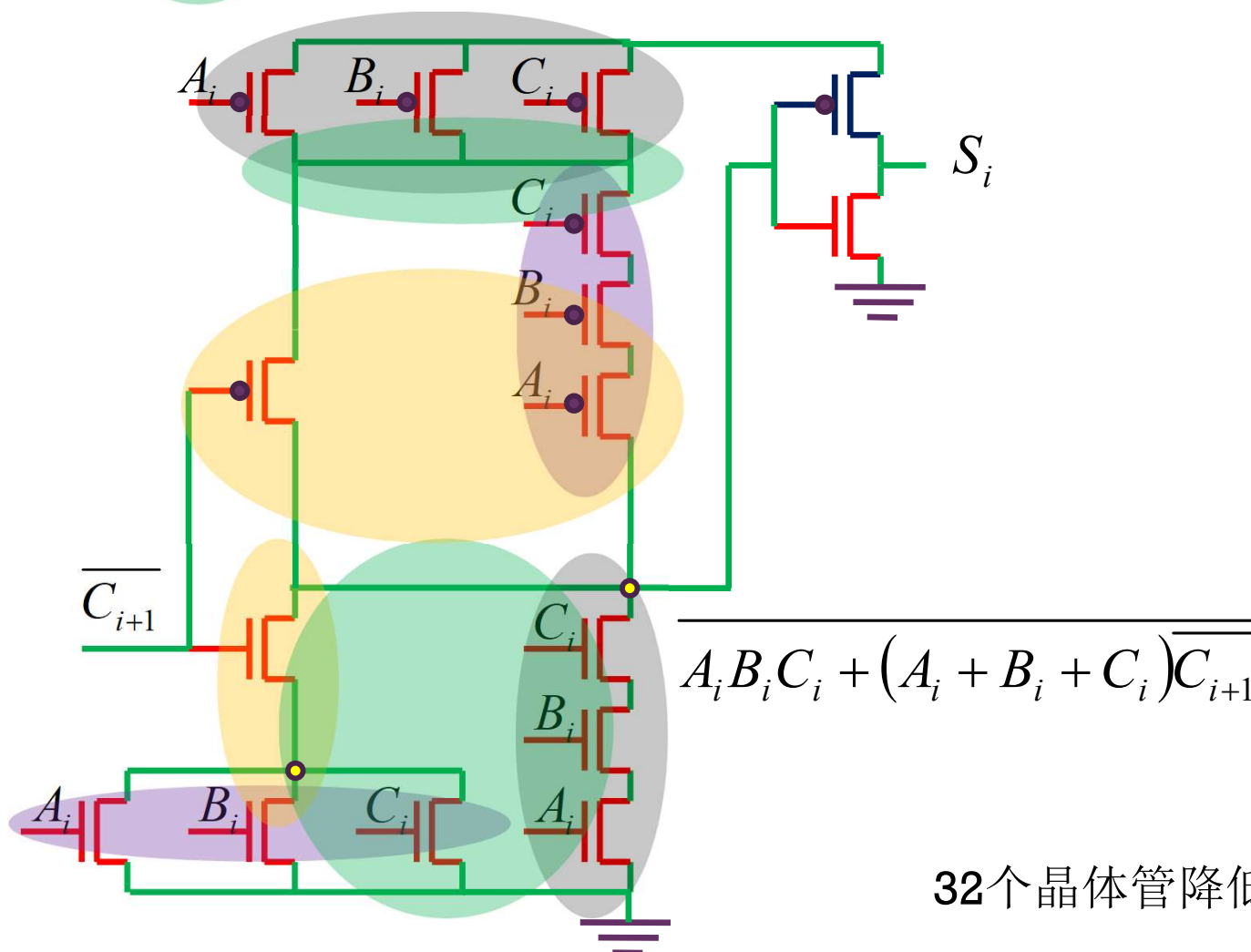
32个晶体管，7层垒叠

教材上给出的逻辑化简

$$S_i = \overline{A_i} \cdot \overline{B_i} \cdot C_i + \overline{A_i} \cdot B_i \cdot \overline{C_i} + A_i \cdot \overline{B_i} \cdot \overline{C_i} + A_i \cdot B_i \cdot C_i$$

$$= A_i \cdot B_i \cdot C_i + (A_i + B_i + C_i) \cdot \overline{C_{i+1}}$$

把12个输入降低为7个输入



32个晶体管降低为16个晶体管

简化逻辑的证明

$$S_i = \overline{A_i} \cdot \overline{B_i} \cdot C_i + \overline{A_i} \cdot B_i \cdot \overline{C_i} + A_i \cdot \overline{B_i} \cdot \overline{C_i} + A_i \cdot B_i \cdot C_i$$

$$= A_i \cdot B_i \cdot C_i + (\overline{A_i} + \overline{B_i} + \overline{C_i}) \cdot \overline{C_{i+1}}$$

A_i	B_i	C_i	C_{i+1}	S_i	$\overline{C_{i+1}}$	$(A_i + B_i + C_i)$		$A_i B_i C_i$	
0	0	0	0	0	1	0	0	0	0
0	0	1	0	1	1	1	1	0	1
0	1	0	0	1	1	1	1	0	1
0	1	1	1	0	0	1	0	0	0
1	0	0	0	1	1	1	1	0	1
1	0	1	1	0	0	1	0	0	0
1	1	0	1	0	0	1	0	0	0
1	1	1	1	1	0	1	0	1	1

两个1或三个1 一个1或三个1 没有1或一个1 有1即可 一个1 三个1 一个1或三个1

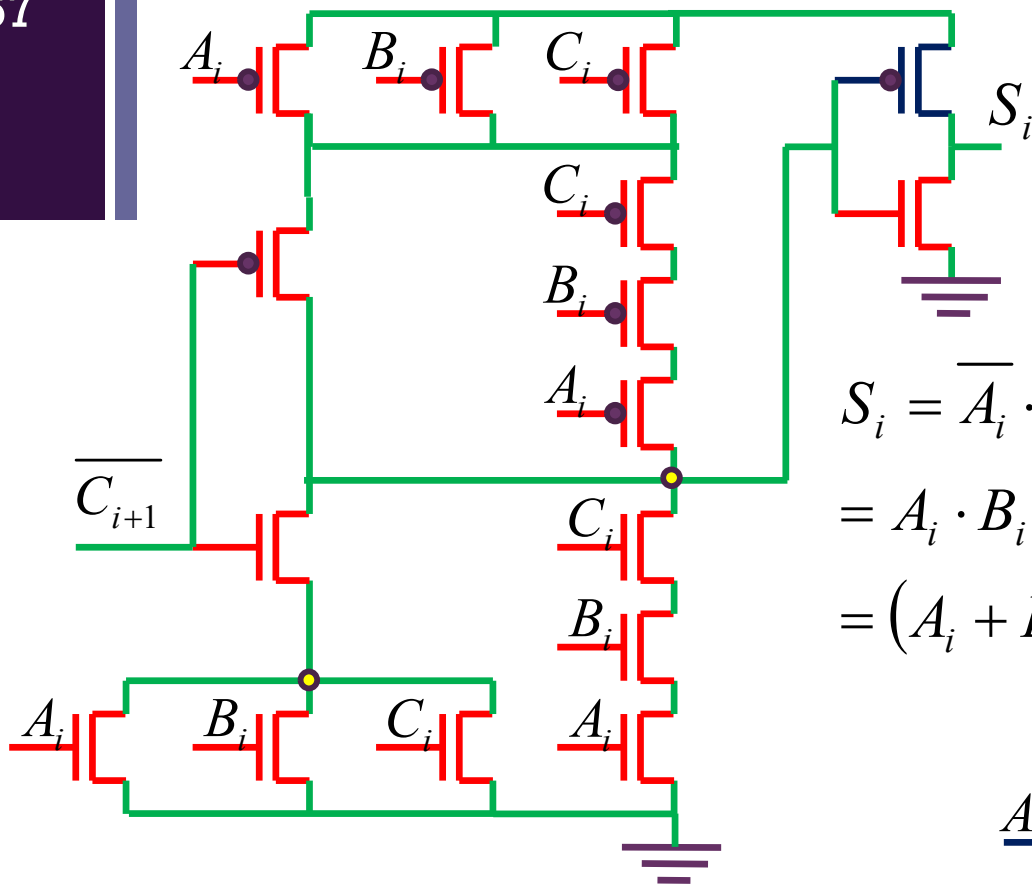
另一个简化逻辑

$$\begin{aligned}
 S_i &= \overline{A_i} \cdot \overline{B_i} \cdot C_i + \overline{A_i} \cdot B_i \cdot \overline{C_i} + A_i \cdot \overline{B_i} \cdot \overline{C_i} + A_i \cdot B_i \cdot C_i \\
 &= A_i \cdot B_i \cdot C_i + (\overline{A_i} + \overline{B_i} + \overline{C_i}) \cdot C_{i+1} \\
 &= (A_i + B_i + C_i) \cdot (A_i \cdot B_i \cdot C_i + \overline{C_{i+1}})
 \end{aligned}$$

A_i	B_i	C_i	C_{i+1}	S_i	$\overline{C_{i+1}}$	$A_i B_i C_i$		$(A_i + B_i + C_i)$	
0	0	0	0	0	1	0	1	0	0
0	0	1	0	1	1	0	1	1	1
0	1	0	0	1	1	0	1	1	1
0	1	1	1	0	0	0	0	1	0
1	0	0	0	1	1	0	1	1	1
1	0	1	1	0	0	0	0	1	0
1	1	0	1	0	0	0	0	1	0
1	1	1	1	1	0	1	1	1	1

两个1或三个1 一个1或三个1 没有1或一个1 三个1 没有1或一个1或三个1 有1即可 一个1或三个1

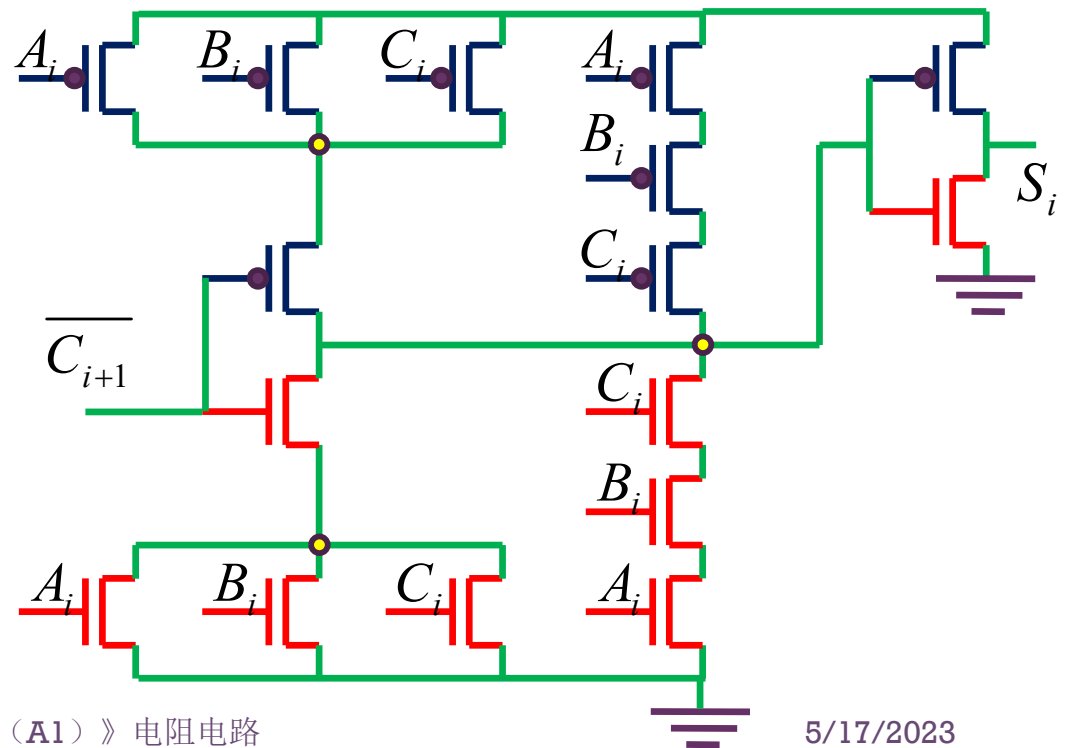
等同逻辑合用



$$\begin{aligned}
 S_i &= \overline{A_i} \cdot \overline{B_i} \cdot C_i + \overline{A_i} \cdot B_i \cdot \overline{C_i} + A_i \cdot \overline{B_i} \cdot \overline{C_i} + A_i \cdot B_i \cdot C_i \\
 &= A_i \cdot B_i \cdot C_i + (A_i + B_i + C_i) \cdot \overline{C_{i+1}} \\
 &= (A_i + B_i + C_i) \cdot (A_i \cdot B_i \cdot C_i + \overline{C_{i+1}})
 \end{aligned}$$

功能完全一样

但后者更对称，更紧致
结构不具互补特征，但逻辑
功能是互补的，故而可点接

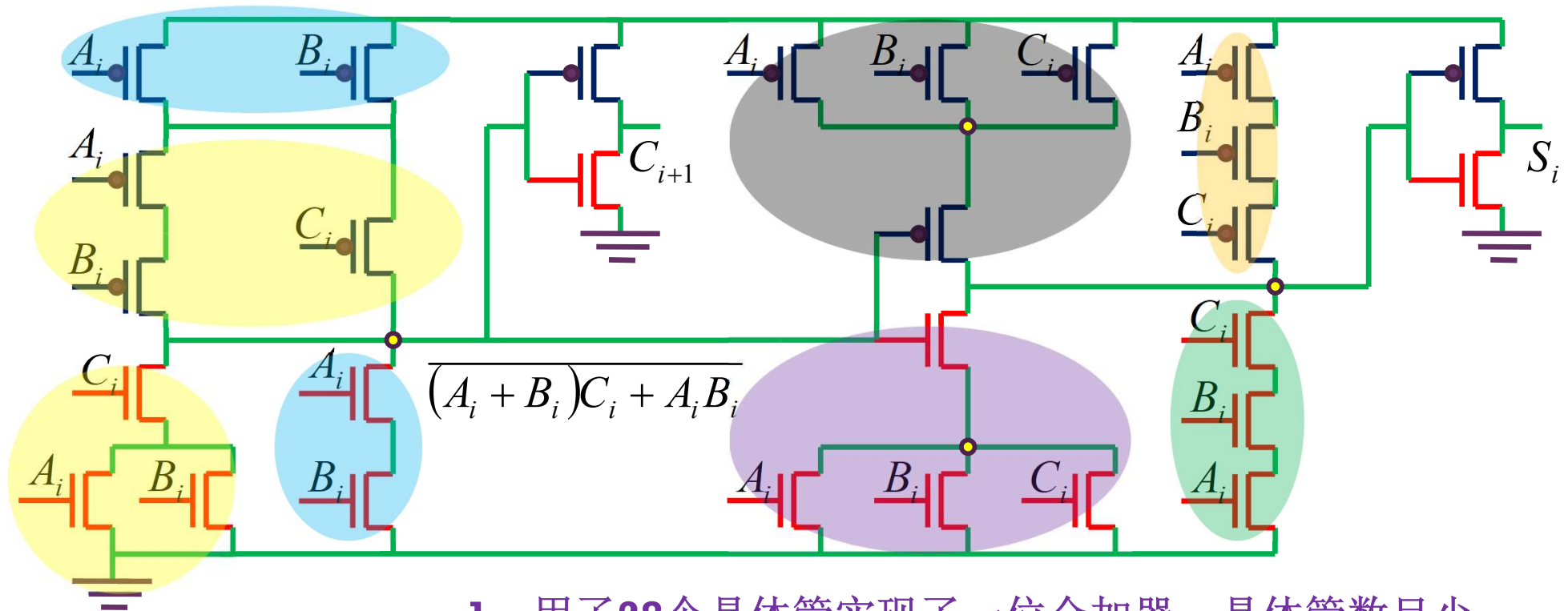


一位全加器的CMOS晶体管级实现

$$C_{i+1} = B_i C_i + A_i C_i + A_i B_i = (A_i + B_i) C_i + A_i B_i$$

$$S_i = \overline{A_i} \cdot \overline{B_i} \cdot C_i + \overline{A_i} \cdot B_i \cdot \overline{C_i} + A_i \cdot \overline{B_i} \cdot \overline{C_i} + A_i \cdot B_i \cdot C_i = A_i \cdot B_i \cdot C_i + (A_i + B_i + C_i) \cdot \overline{C_{i+1}}$$

$$= (A_i + B_i + C_i) \cdot (A_i \cdot B_i \cdot C_i + \overline{C_{i+1}})$$



- 1、用了**28**个晶体管实现了一位全加器；晶体管数目少
- 2、输出用反相器，犹如缓冲器，输出逻辑电平更稳定

用标准CMOS结构实现如下逻辑表达式，至少需要多少个晶体管？

$$Z = AB + BC$$

- ☐ A 4
- ☐ B 6
- ☒ C 8
- ☐ D 10

提交

作业12.2 卡诺图化简

- 1、化简卡诺图，写出输出Z用ABCD表述的逻辑表达式
- 2、用PMOS互补NMOS的CMOS电路形态（上P下N，形式互补）实现这些逻辑运算，画出CMOS晶体管级电路图
- 3、如果有与门、或门、非门电路可供选用，用与、或、非门实现上述逻辑
- 4、如果只有二输入的与非门可供选用，如何用与非门实现前述逻辑？

AB \ CD	00	01	11	10
00	1	0	0	1
01	0	1	*	*
11	1	1	*	1
10	1	*	0	1

卡诺图化简

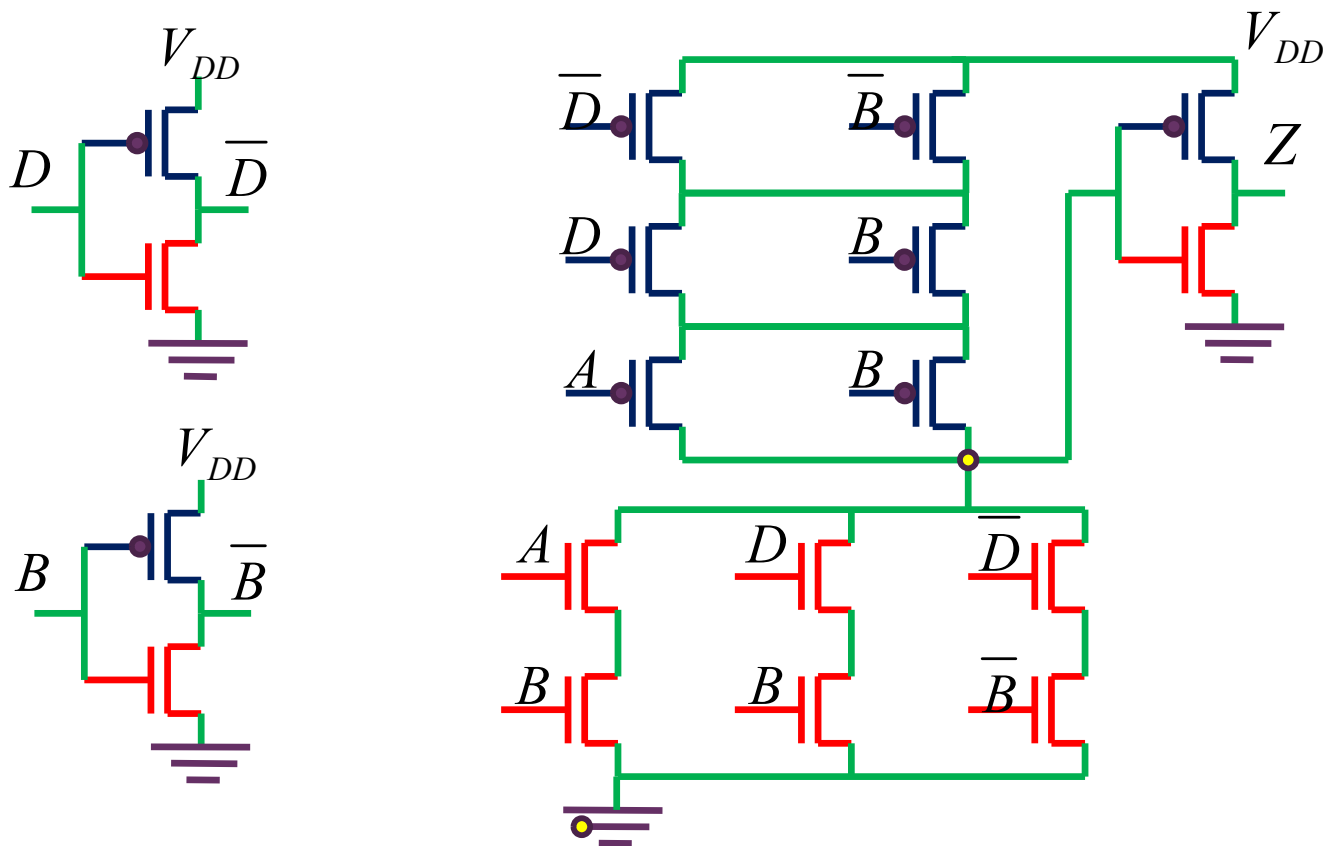
AB \ CD	00	01	11	10
00	1	0	0	1
01	0	1	*	*
11	1	1	*	1
10	1	*	0	1

$$Z = A \cdot B + B \cdot D + \overline{B} \cdot \overline{D}$$

表述不唯一

CMOS晶体管级实现

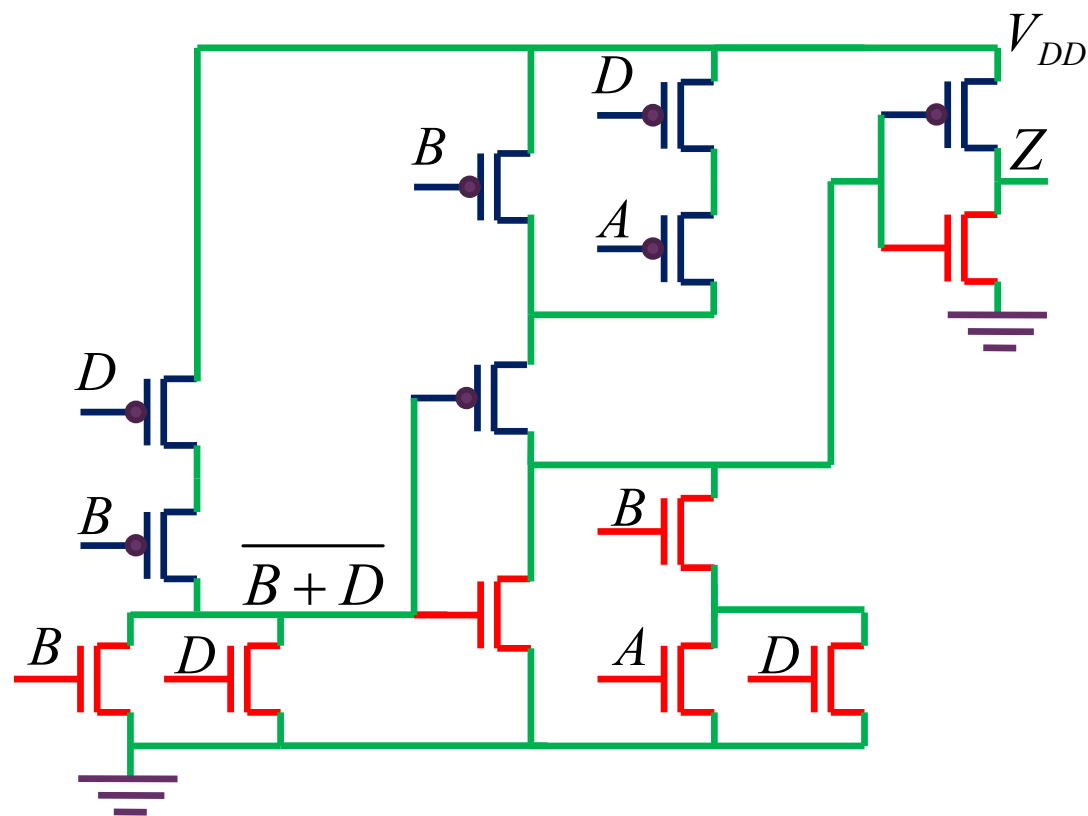
$$Z = A \cdot B + B \cdot D + \overline{B} \cdot \overline{D}$$



18个晶体管，5层垒叠

晶体管数目减少的措施

$$Z = A \cdot B + B \cdot D + \overline{B} \cdot \overline{D} = B \cdot (A + D) + \overline{B + D}$$

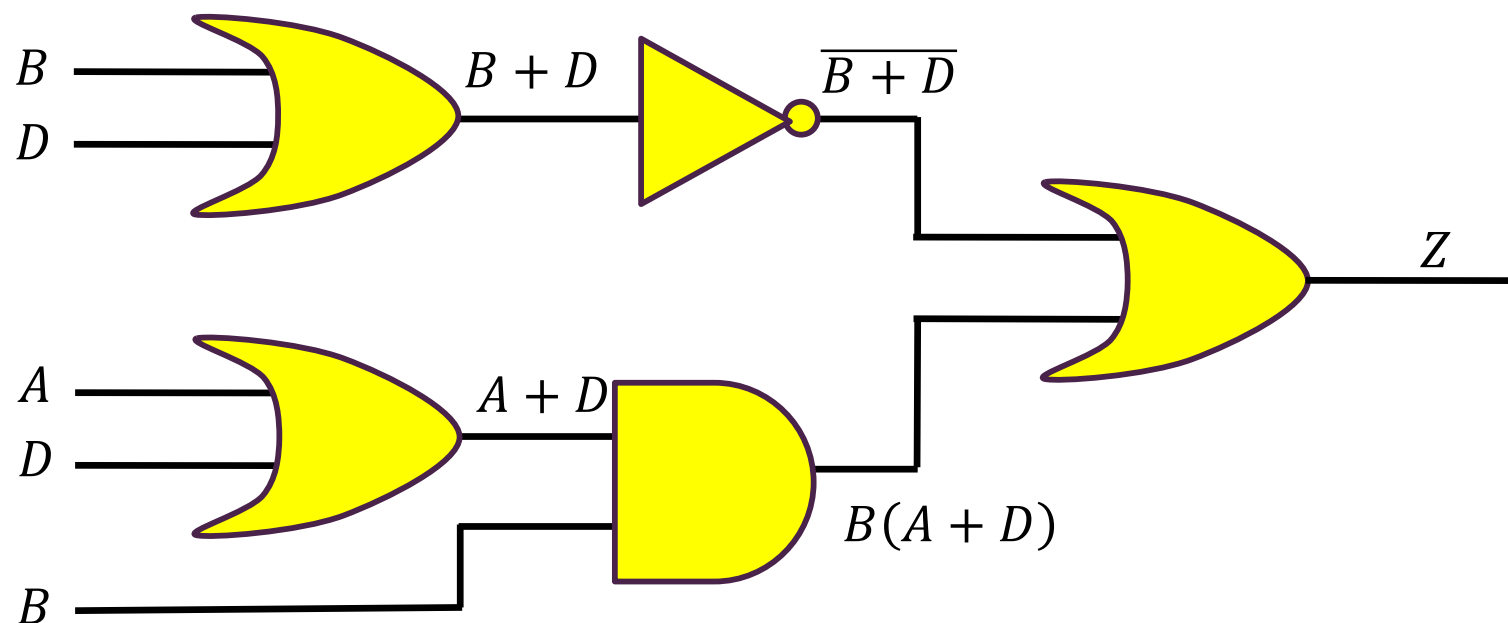


14个晶体管，5层垒叠

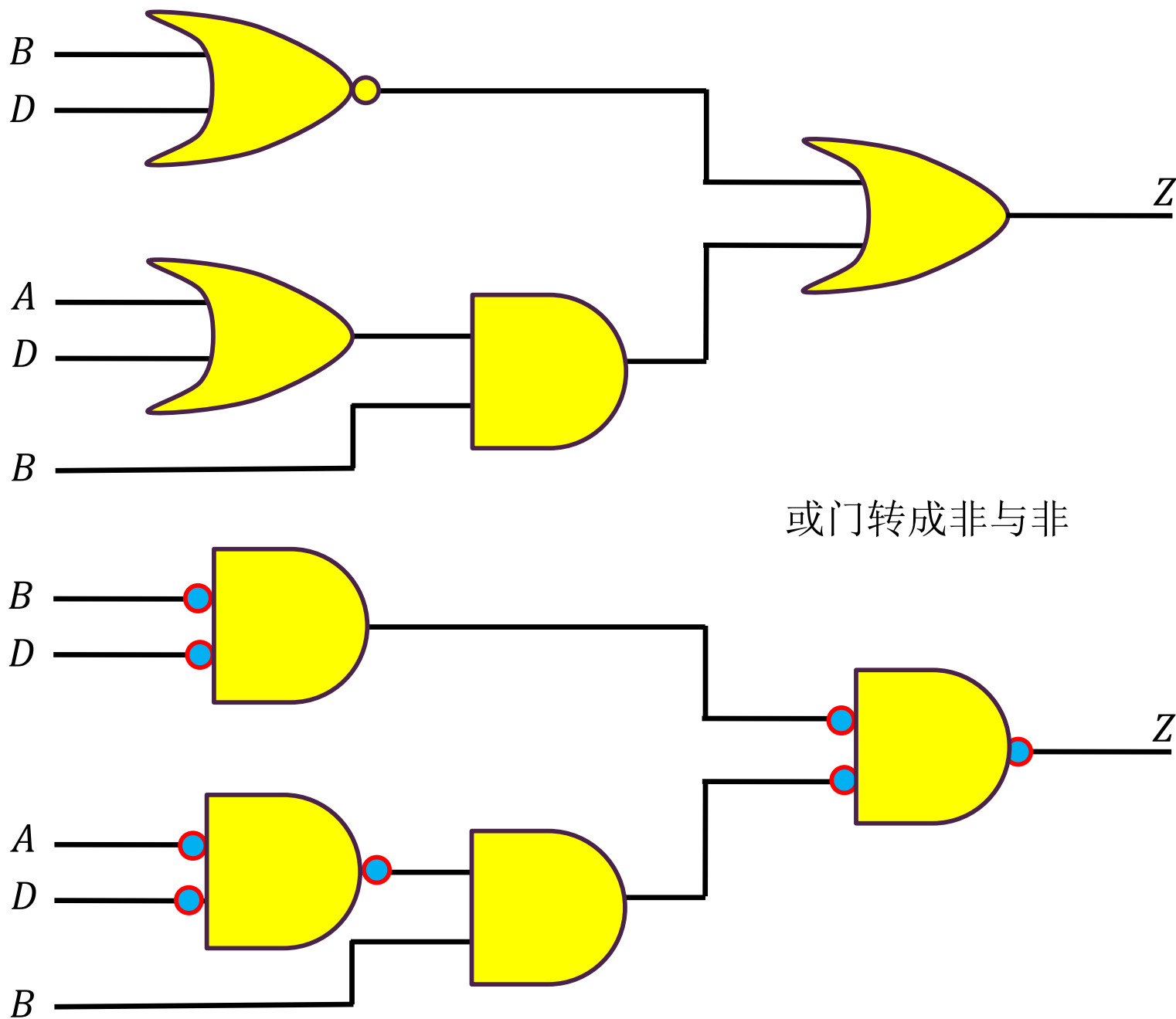
逻辑级实现

如果有与门、或门、非门电路可供选用，用与、或、非门实现上述逻辑

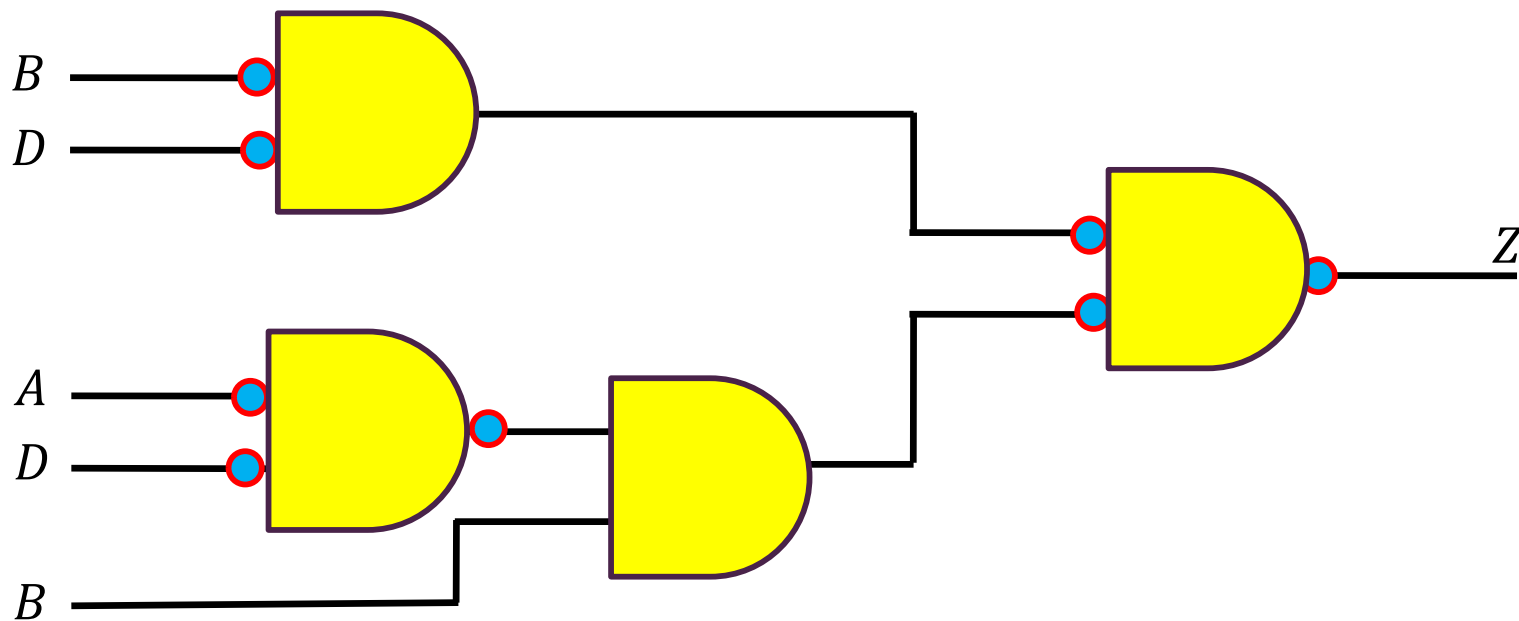
$$Z = A \cdot B + B \cdot D + \overline{B} \cdot \overline{D} = B \cdot (A + D) + \overline{B + D}$$



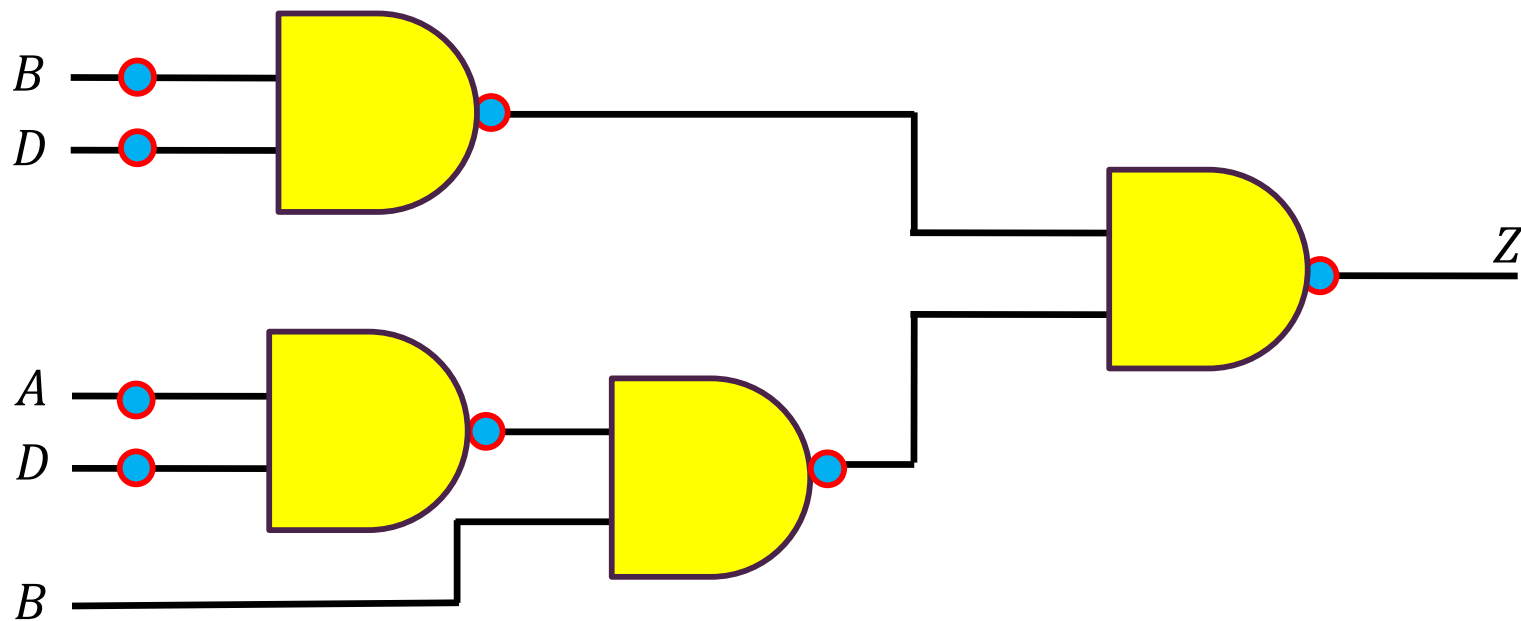
$$Z = A \cdot B + B \cdot D + \overline{B} \cdot \overline{D} = B \cdot (A + D) + \overline{B + D}$$



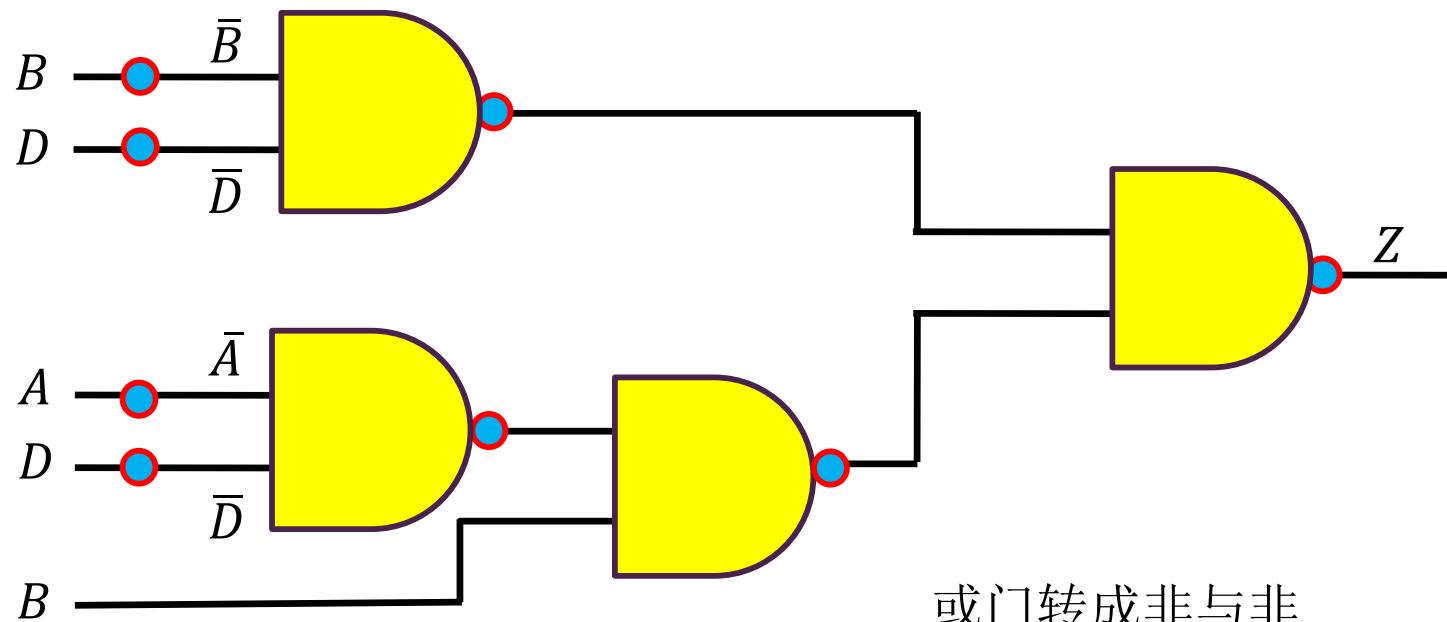
与非门



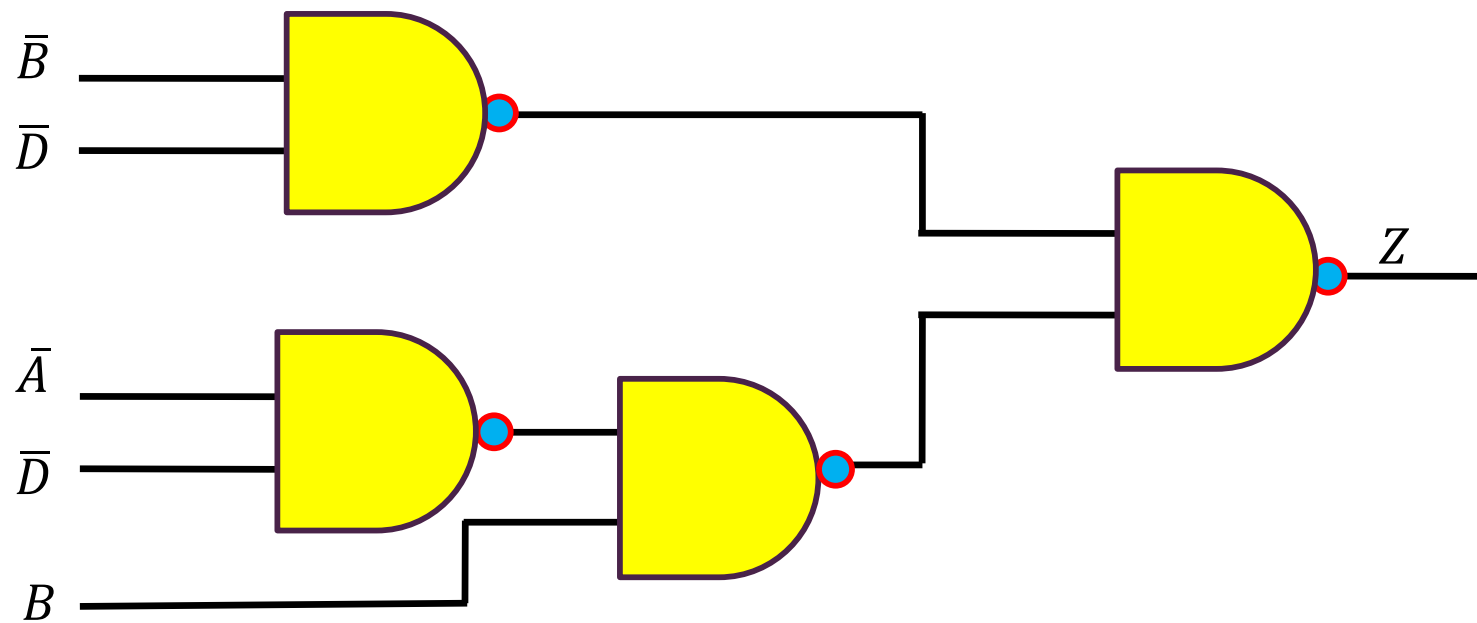
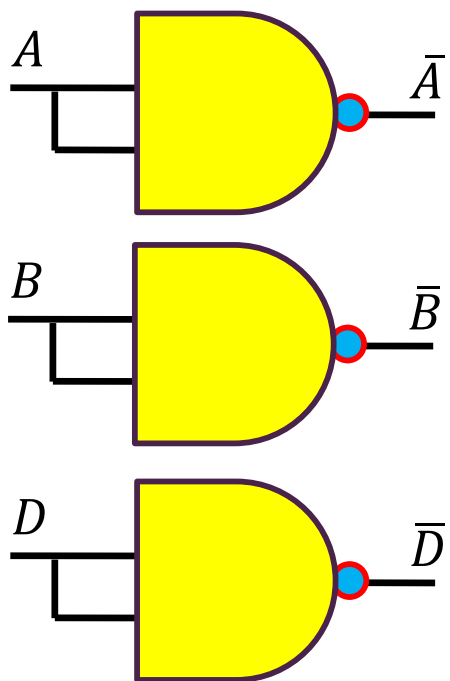
与非之外看剩非



非门实现

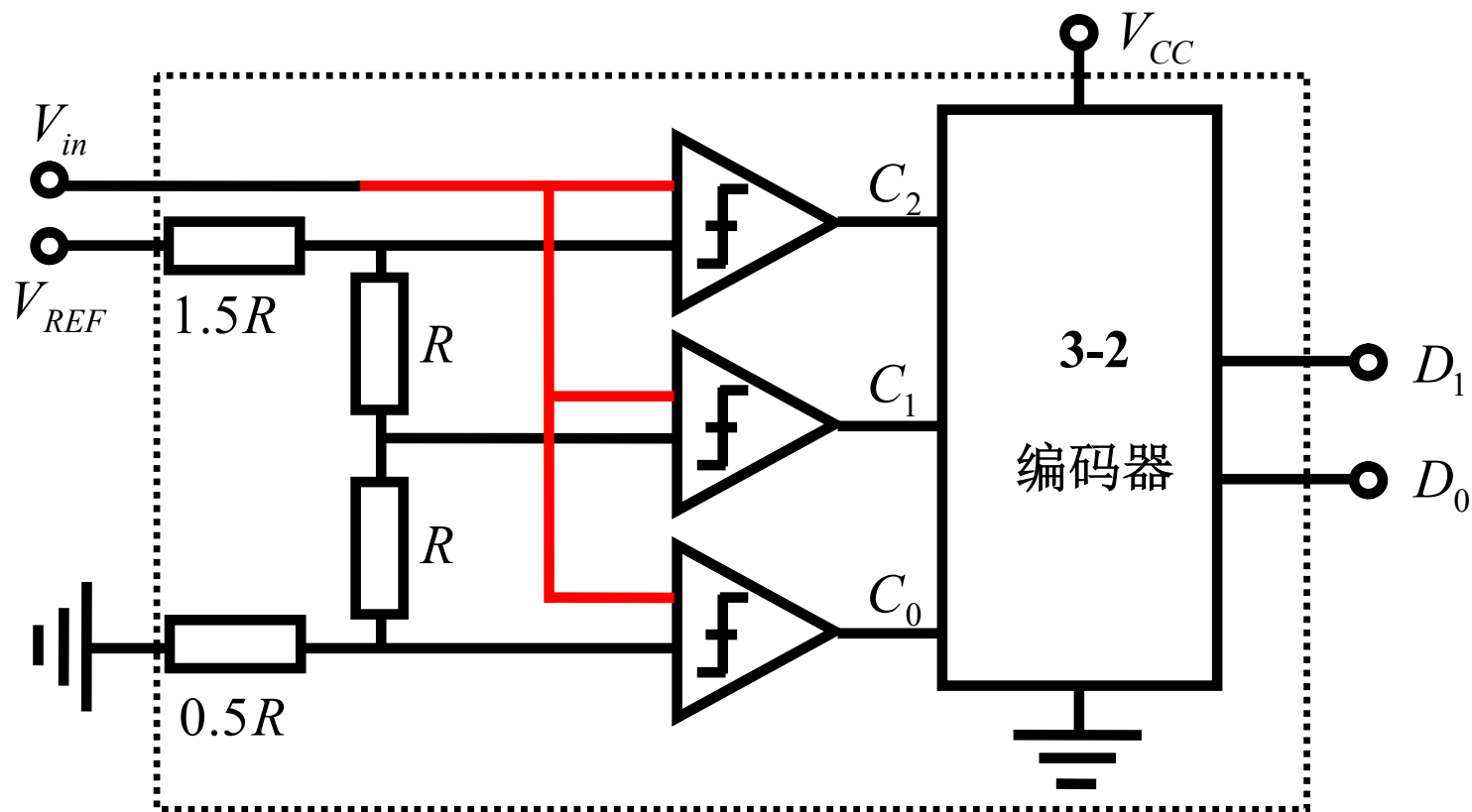


与非之外看剩非



剩非非门用与非

作业13.3 Flash ADC 编码器设计



模拟输入电压	$C_2C_1C_0$	数字输出码 D_1D_0
$\dots \sim 1/8V_{REF}$	000	00
$1/8V_{REF} \sim 3/8V_{REF}$	001	01
$3/8V_{REF} \sim 5/8V_{REF}$	011	10
$5/8V_{REF} \sim \dots$	111	11

编码器设计

- 已知flash-ADC的码表如左图
- 给出逻辑表达式，用与非门实现上述逻辑表达式
- 画出编码器的CMOS实现方案

$C_2C_1C_0$	数字输出码 D_1D_0
000	00
001	01
011	10
111	11

三个输入变量，共**8**种情况，这里仅有**4**种情况

剩下**4**种情况真值表中都是*(是**0**是**1**不在乎)，因为在实际电路中，这**4**种情况不会也不应该出现

从码表到卡诺图

$C_2C_1C_0$	数字输出码 D_1D_0
000	00
001	01
011	10
111	11

D_1

$C_2 \backslash C_1C_0$	00	01	11	10
0	0	0	1	*
1	*	*	1	*

$$D_1 = C_1$$

D_0

$C_2 \backslash C_1C_0$	00	01	11	10
0	0	1	0	*
1	*	*	1	*

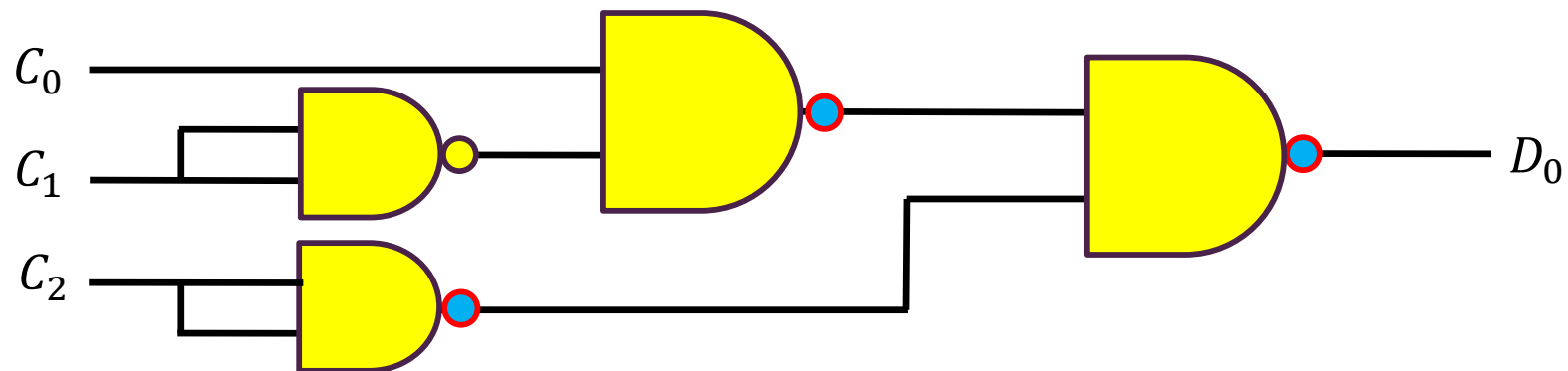
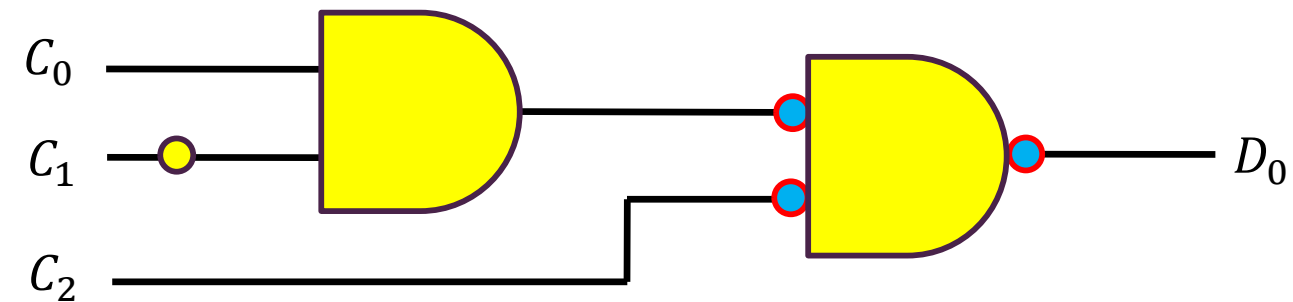
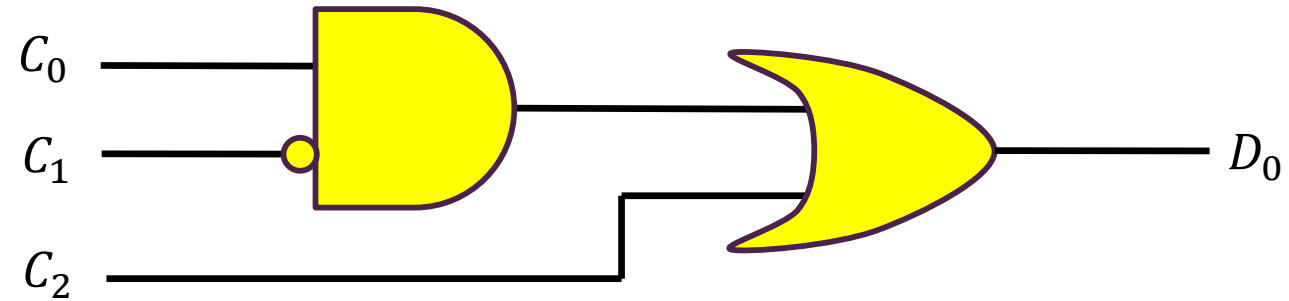
$$D_0 = C_2 + \overline{C_1} \cdot C_0$$

与非门实现

$$D_1 = C_1$$



$$D_0 = C_2 + \overline{C_1} \cdot C_0$$

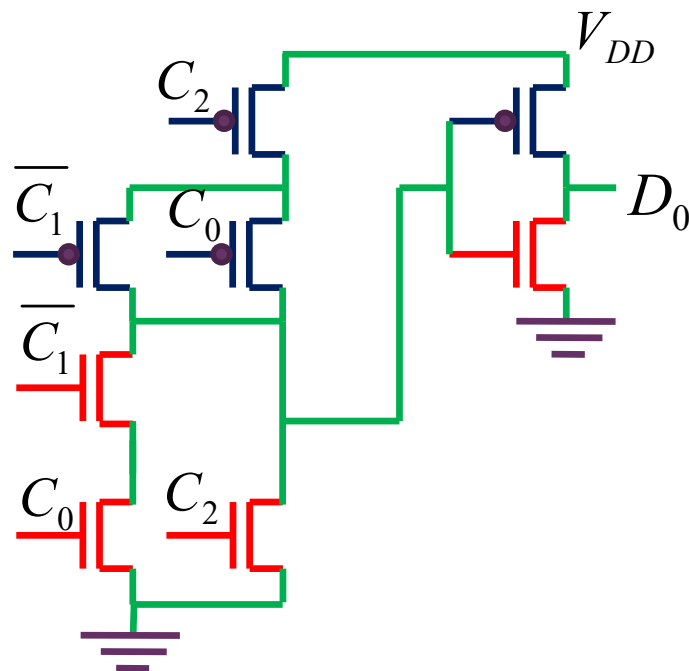
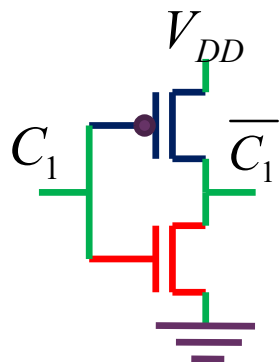


CMOS晶体管级实现

$$D_1 = C_1$$



$$D_0 = C_2 + \overline{C_1} \cdot C_0$$



用与非门可以实现与、或、非三种逻辑运算功能。首先用正常的与、或、非实现逻辑关系，之后按如下口诀，可将其中的所有门电路全部转换为与非门电路连接关系：【填空1】转成【填空2】，与非之外看【填空3】。

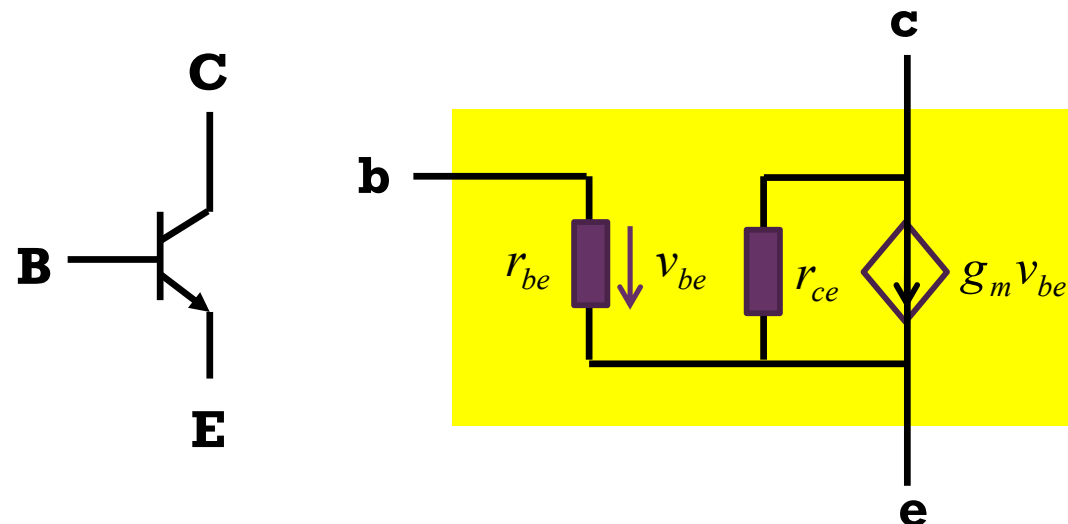
正常使用填空题需3.0以上版本雨课堂

作答

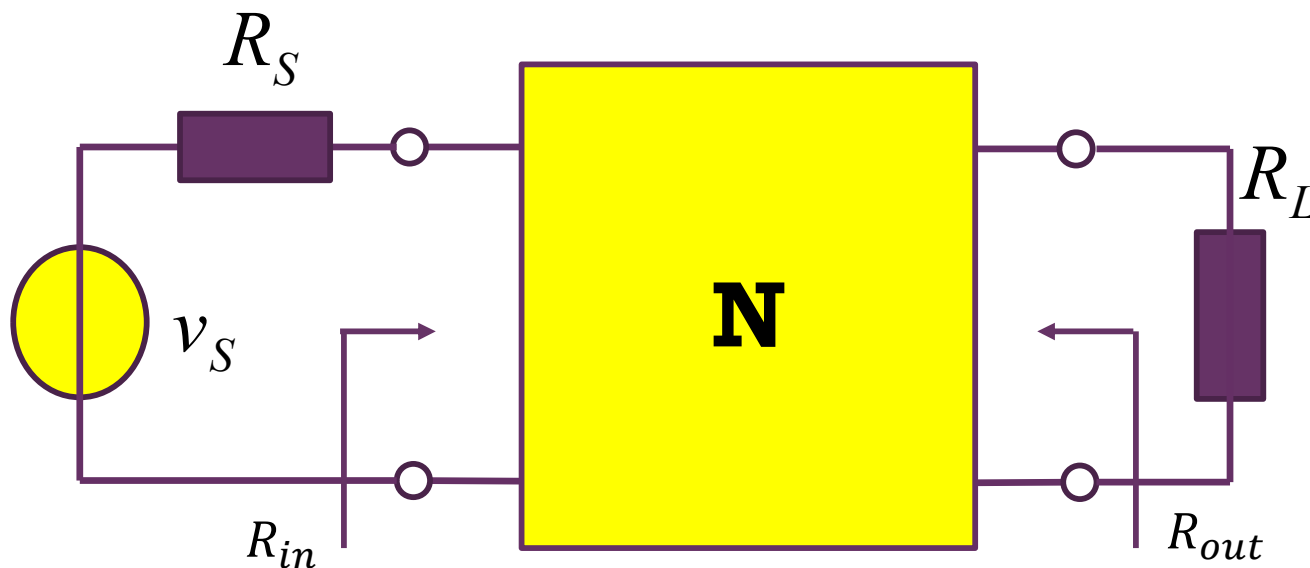
第13讲 三种组态 作业

作业1 输入电阻和输出电阻的变化规律

- 晶体管模型如图所示，研究三种组态的输入电阻 R_{in} 随负载电阻 R_L 变化的特性曲线示意图，输出电阻 R_{out} 随信源内阻 R_S 变化的特性曲线示意图，并在特性曲线上进一步理解理想晶体管的输入、输出阻抗

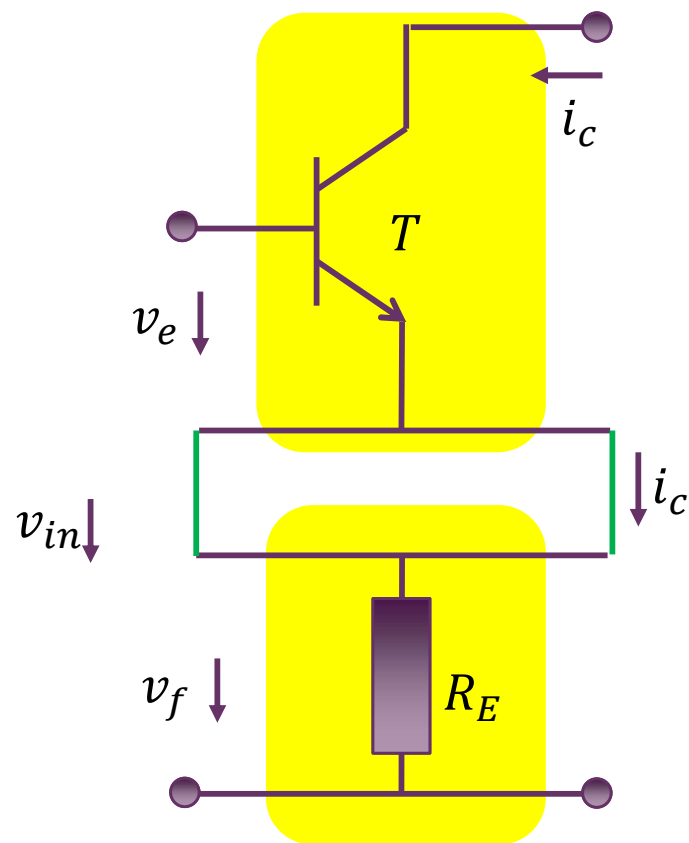
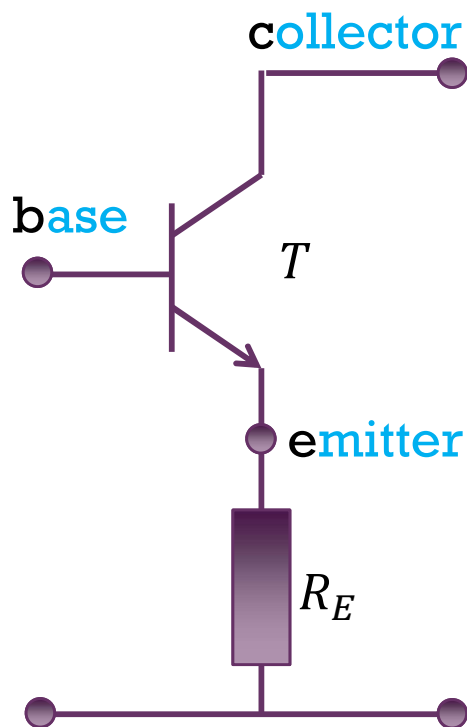


- 选作：如果 $R_S = R_{in}$ ， $R_L = R_{out}$ ，两个端口同时匹配，该二端口网络将具有最大功率增益，求三种组态晶体管放大器的最大功率增益



作业2 非理想晶体管射极串联负反馈

- 理想晶体管添加射极串联负反馈电阻后，整体仍然可以视为理想晶体管，仅仅是跨导增益下降了
- 对于非理想晶体管，求其 y 参量矩阵，并确认 y 参量等效电路，分析和理想晶体管比较，有何差别？和原始晶体管比，有何差别？



CAD (选作)

- 以晶体管放大器为例，假设二端口网络（晶体管）网络参量/等效电路已经确定，通过对 R_S 、 R_L 扫描，仿真验证存在最佳的信源内阻和负载电阻，使得该网络具有最大功率增益
- 如果二端口网络网络参量/等效电路已经确定，那么最大功率增益对应的信源内阻和负载电阻可以理论上确定
- 对于图示晶体管等效电路，三种组态晶体管的最大功率传输匹配负载分别是多少？理论和仿真结果对应。

